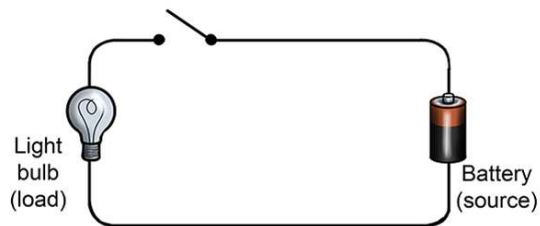
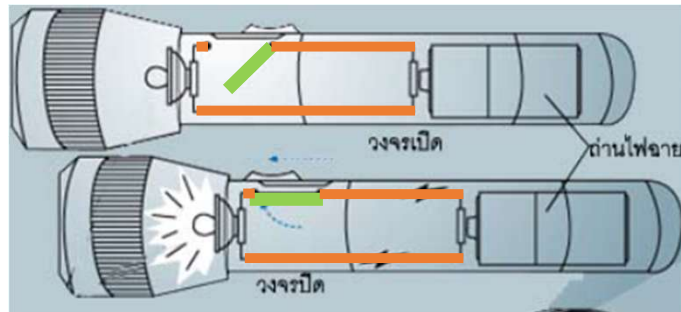
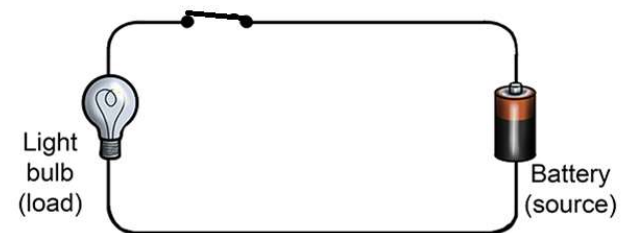
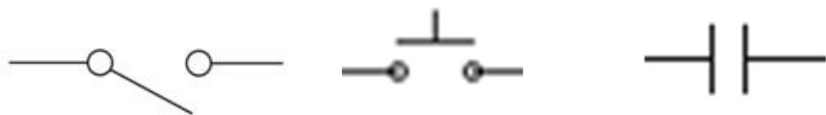


การควบคุมสวิตช์ Switch Control



วงจรเปิด (Open Circuit)

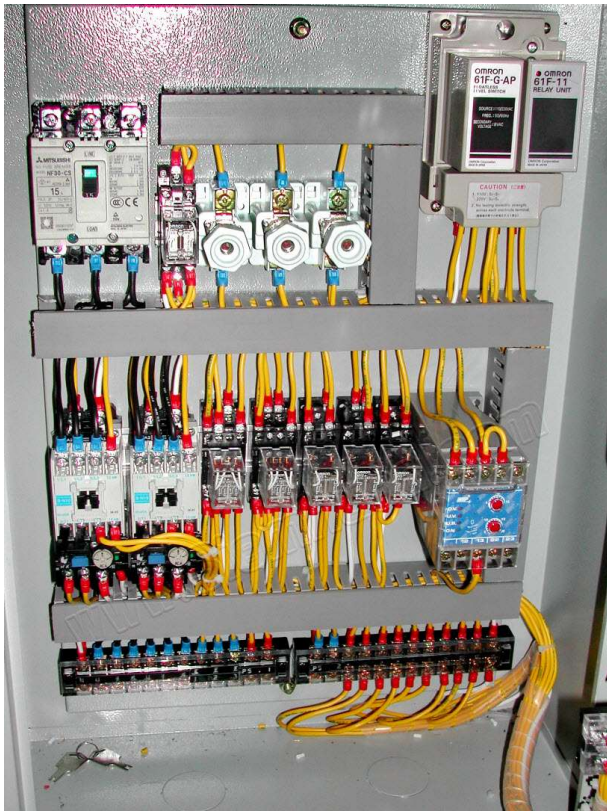
- กระแสไฟฟ้าไม่สามารถไหลผ่านได้ ทำให้อุปกรณ์ไม่ทำงาน
- หน้าสัมผัสไม่เชื่อมต่อกัน (Open Contact)



วงจรปิด (Close Circuit)

- หน้าสัมผัสเชื่อมต่อกัน (Close Contact)
- กระแสไฟฟ้าไหลในวงจรได้ ทำให้อุปกรณ์ทำงาน





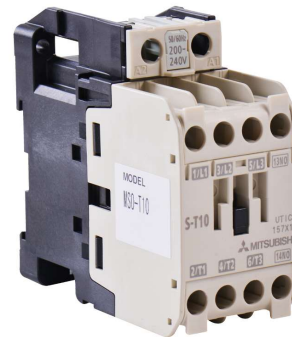
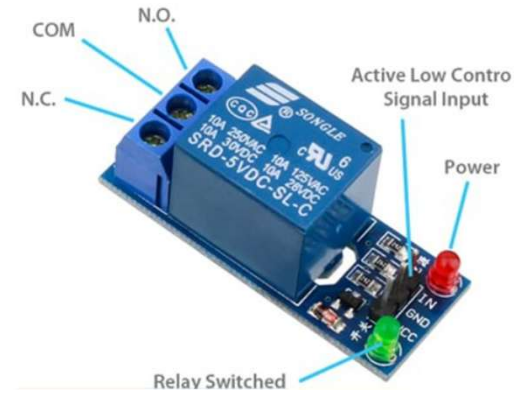
Normally Open Contact



Normally Closed Contact



Changeover Contact

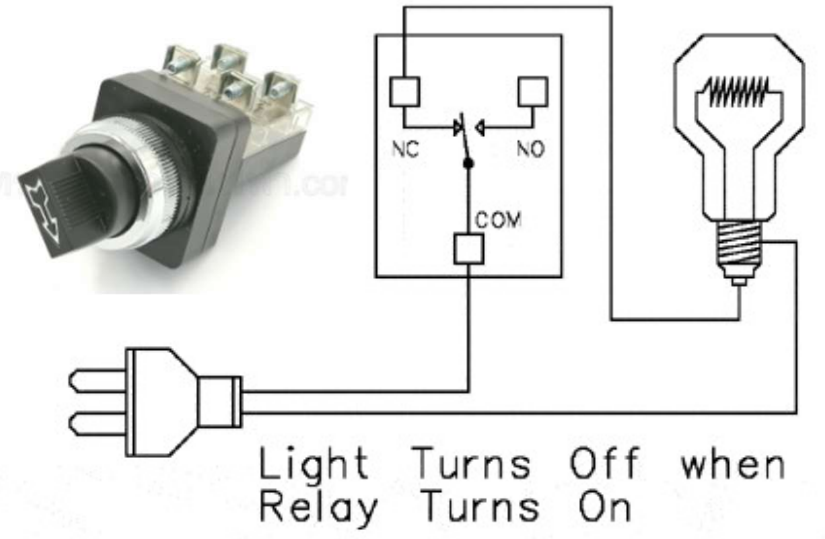
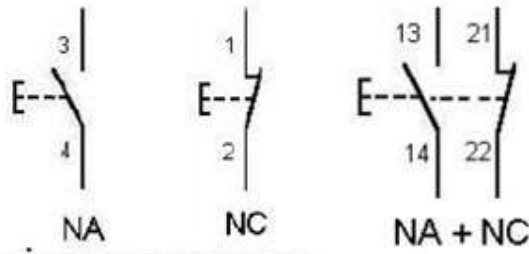
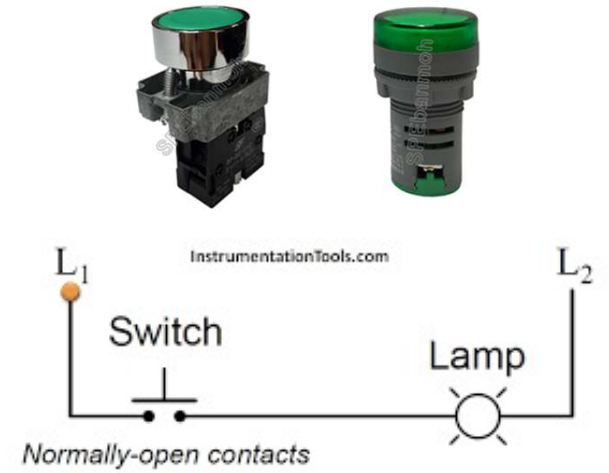
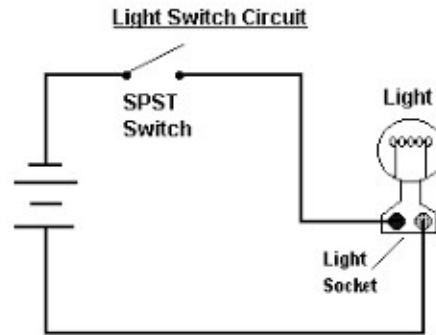




หมุนปุ่มมาทาง MAN ตั้งเปิดปิด จากสวิตช์หน้าตู้

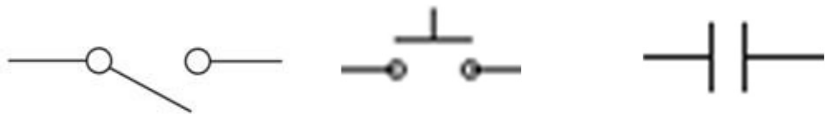


หมุนปุ่มมาทาง AUTO รอเวลาที่ตั้งไว้ จาก timer



วงจรเปิด (Open Circuit)

- กระแสไฟฟ้าไม่สามารถไหลผ่านได้ ทำให้อุปกรณ์ไม่ทำงาน
- หน้าสัมผัสไม่เชื่อมต่อกัน (Open Contact)



A Normal Open contact (NO)

ปกติหน้าสัมผัสเปิด

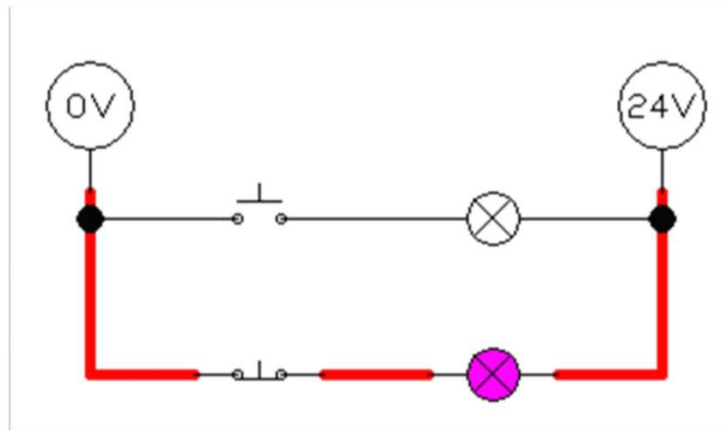
วงจรปิด (Close Circuit)

- หน้าสัมผัสเชื่อมต่อกัน (Close Contact)
- กระแสไฟฟ้าไหลในวงจรได้ ทำให้อุปกรณ์ทำงาน



A Normal Closed contact (NC)

ปกติหน้าสัมผัสปิด



Pressure se...

Analog pres...

Analog flow ...

Pneumatic t...

Displaceme...

Voltmeter

Ampere meter

Indicator light

Buzzer

Relay

Relay with ...

Relay with ...

Relay counter

Valve sole...

Proportional...

Starting curr...

Electrical co...

Electrical co...

Function ge...

Setpoint val...

Pushbutton ...

Detent swit...

Pushbutton ...

Detent swit...

Detent swit...

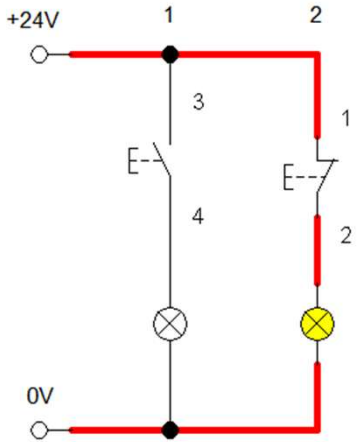
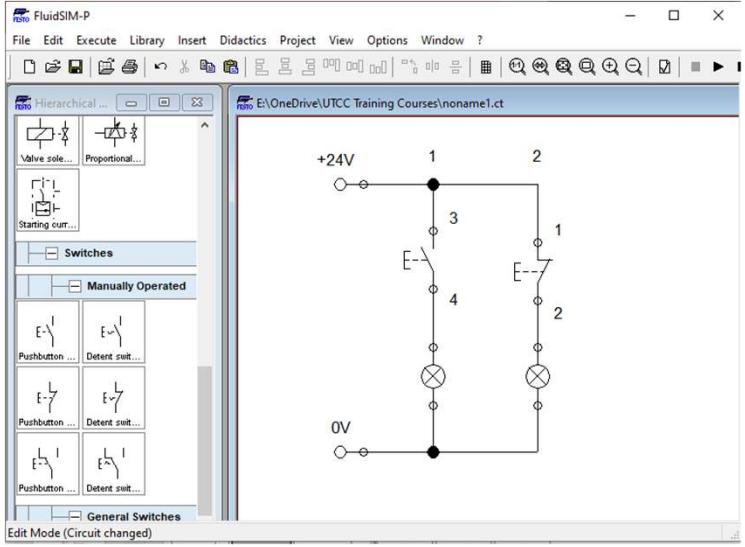
Pushbutton ...

Detent swit...

Make switch

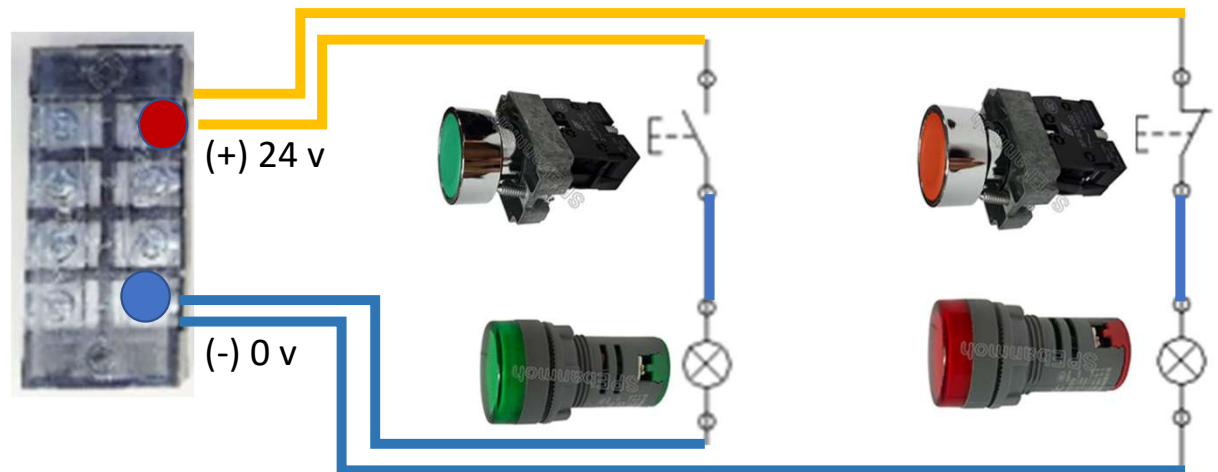
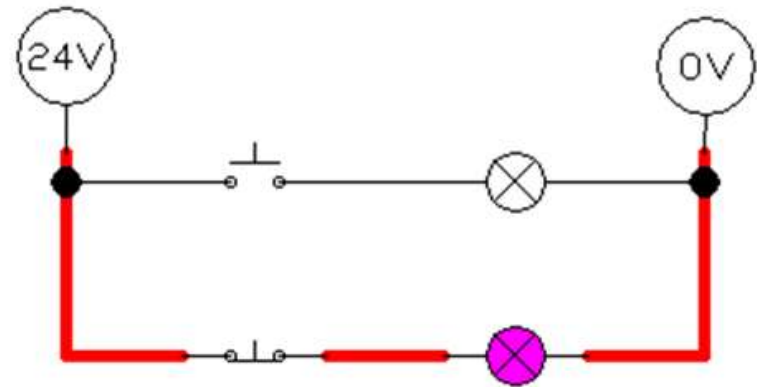
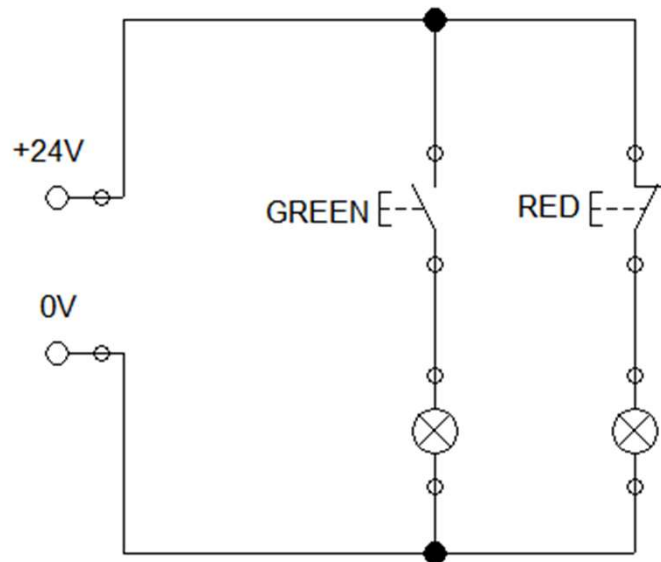
Break switch

Changeover...

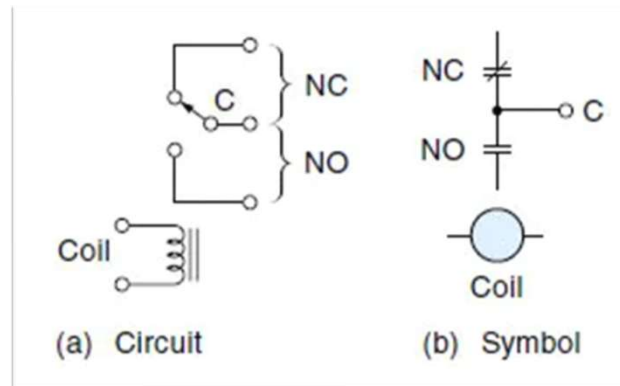
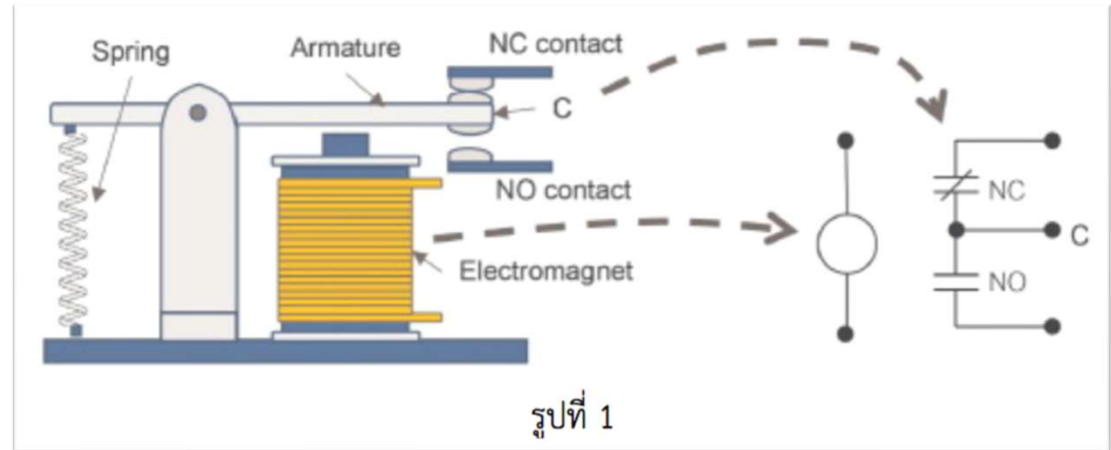
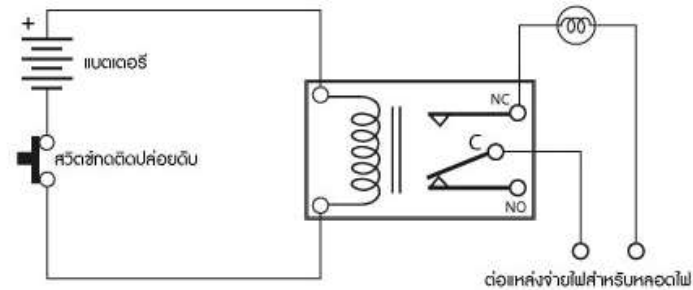
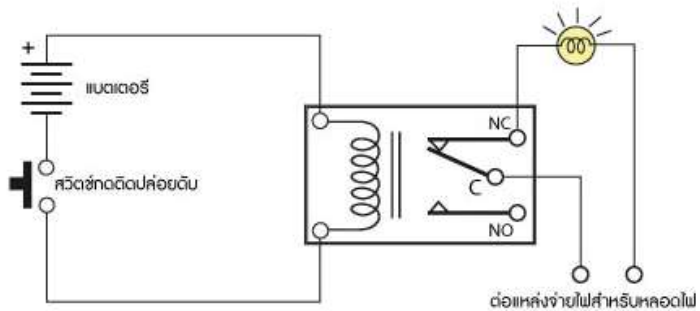


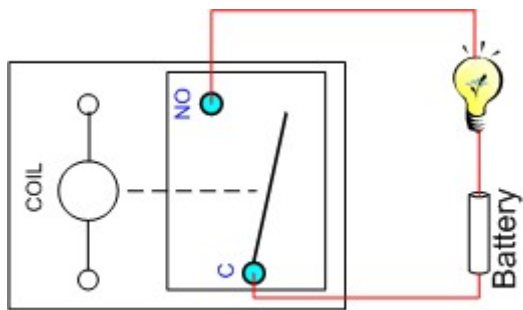
Lab 1

Basic 01 - Switch Control (NO-NC)

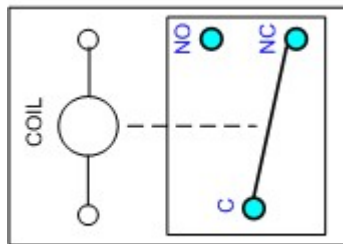


รีเลย์ Relay (สวิตช์ ที่ใช้กระแสไฟฟ้าในสั้งงาน หน้าสัมผัสเปิดปิด)

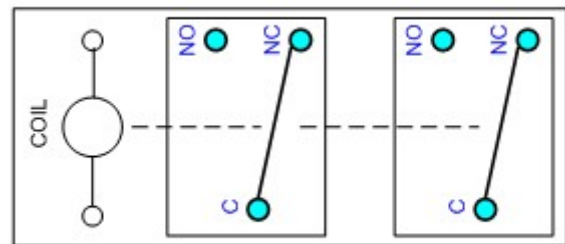




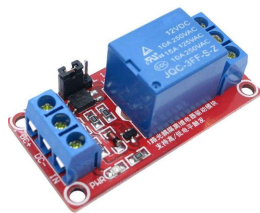
Single Pole, Single Throw (SPST)



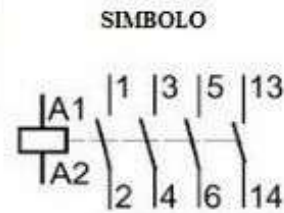
Single Pole, Double Throw (SPDT)



Double Pole, Double Throw (DPDT)

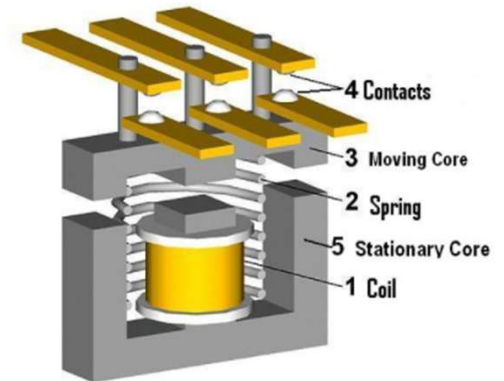


CONTACTOR

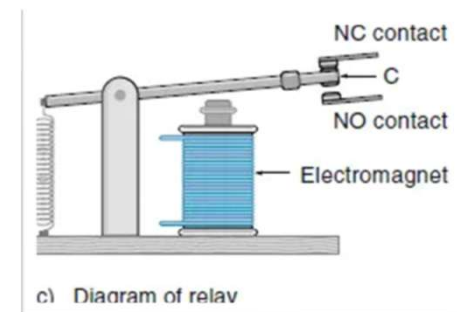


SIMBOLO

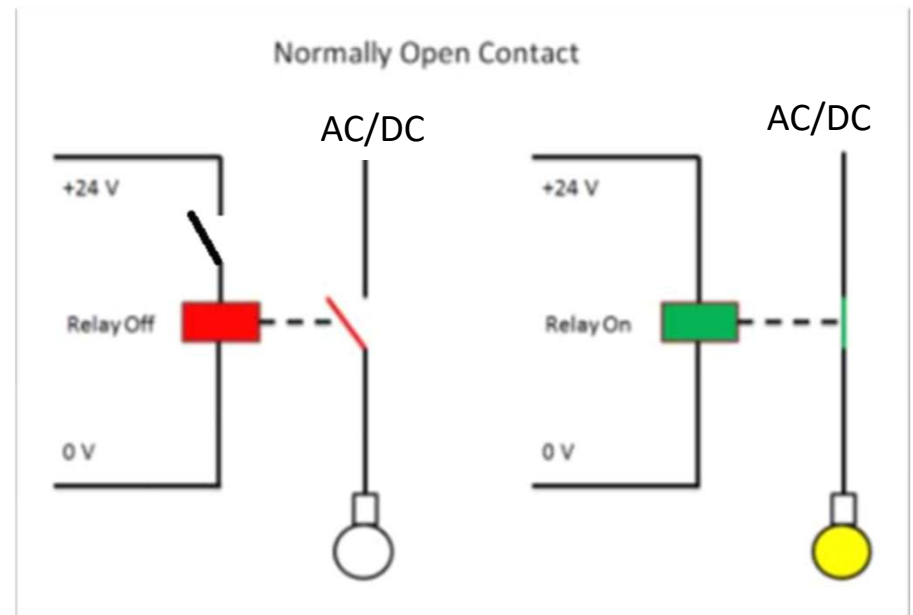
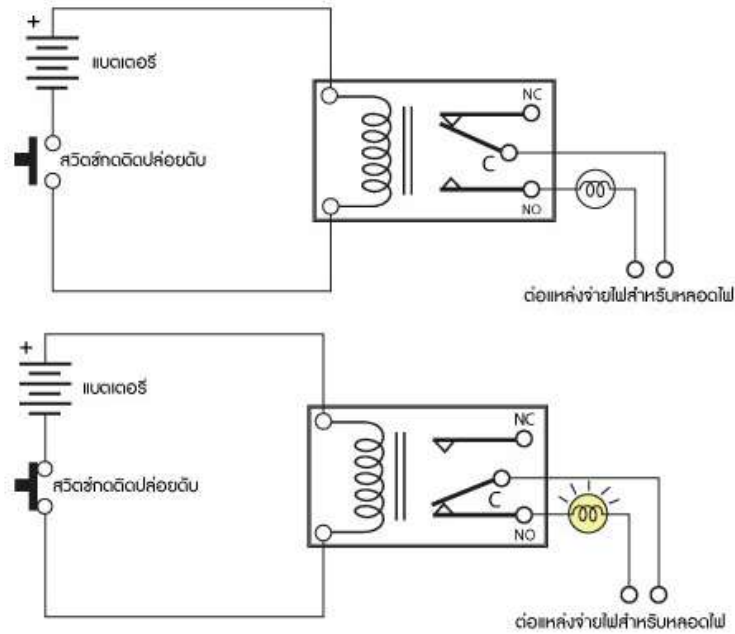
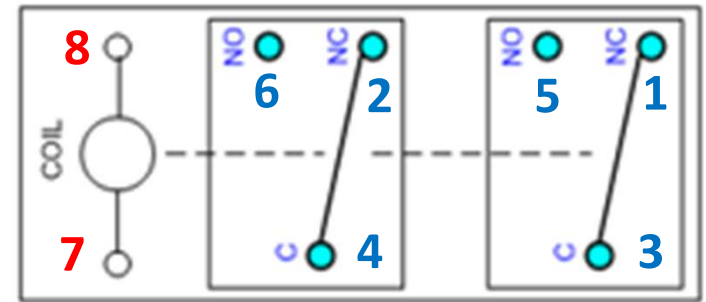
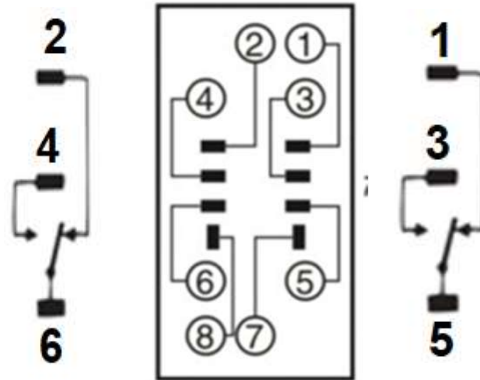
www.areatecnologia.com

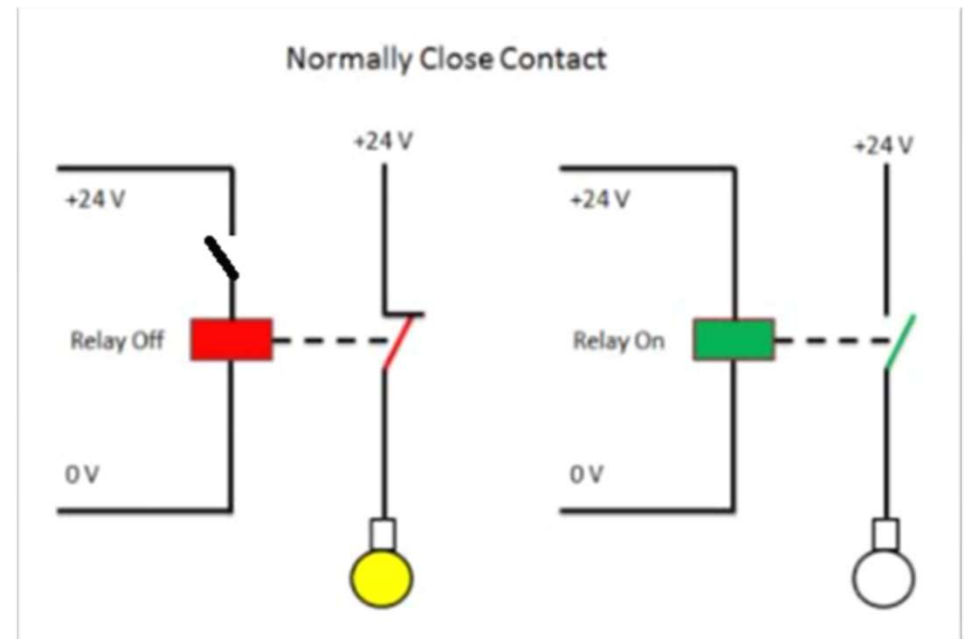
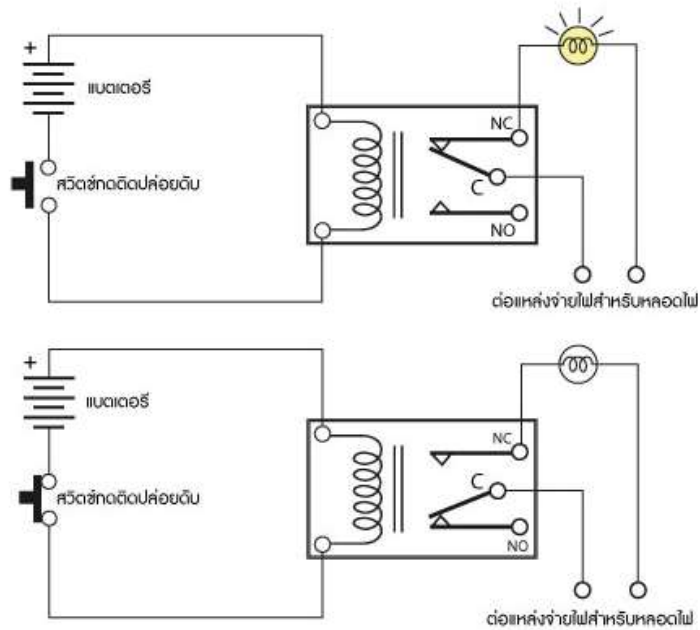
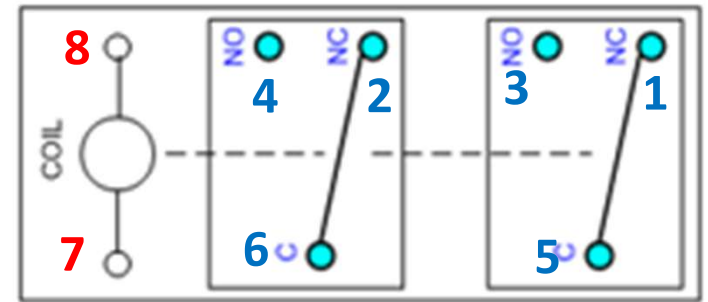
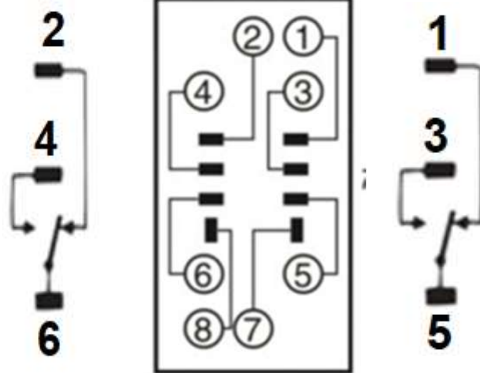


รูปที่ 2

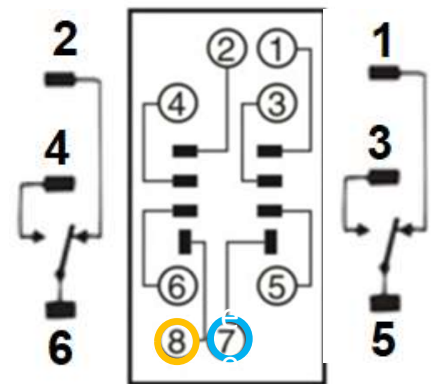
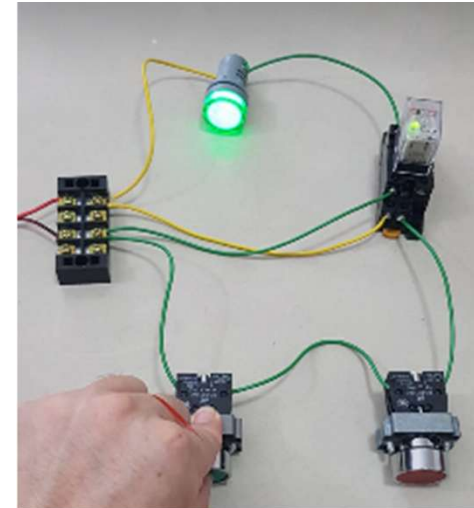
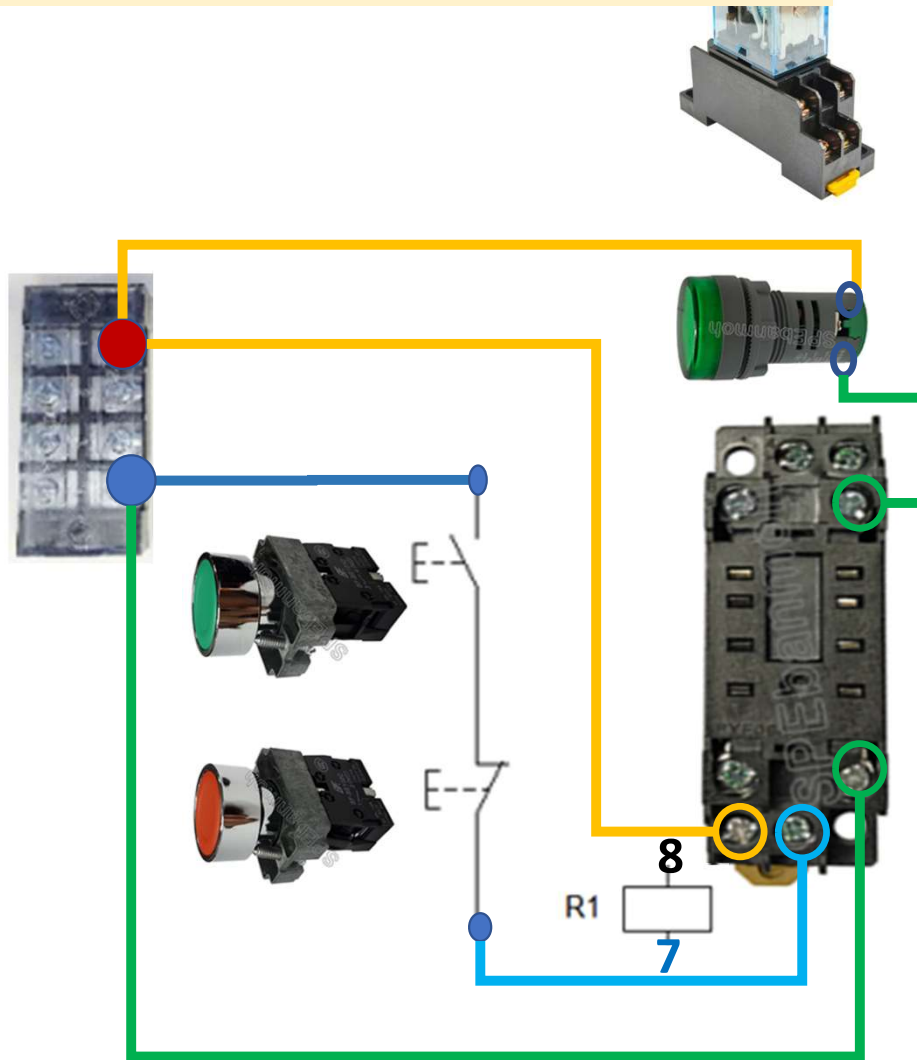
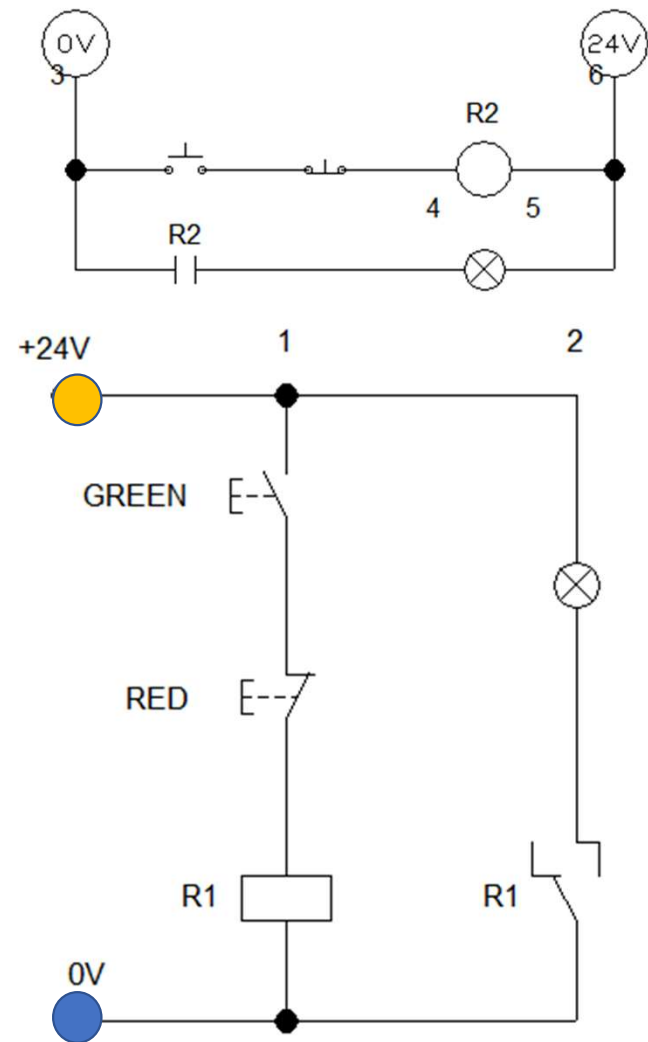


รีเลย์ Relay Switch

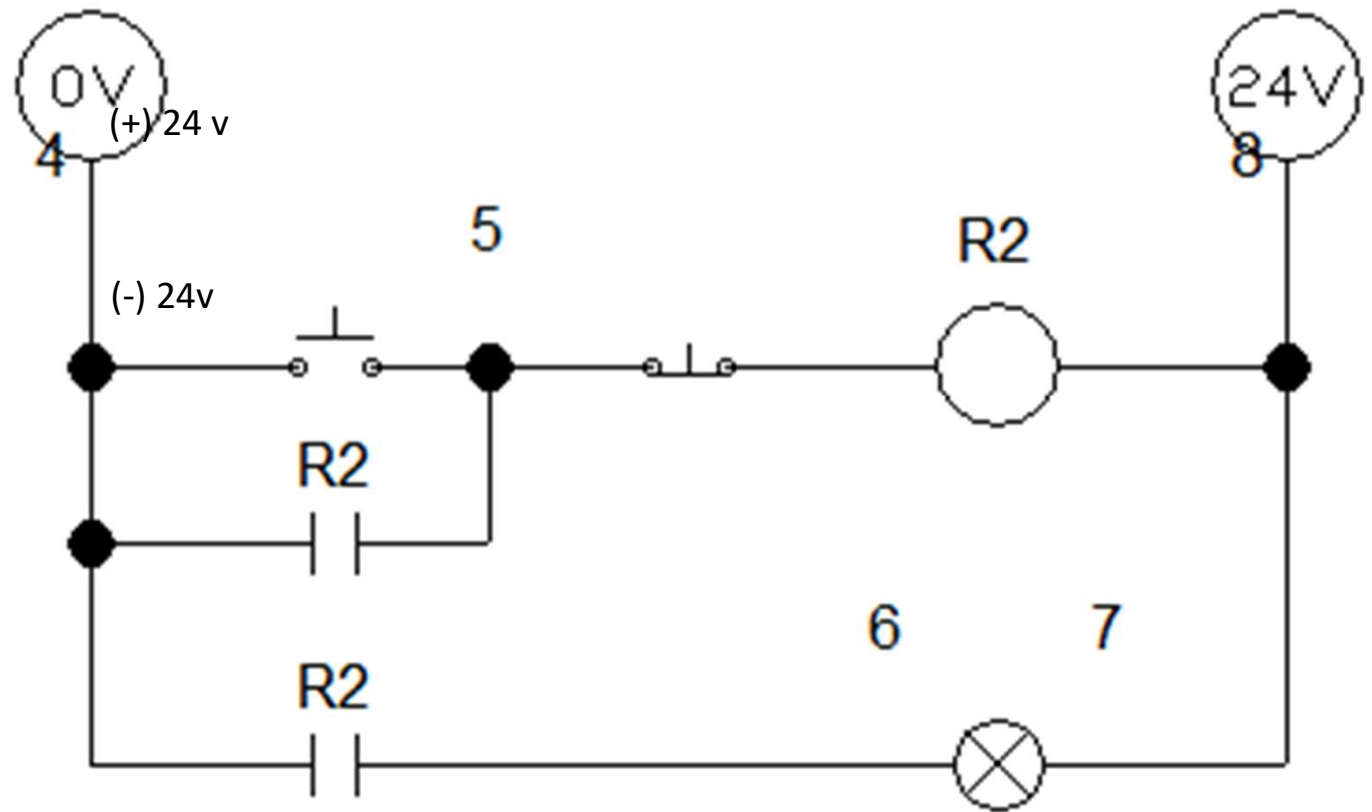
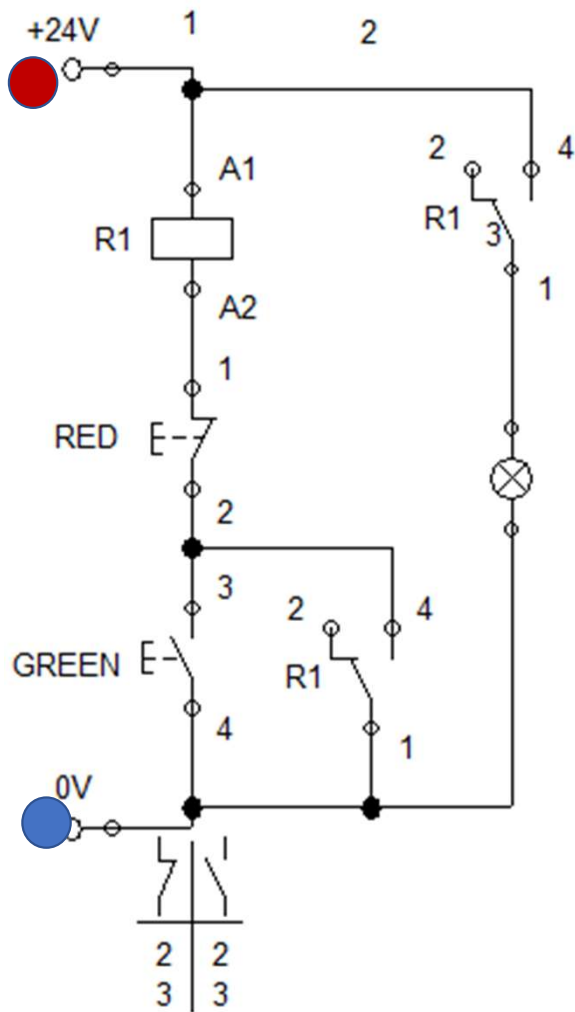


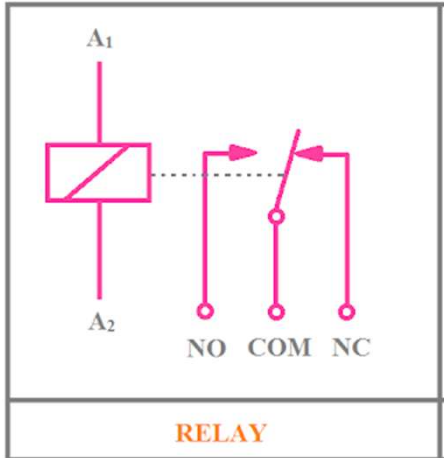


Lab 2 Basic 02 - Relay Control



Lab 3 Relay Control (Self Holding) [youtube](#)





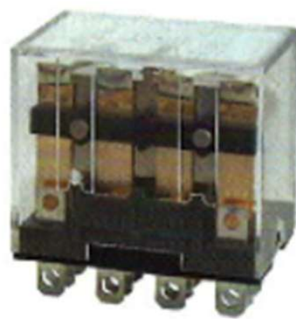
MY2 N
Price:



MY4 N
Price:



LY2N
Price:



LY4N
Price:



MK2P-I
Price:



MK3P-I
Price:



PYF08A
Price:



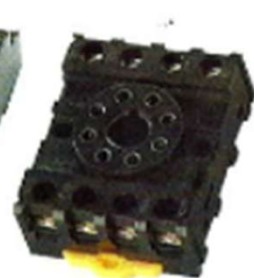
PYF14A
Price:



PYF08A-E
Price:



PYF14A-E
Price:



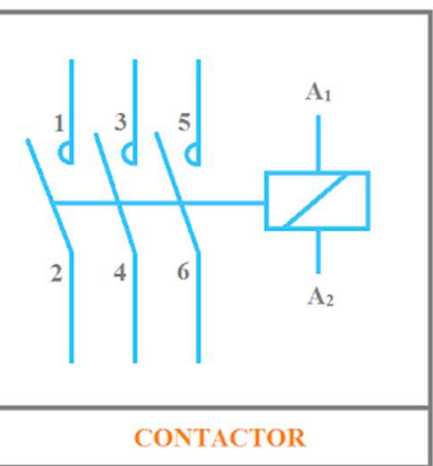
PF083A-E
Price:



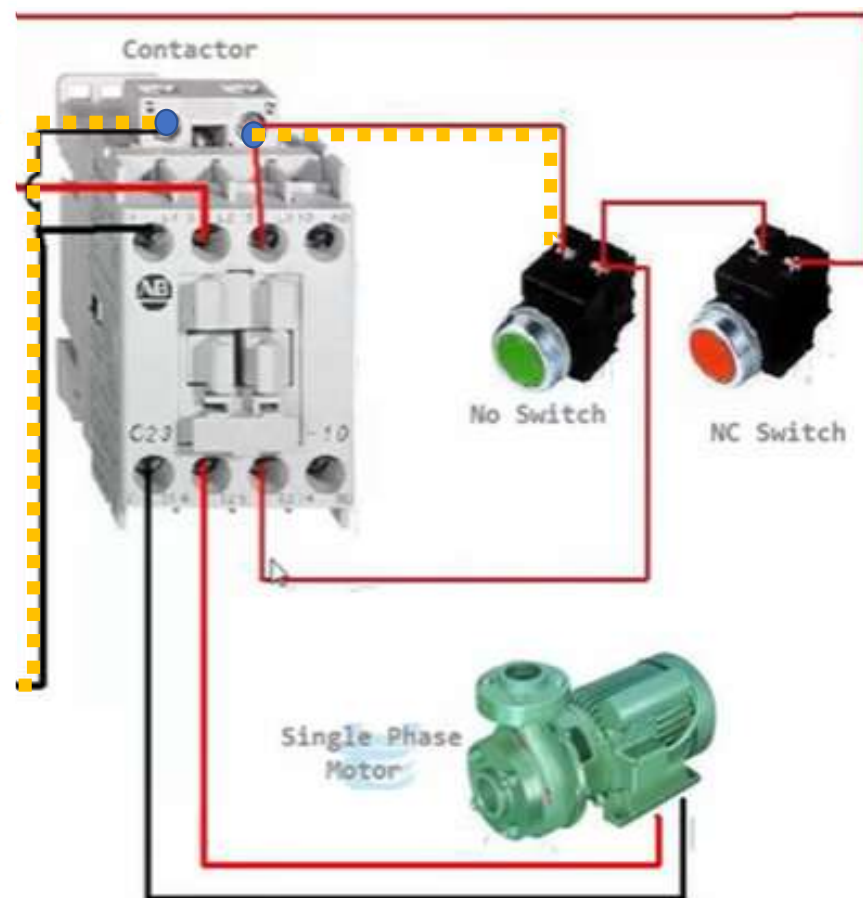
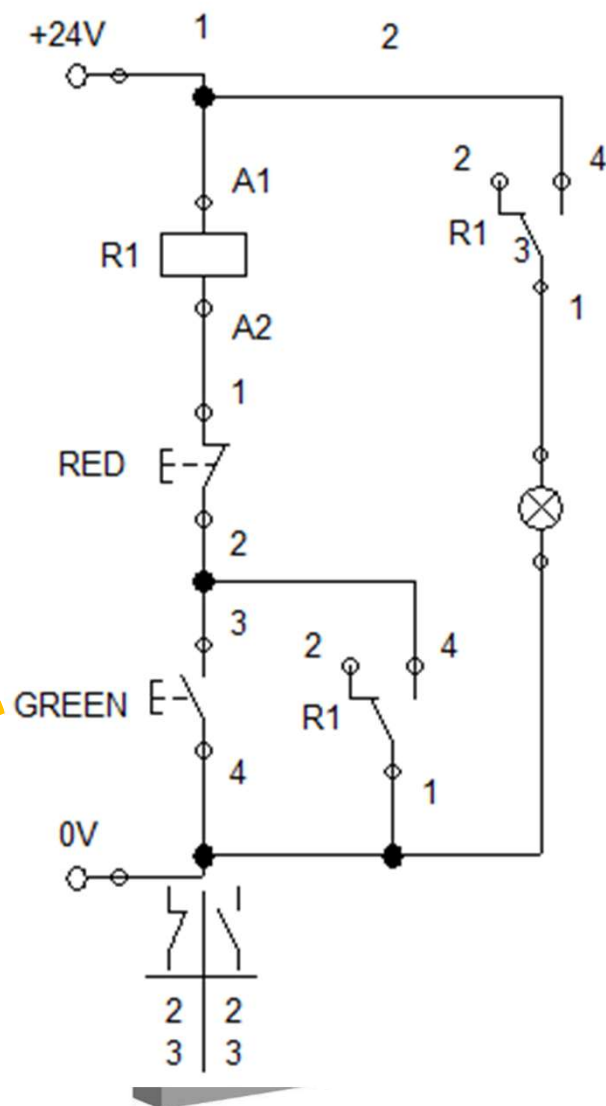
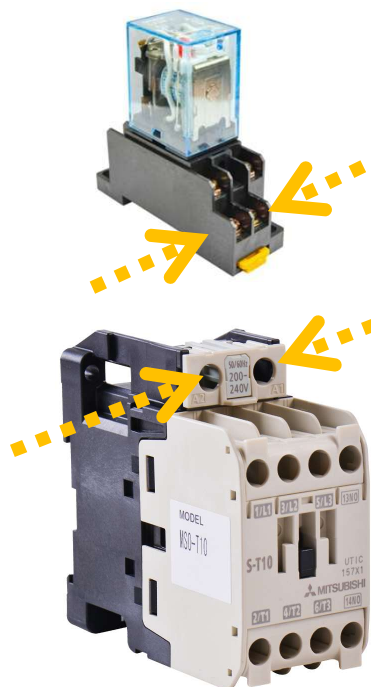
PF113A-E
Price:

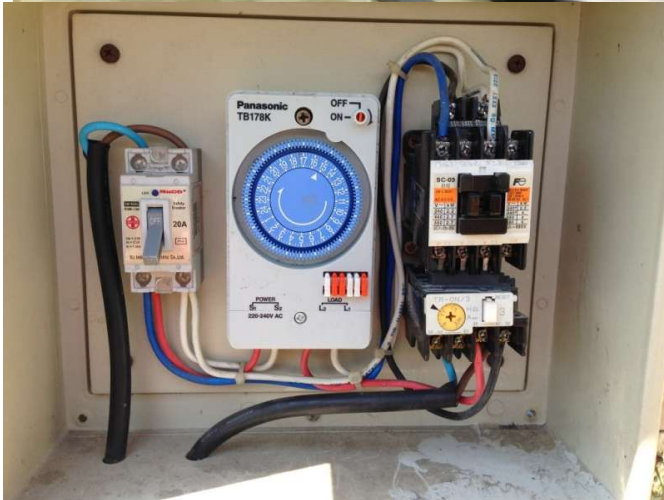
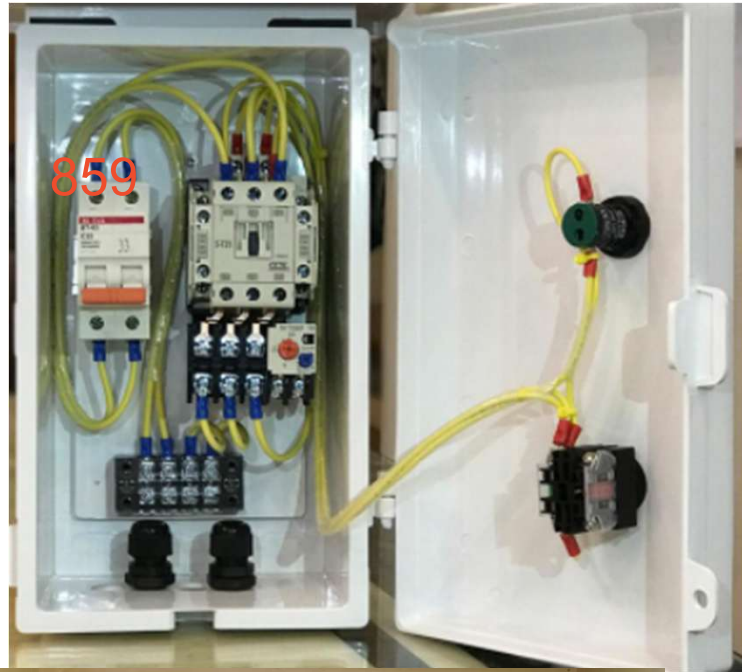


PTF14A-E
Price:

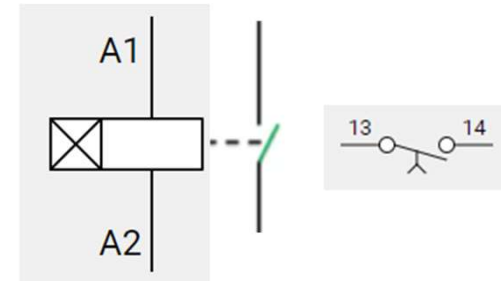


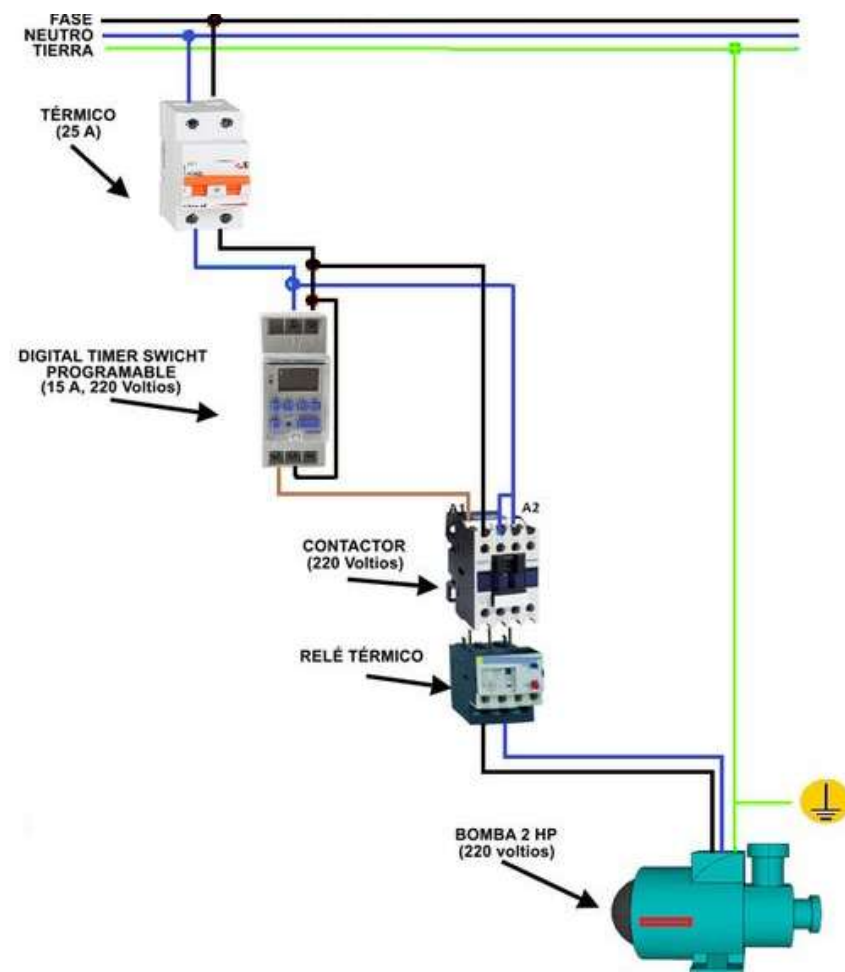
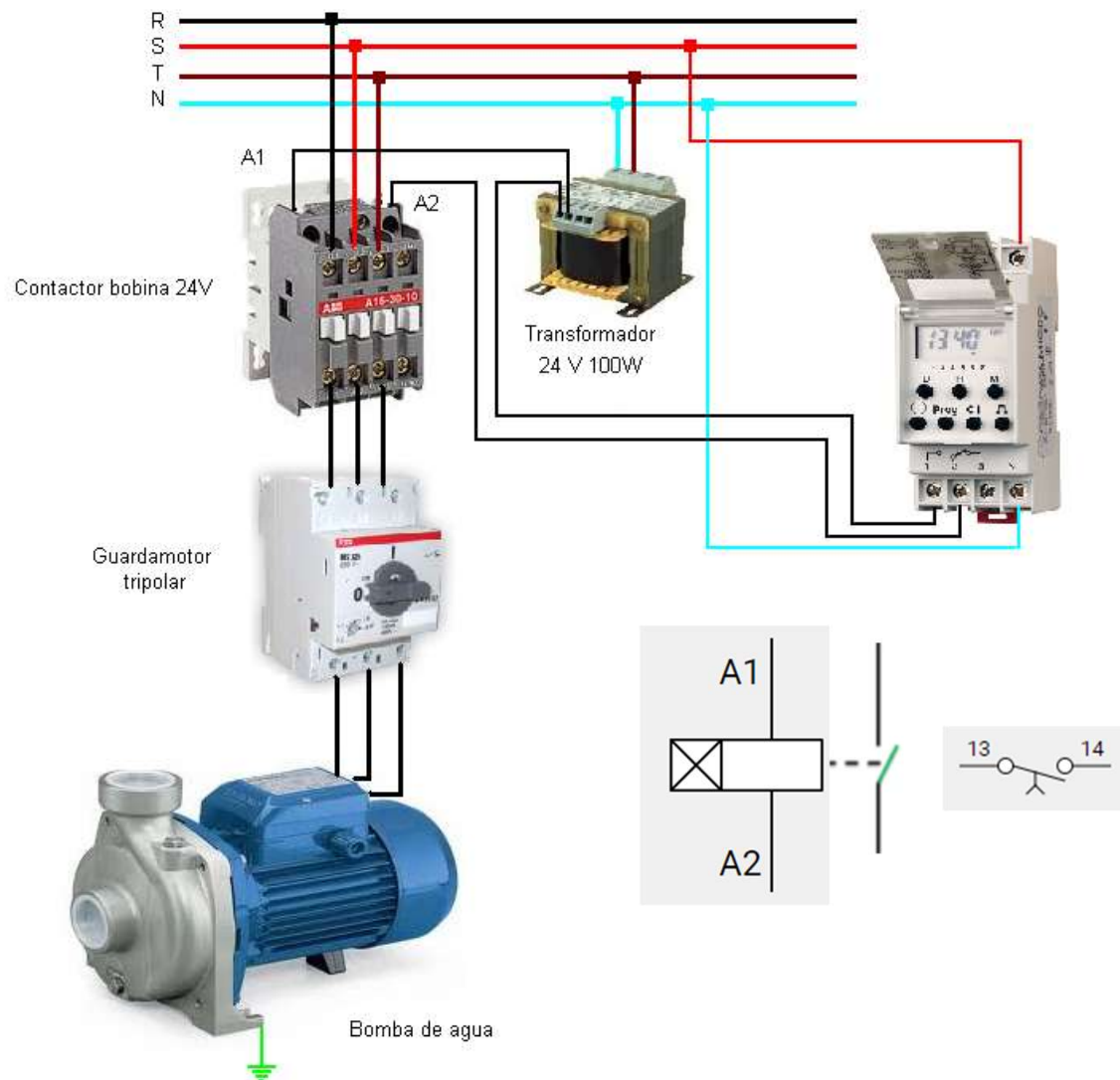
Specifications		Model	MY2 N	MY4 N	LY2	LY4	MK2P-I	MK3P-I
Contact capacity	AC Position		5A 250V	3A 250V	10A 250V	10A 250V	10A 250V	10A 250V
	DC		5A 30V	3A 30V	10A 30V	10A 30V	10A 30V	10A 30V

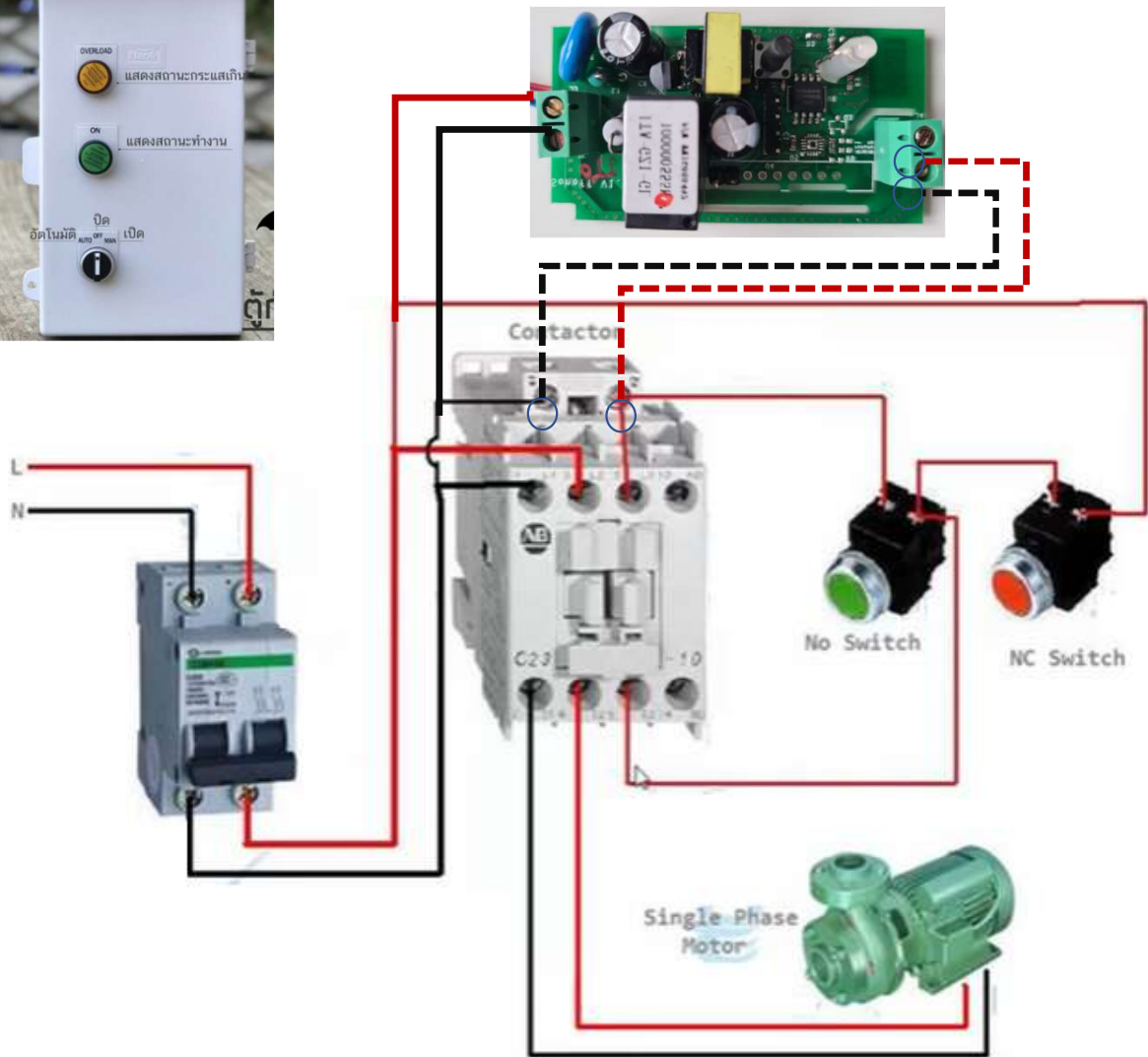
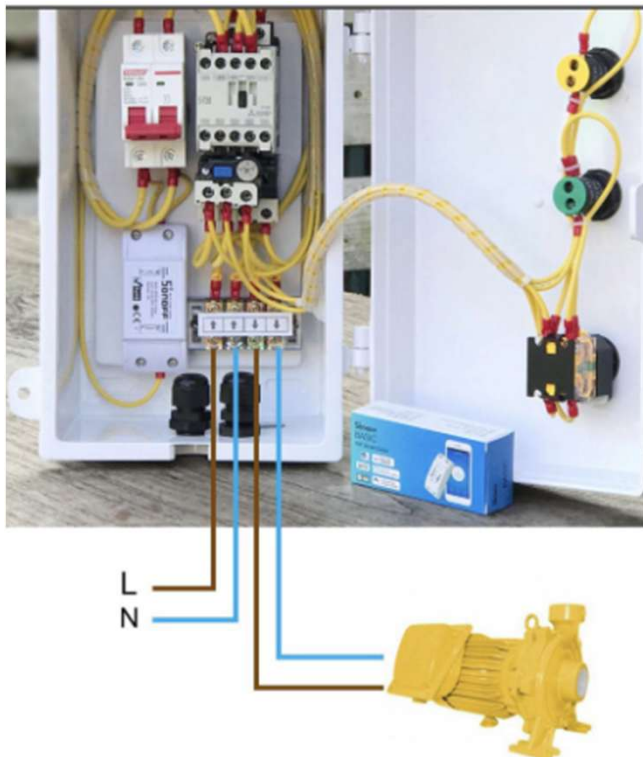


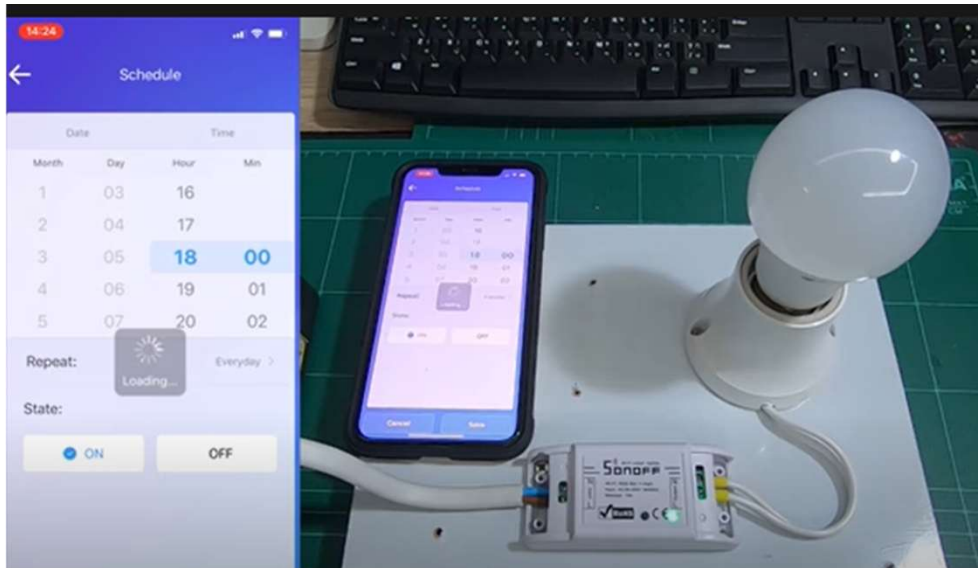


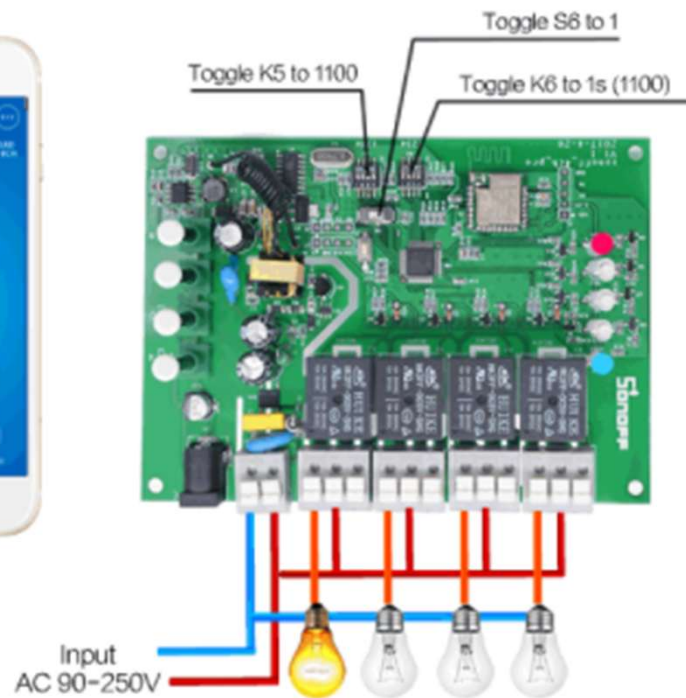
ตัวตั้งเวลา
Timer - 1,495











Sonoff 4CH Pro R3 สวิตช์สั่งงานผ่าน WiFi 4 ช่อง
แอมป์รี รีโมท RF

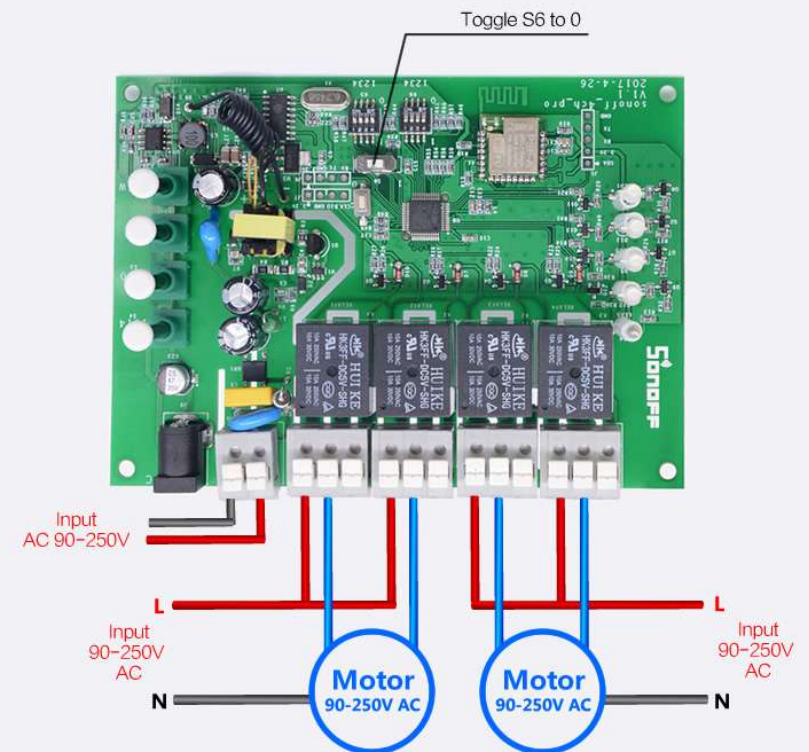
★★★★★ 78 รีวิว | สำเร็จ 46 16 ปัญหา

แบรนด์: Sonoff | เพิ่มแล้ว อุปกรณ์ที่รวมในพัสดุ จาก Sonoff in TH

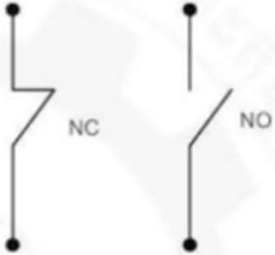
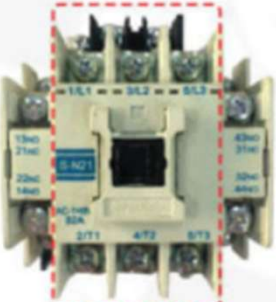
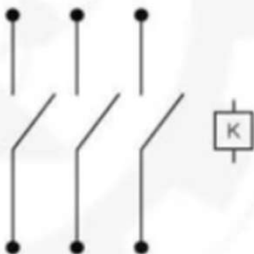
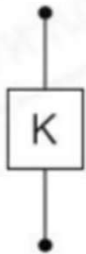
B1,000.00
฿1,200.00 -17%

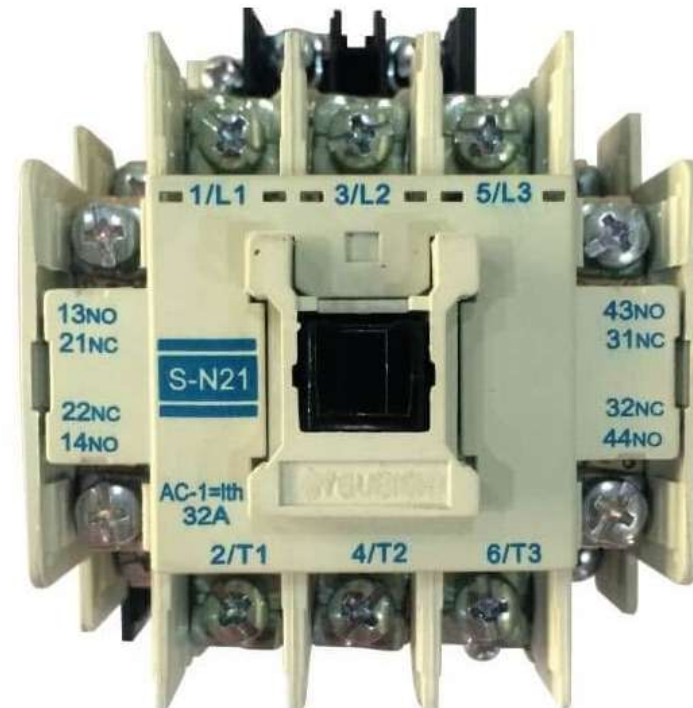
จำนวน 1

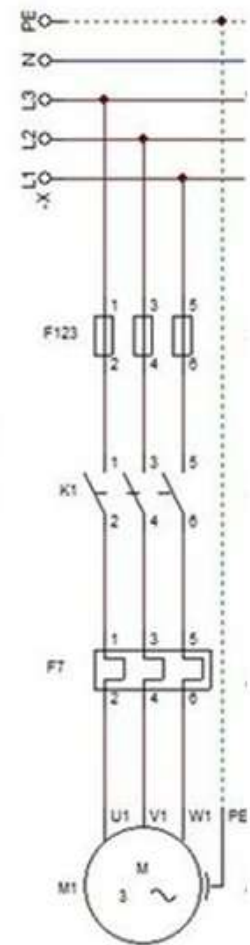
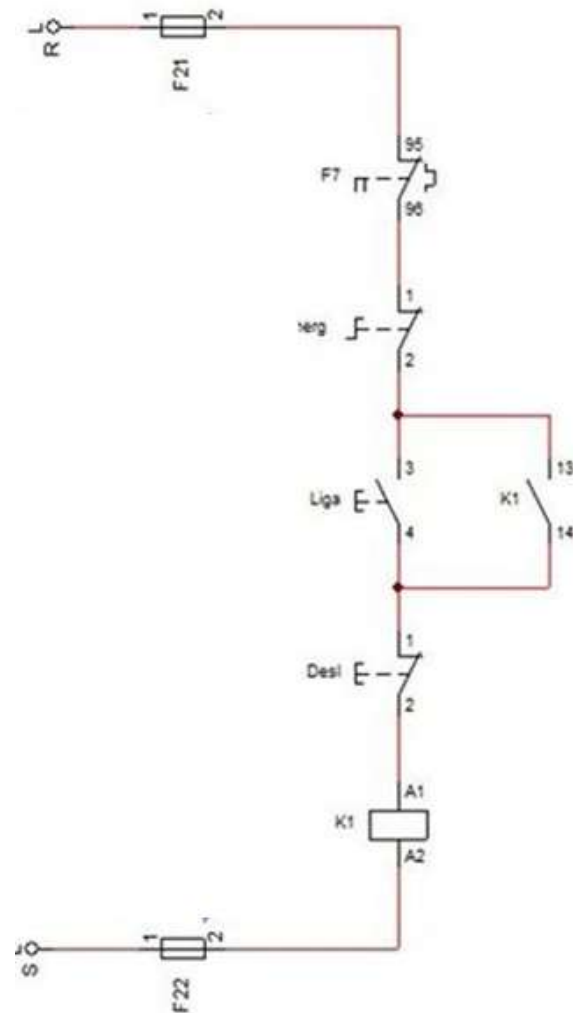
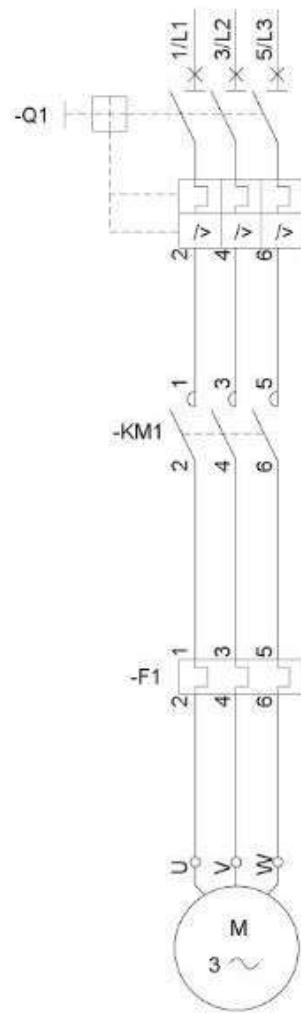
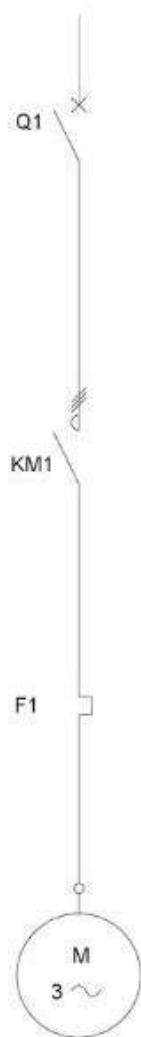
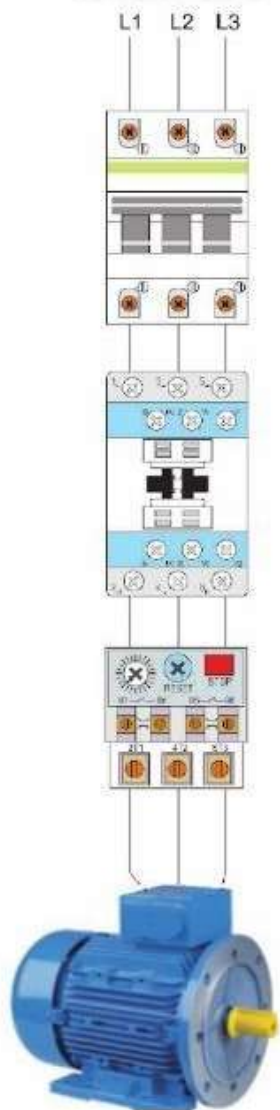
AC Motor Wiring Diagram

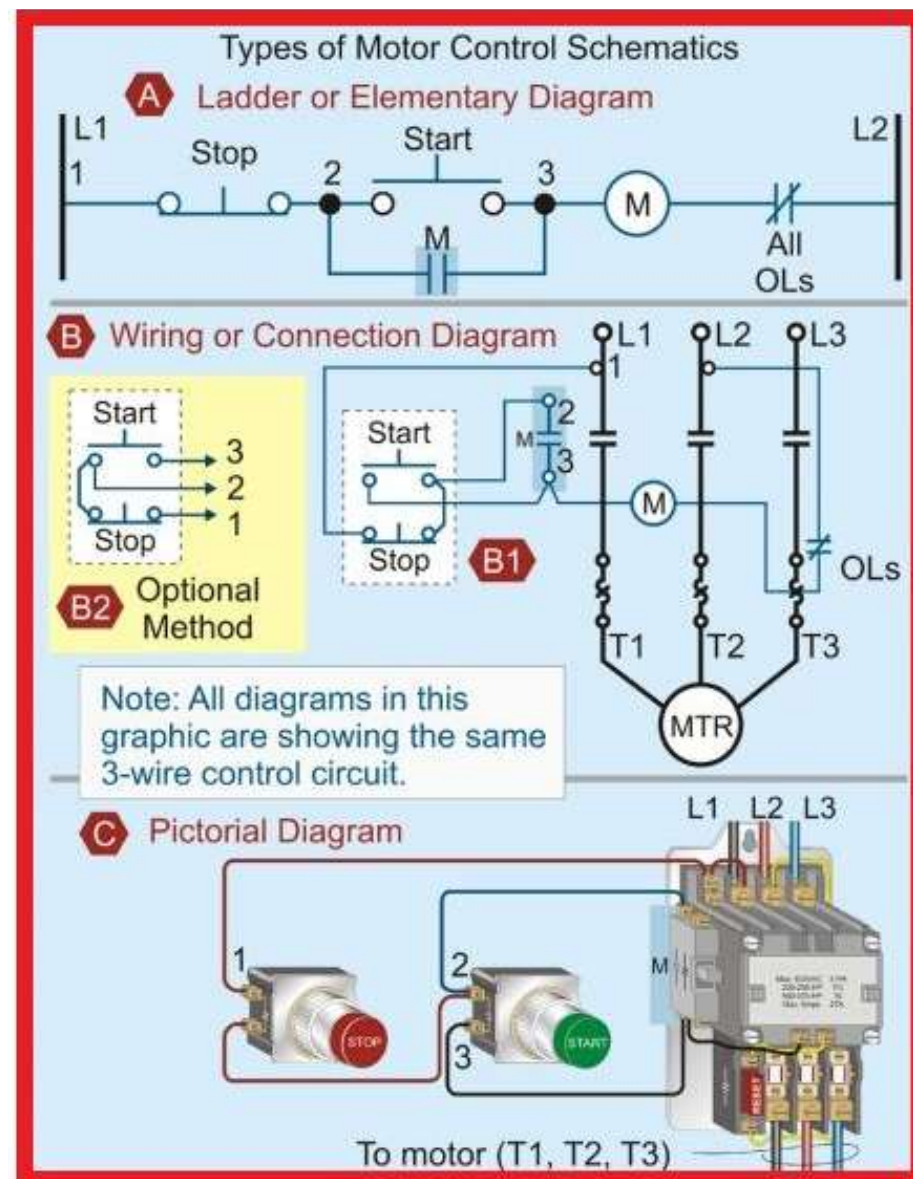
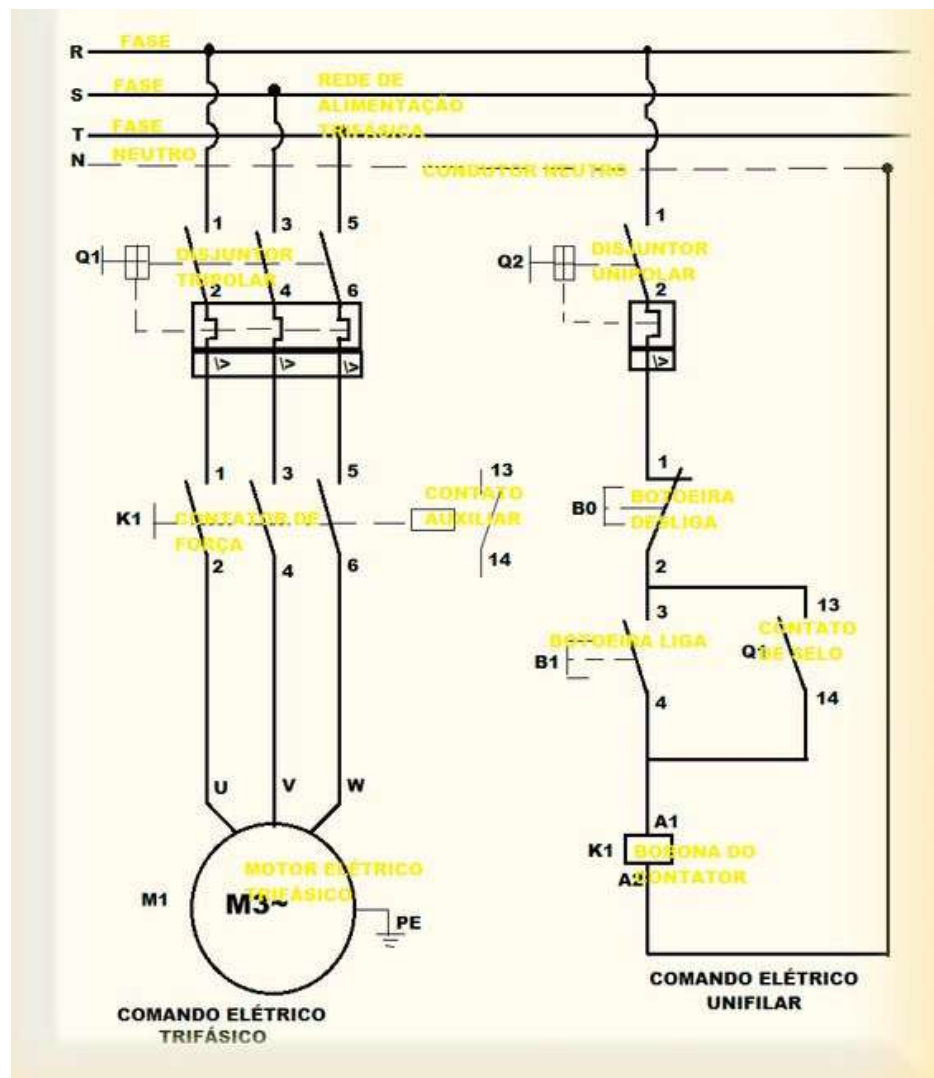


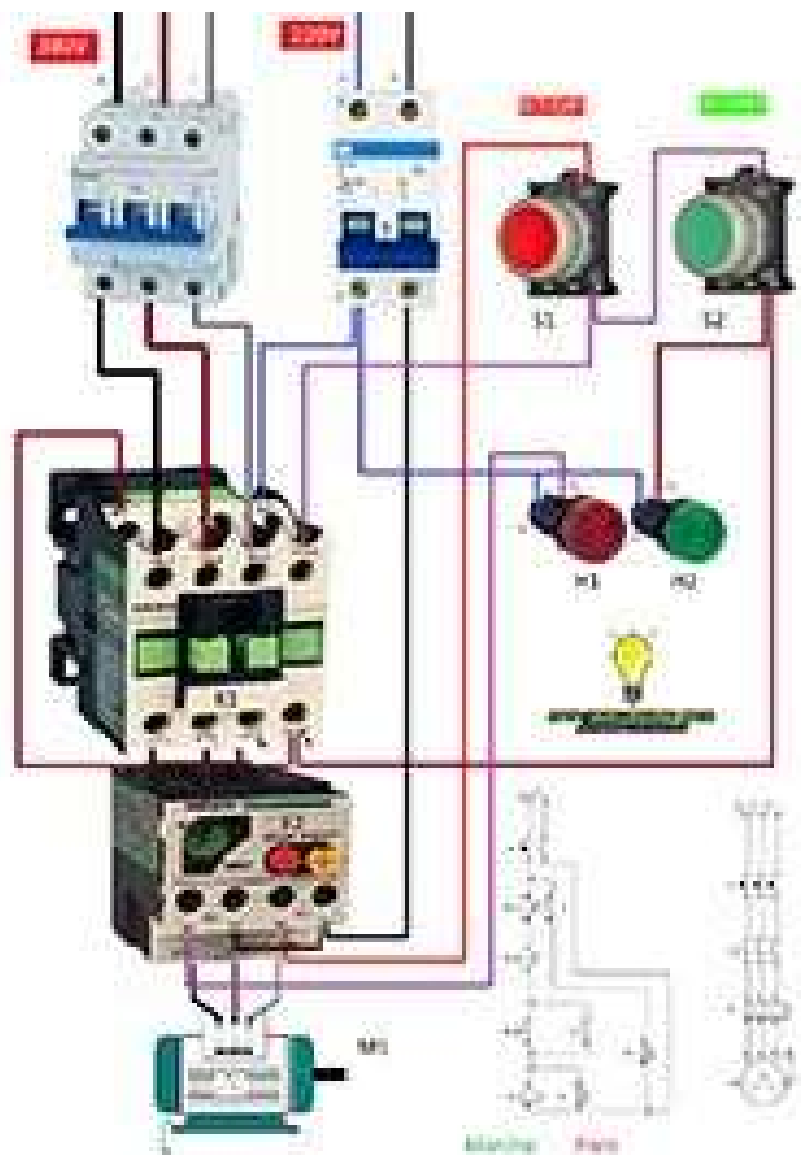
Note : only 1 motor can run each time.

ลักษณะ	วงจรไฟฟ้า
	
	
	

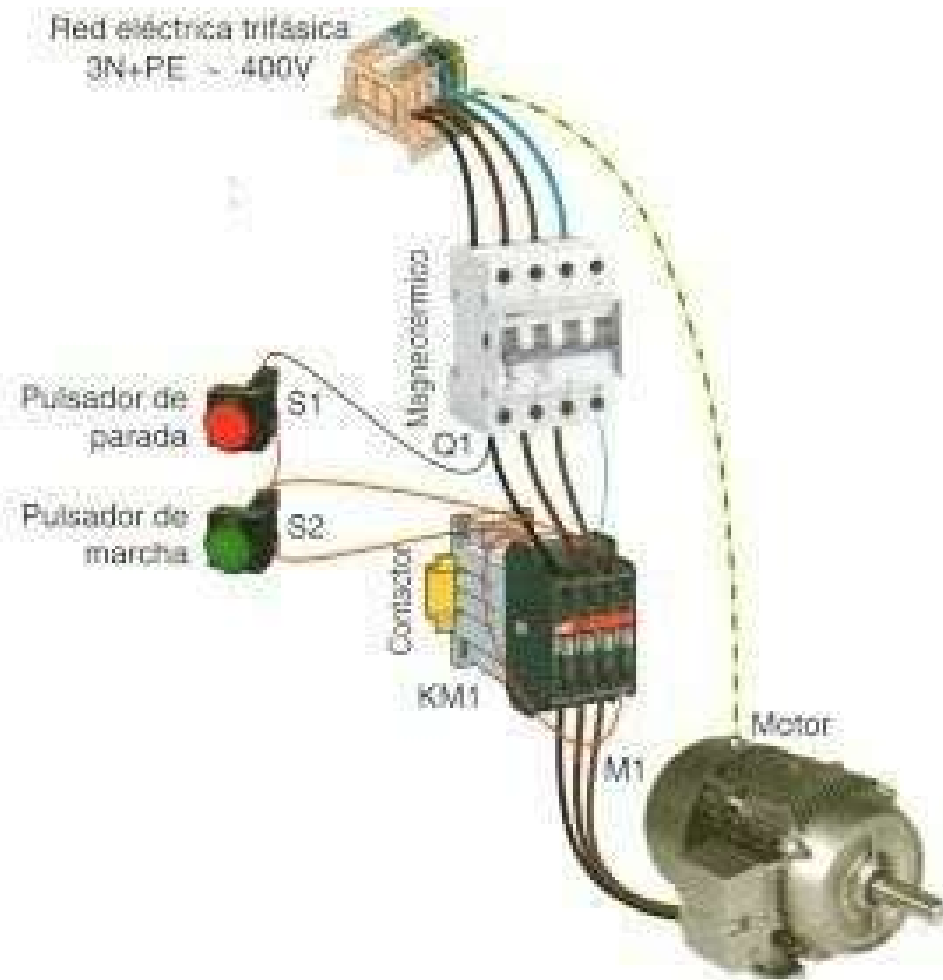


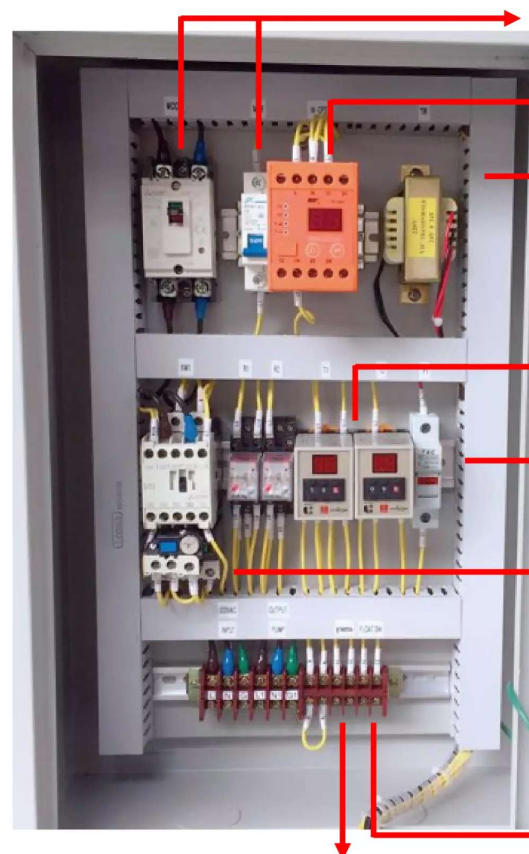
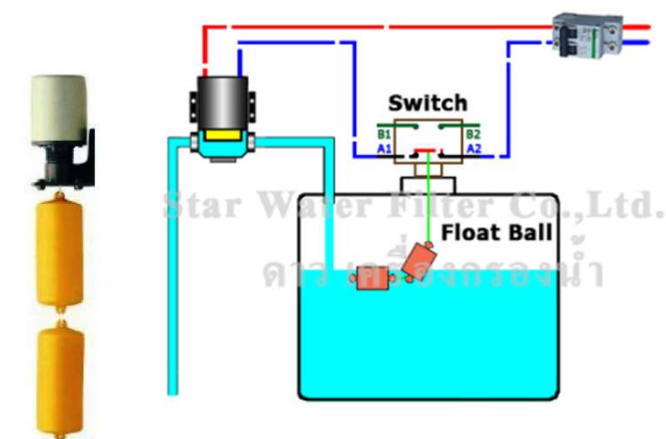
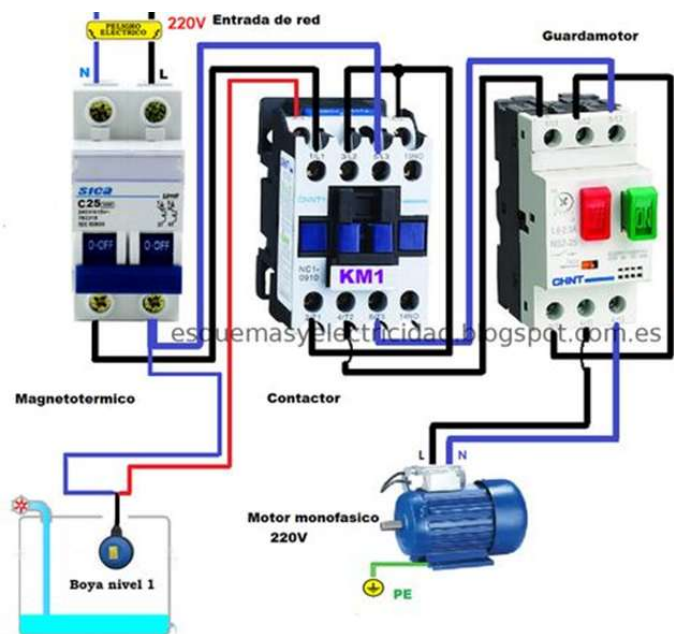






Red eléctrica trifásica
3N+PE ~ 400V





- มี MCCB , MCB ป้องกันการลัดวงจร
- มีระบบป้องกันไฟตกไฟเกิน เพื่อไม่ให้อุปกรณ์ภายในตู้เสียหายได้ รวมถึงปั๊มด้วย
- มีระบบหม้อแปลงไฟจาก 220vac เป็น 24vac เพื่อความปลอดภัยต่อผู้ใช้งาน

- มี Timer หน่วงเวลาเช็คระบบน้ำในเส้นท่อ แล้วกลับมาทำงานของปั๊มน้ำแบบอัตโนมัติ

- มี Fuse control ป้องกันการลัดวงจรแรงดันต่ำ

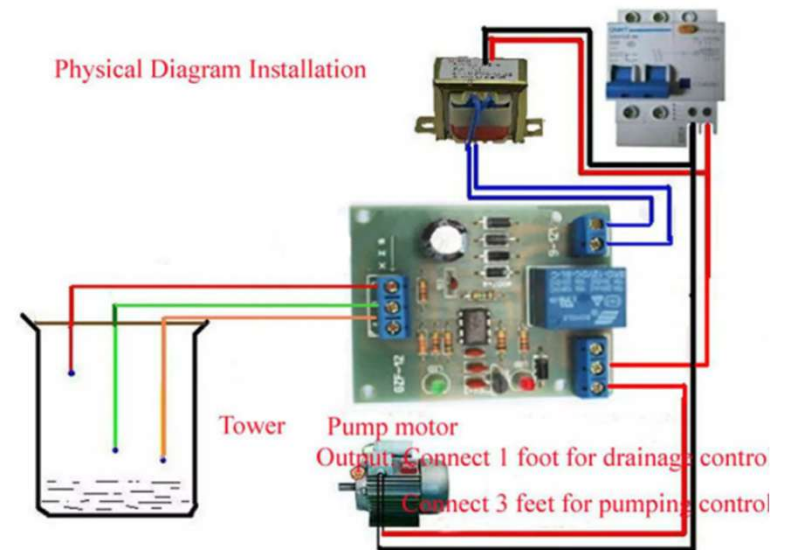
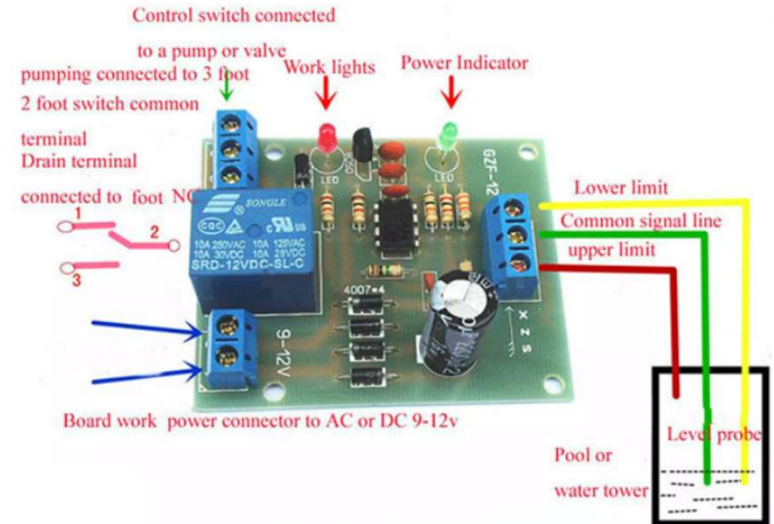
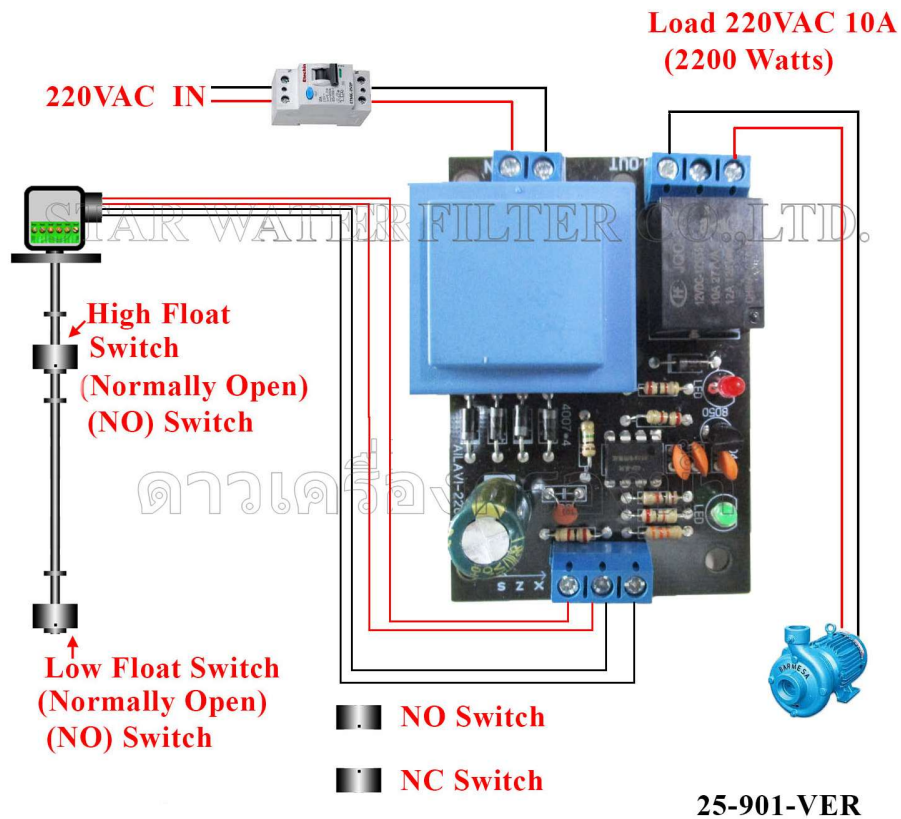
- มีโอเวอร์โวลต์รีเลย์ ป้องกัน การกินเกินกระแสมอเตอร์

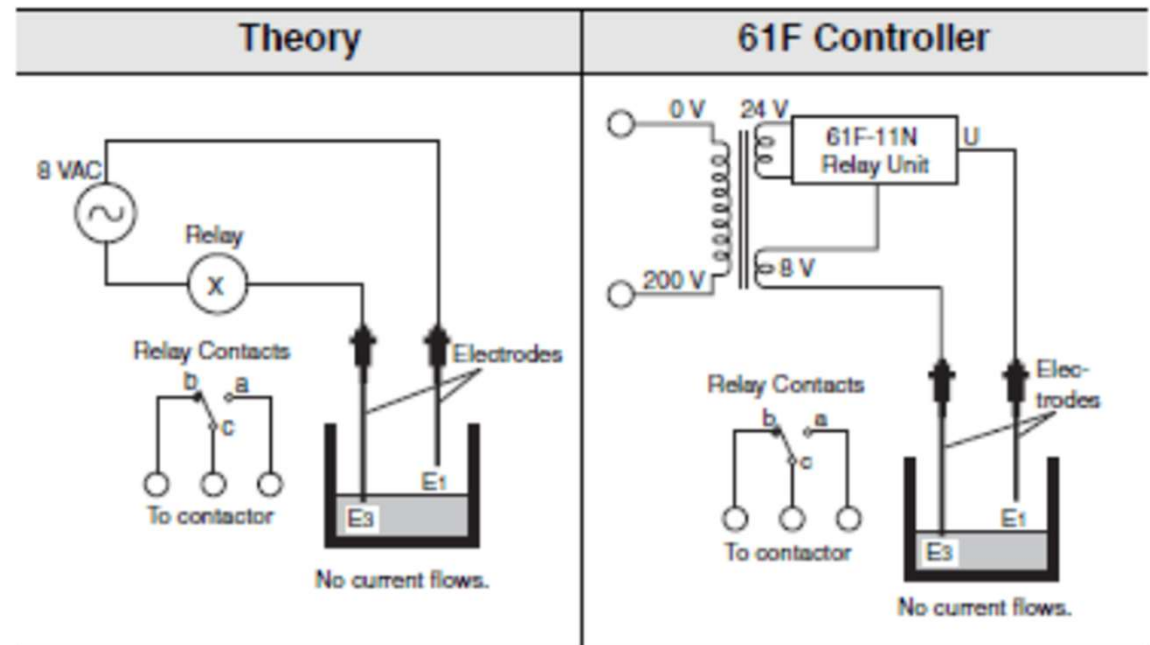
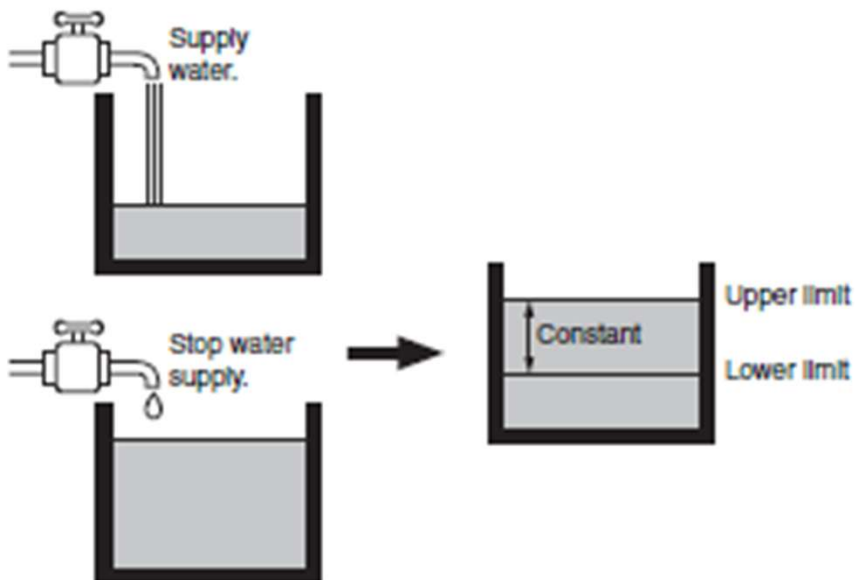


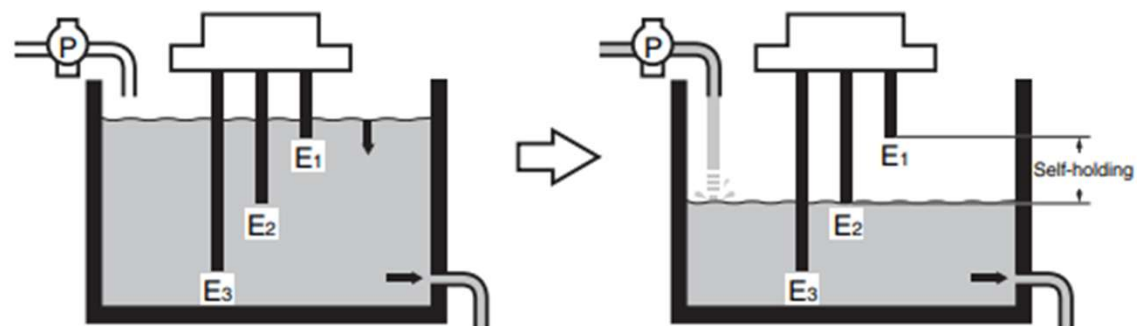
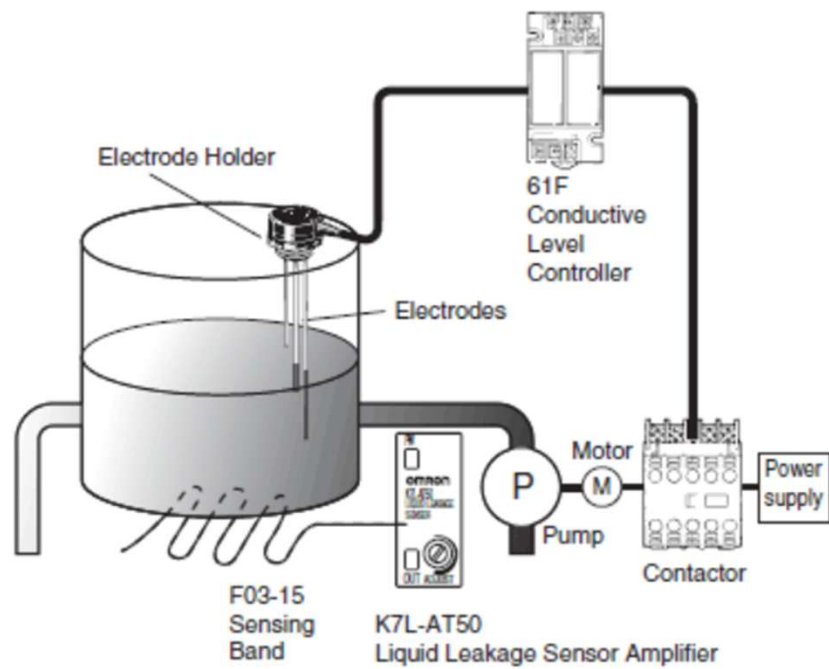
- มีระบบป้องกันปั๊มน้ำ Run dry เพื่อไม่ให้ปั๊มน้ำทำงานตัวเปล่า อาจทำให้ปั๊มไหม้เสียหายได้ ด้วย โฟลสวิทช์

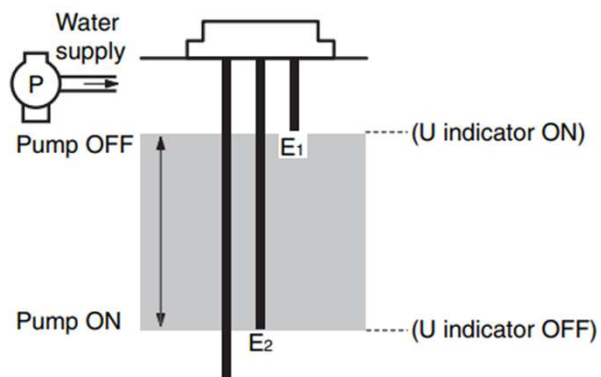
Float Switch

High Low Water Level Controller Module 10A 220VAC





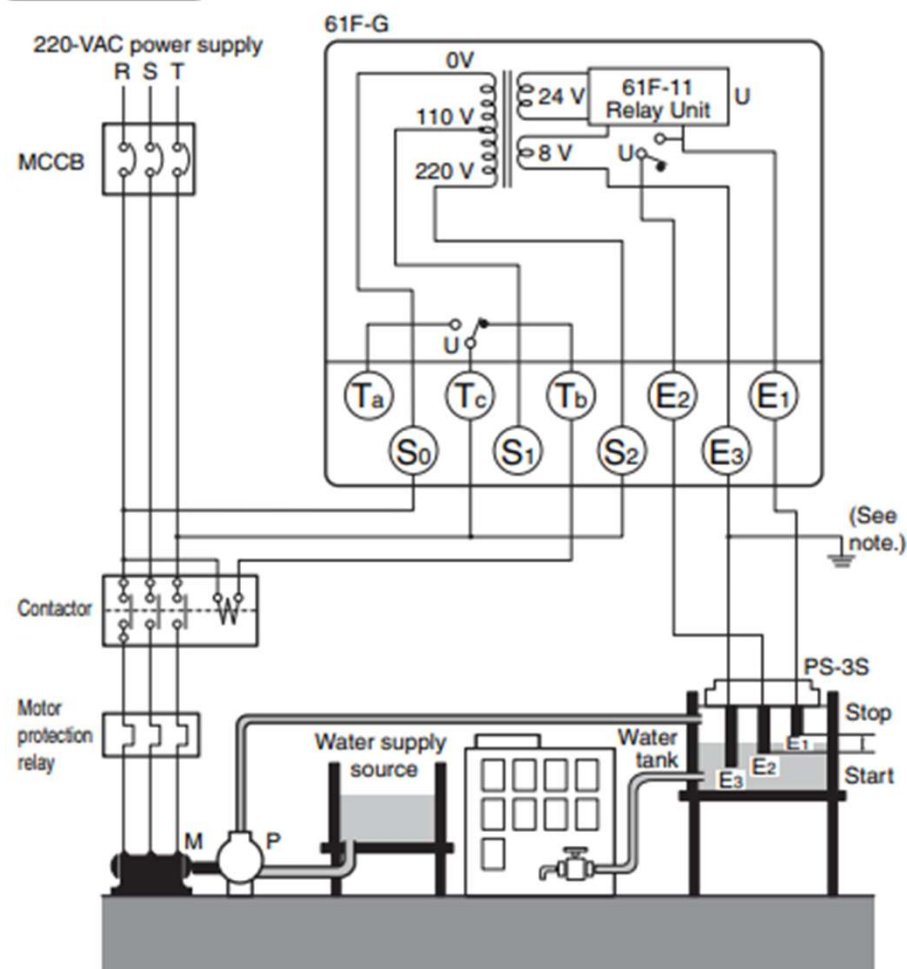




- Connect Tb to the contactor's coil terminal.
- Power Supply Connections (for models with 110/220-V power)
110 VAC: Connect S₀ and S₁.
220 VAC: Connect S₀ and S₂.

Automatic Water Supply Control

Connections

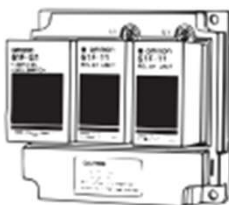


Note: Be sure to ground the common Electrode (the longest Electrode).

Basic Type

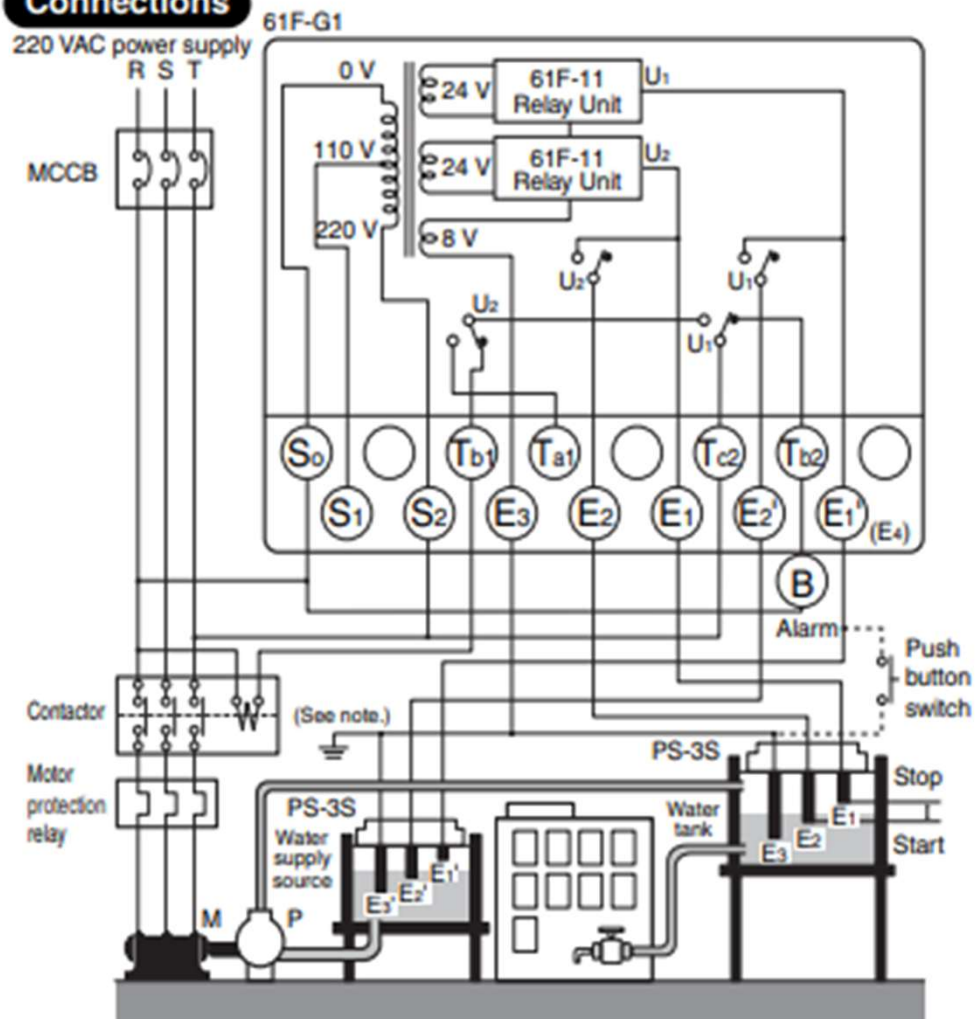
61F-G1

Dimensions:
page 14

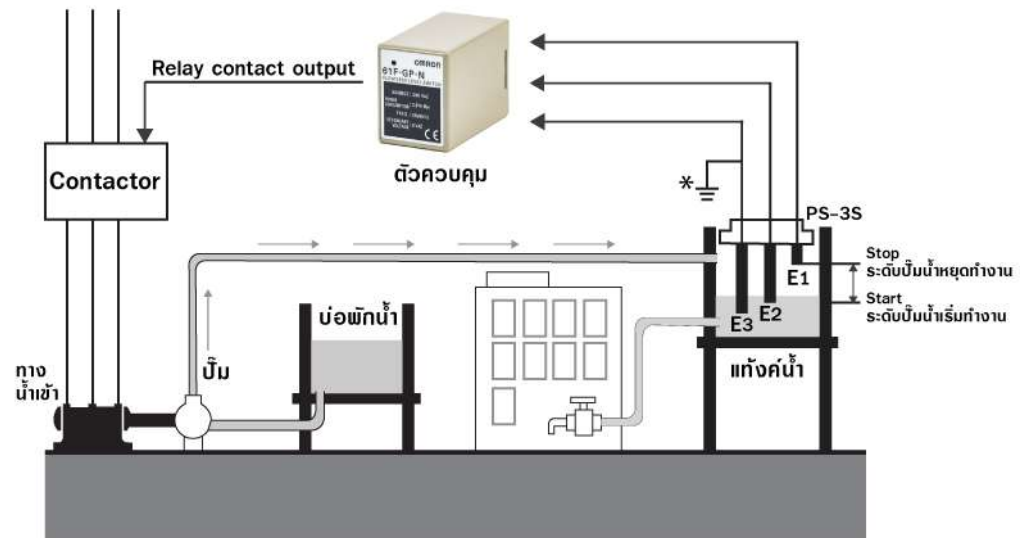
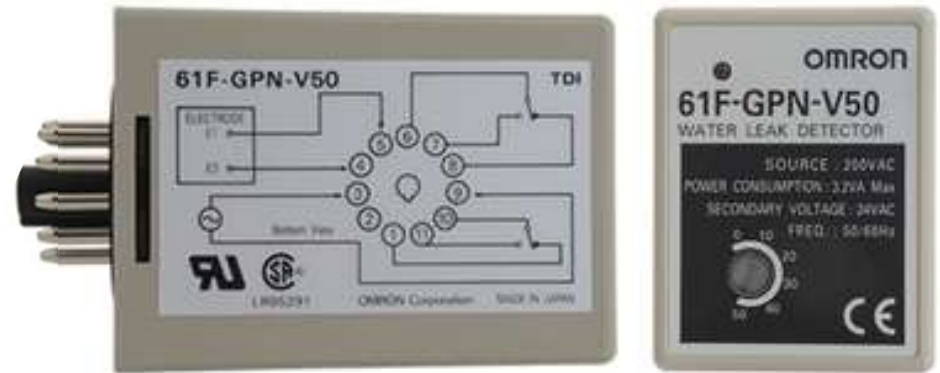
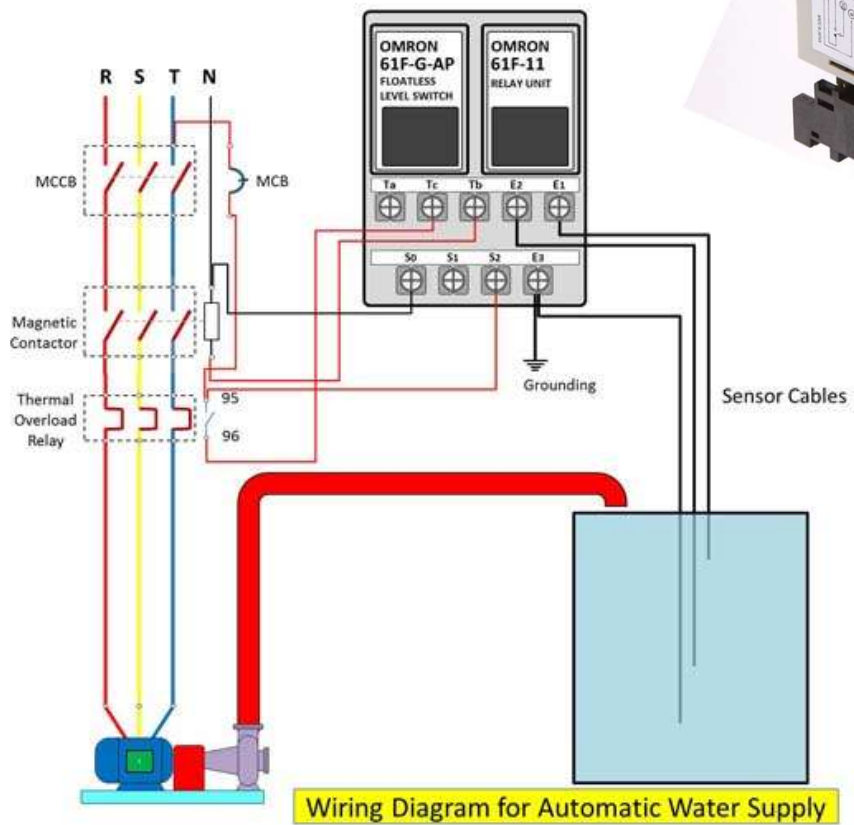


Automatic Water Supply Control with Pump Idling Prevention

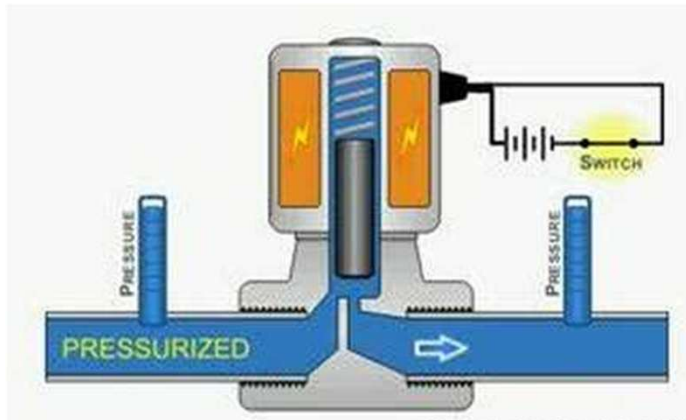
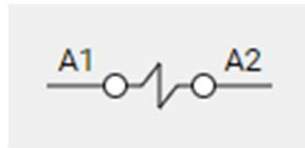
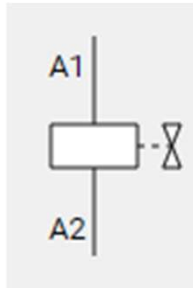
Connections



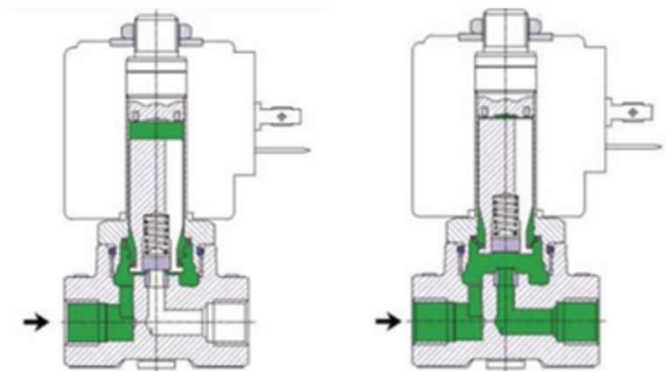
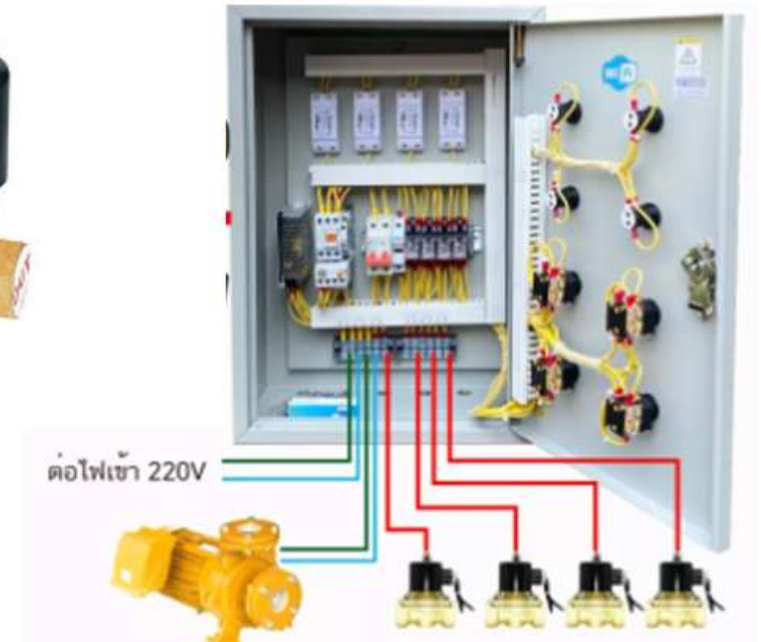
Note: Be sure to ground the common Electrode (the longest Electrode).



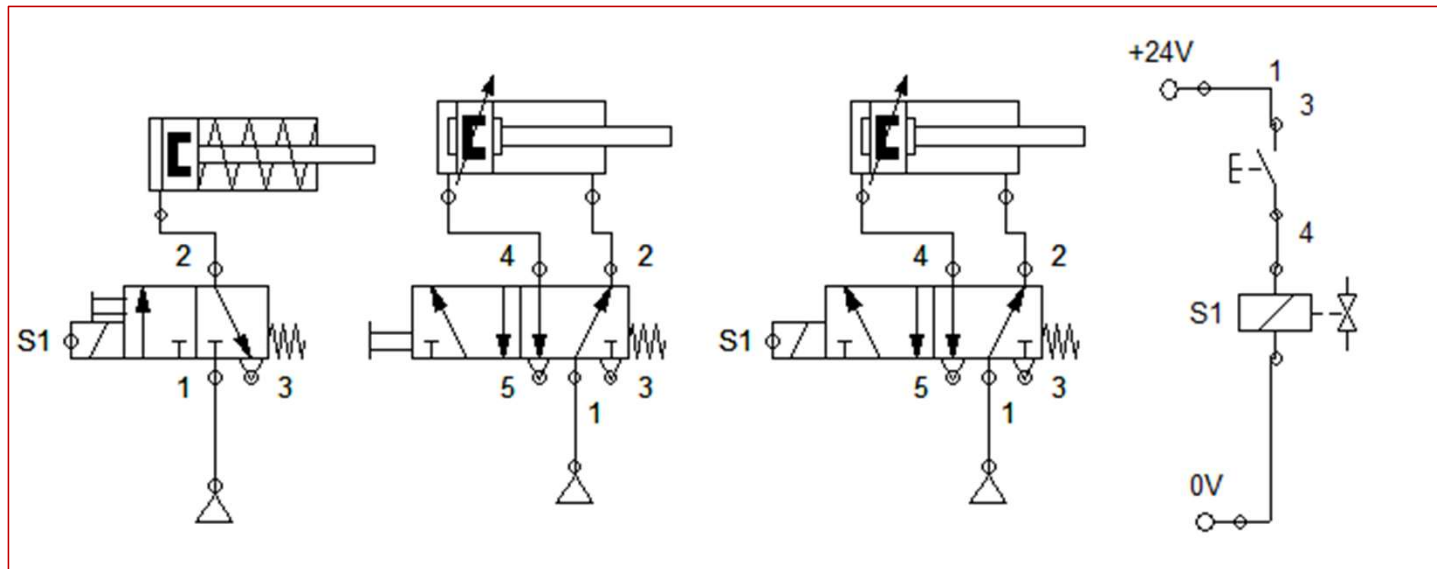
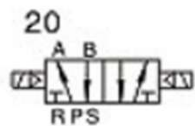
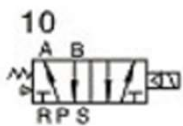
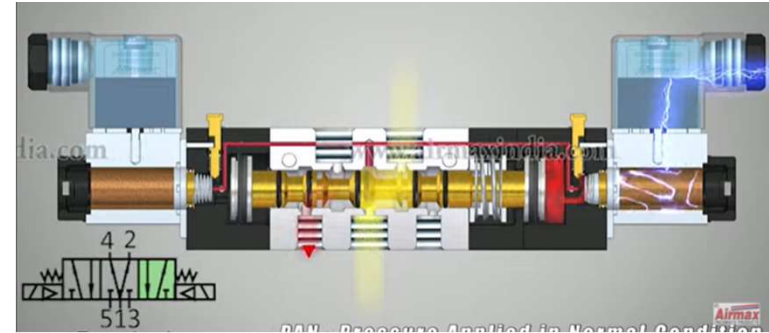
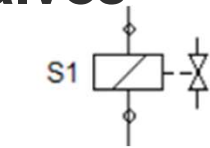
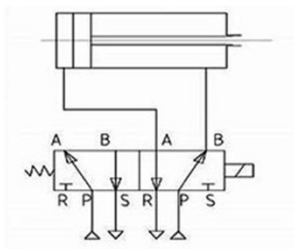
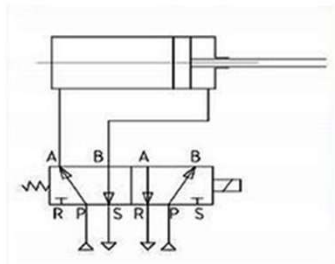
Solenoid Valves



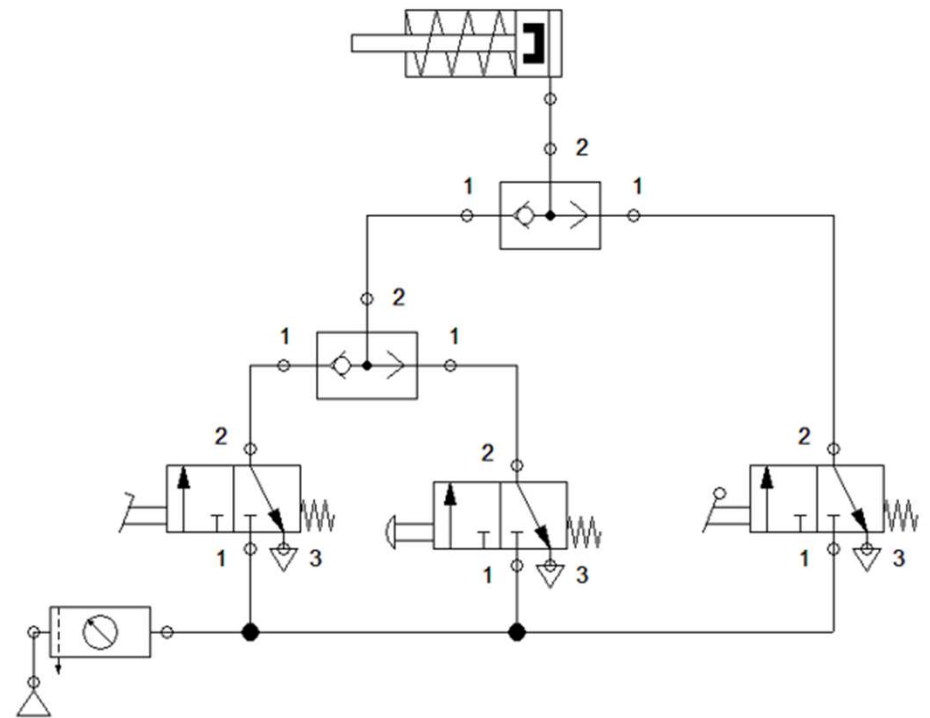
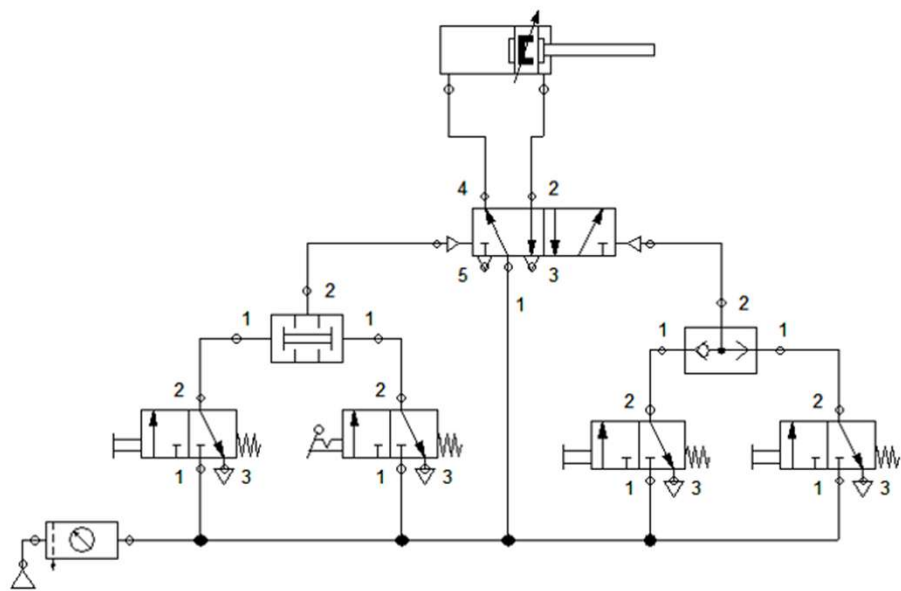
ระบบรดน้ำอัตโนมัติ 4 โซน

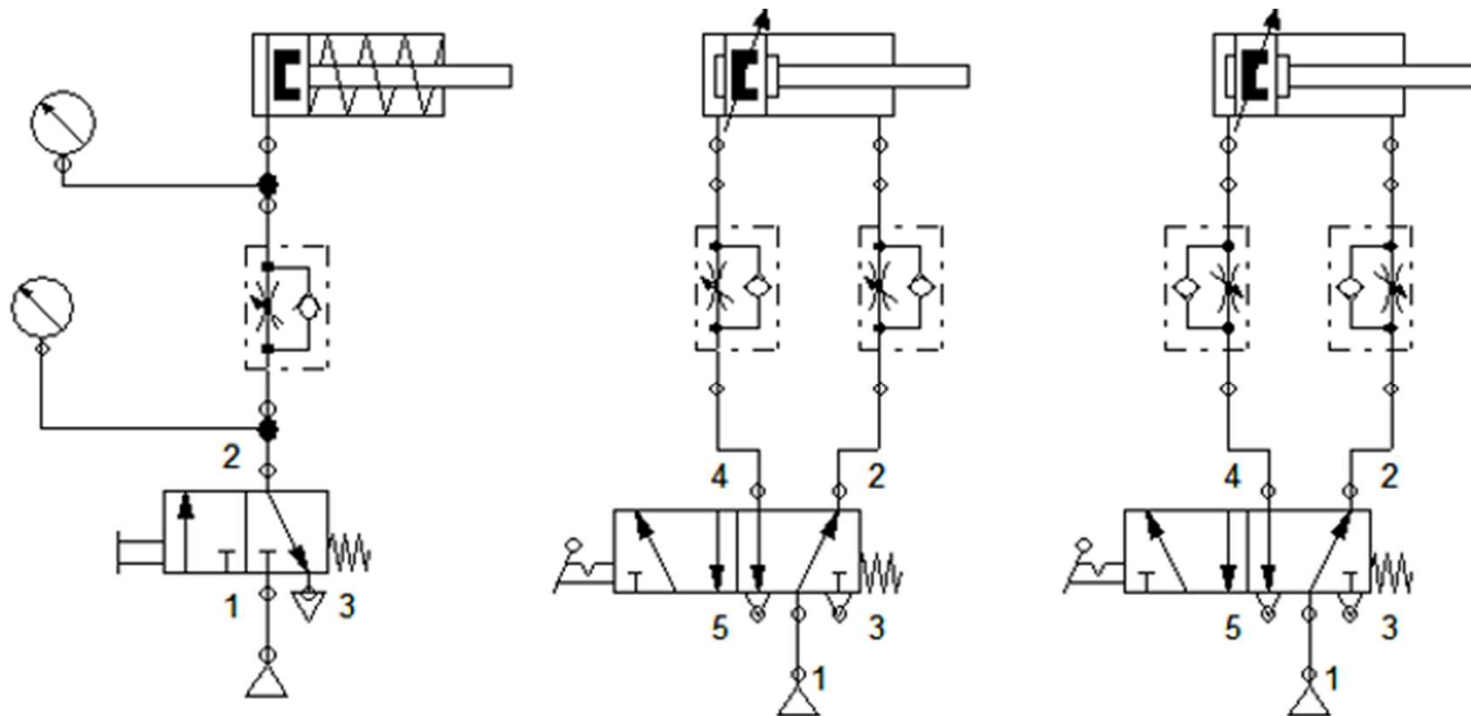


Lab 4 Solenoid Valves

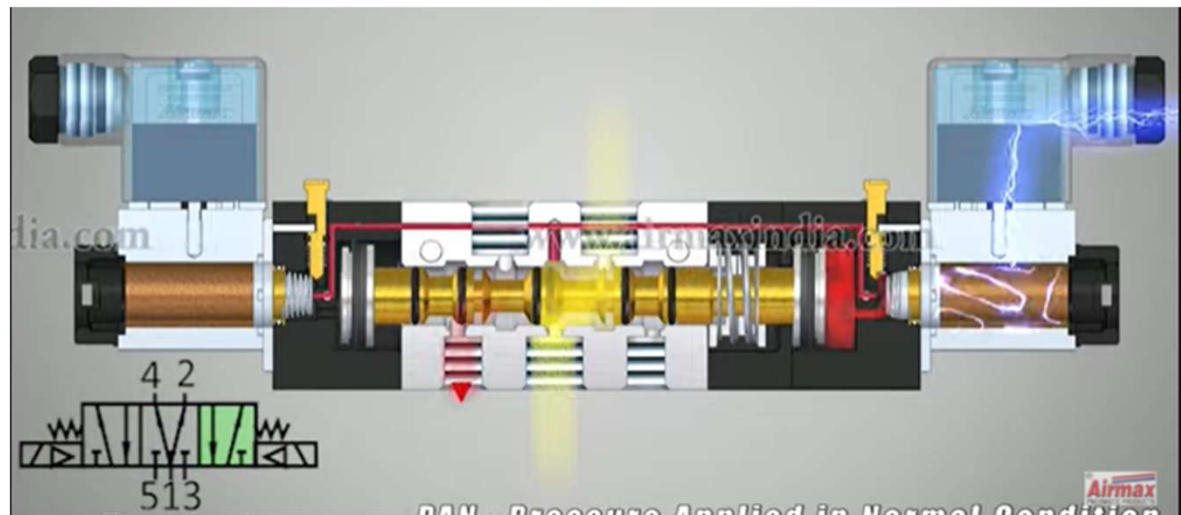
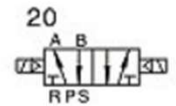
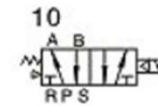
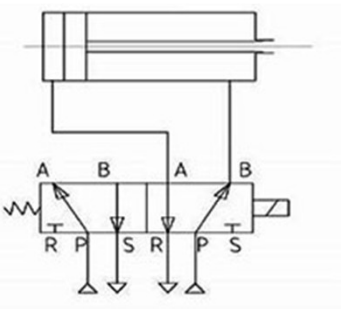
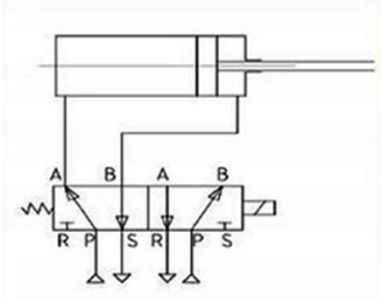


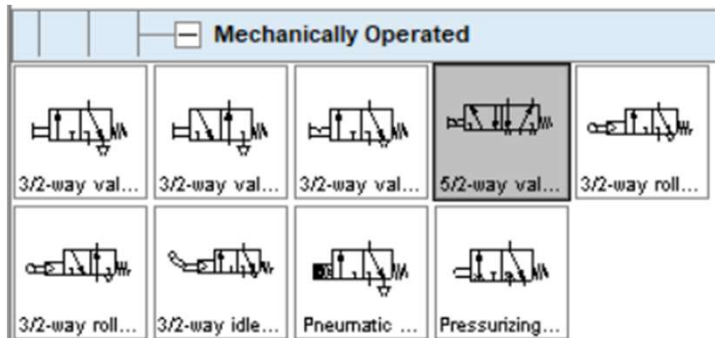
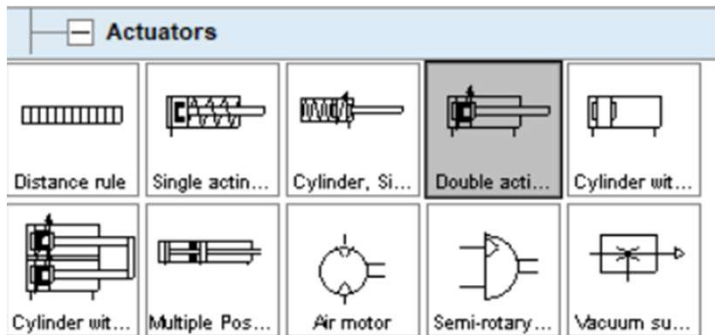
pneumatic cylinder





Solenoid Valves





Configure Way Valve

Left Actuation

☐ Spring-returned

☐ Piloted

☐ External supply

☐ Pneumatic spring

☐ External supply

Manually

Mechanically

Pneumatically/
Electrically

Description

5/2-way valve, with selection switch

Valve Body

☐ Reversible

Initial Position

☐ ☒ ☐ ☐

Right Actuation

☒ Spring-returned

☐ Piloted

☐ External supply

☐ Pneumatic spring

☐ External supply

Manually

Mechanically

Pneumatically/
Electrically

☐ Left ☐ Dominant Signal ☐ Right

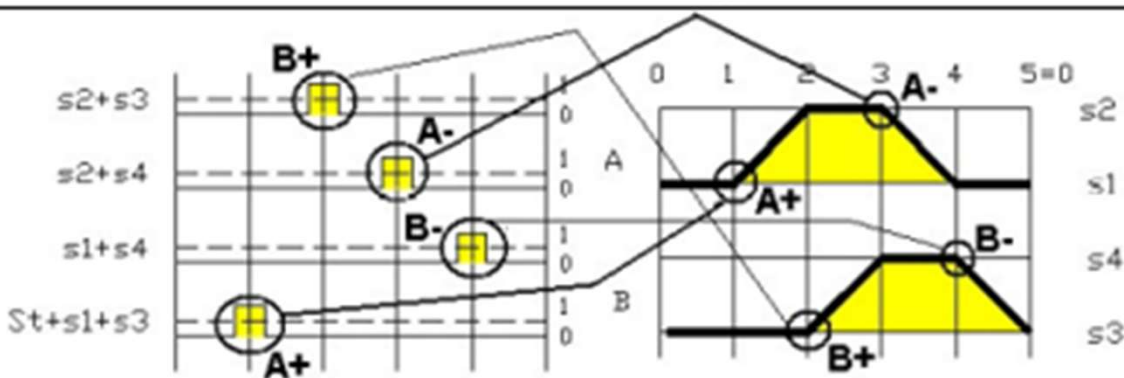
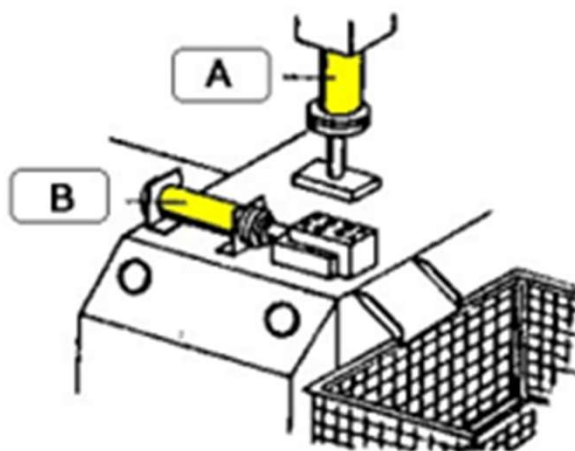
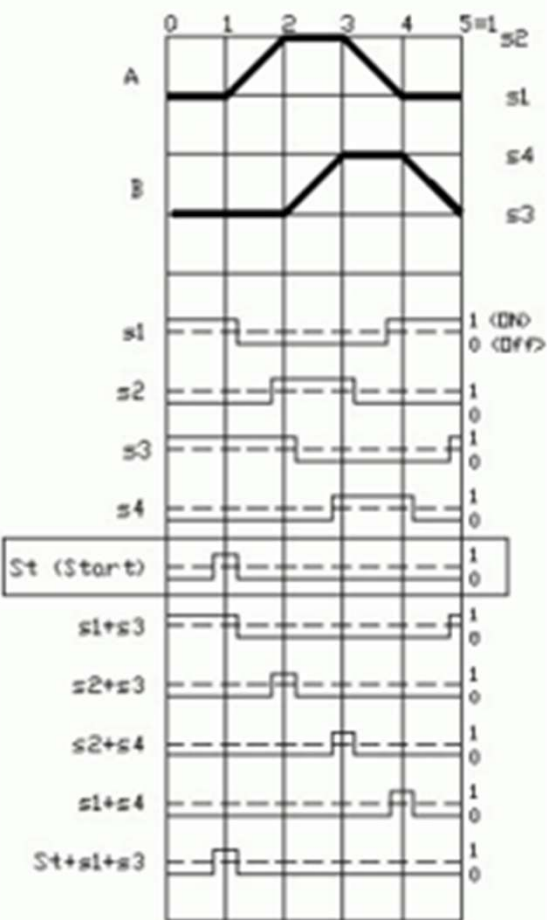
Standard Nominal Flow Rate l/min (0.1..5000)

Mirror

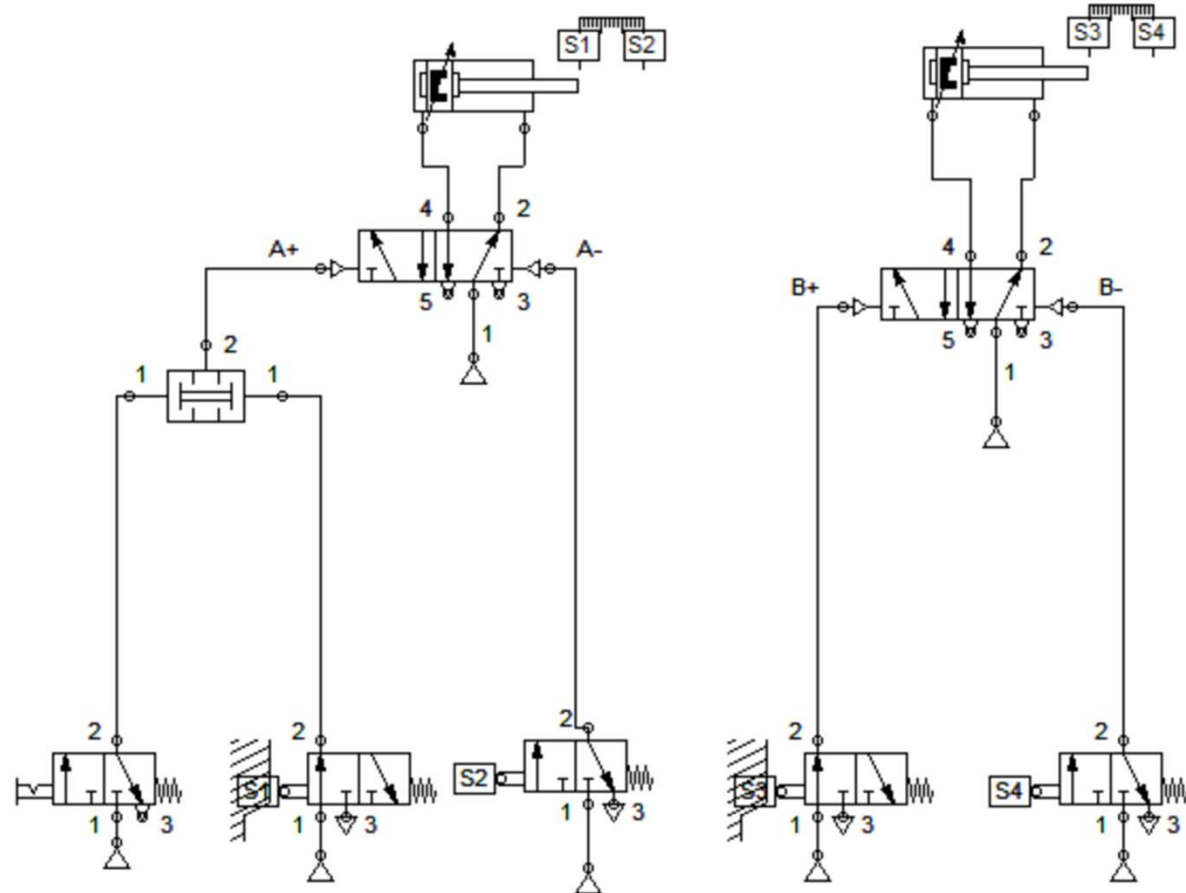
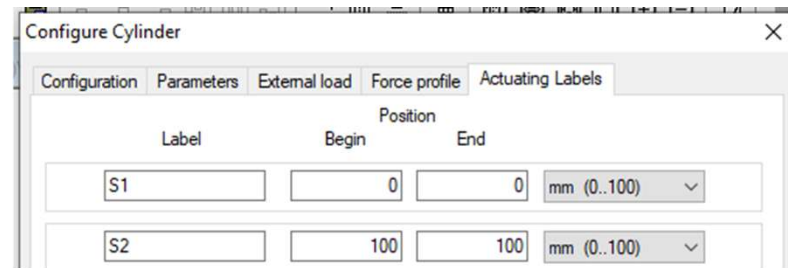
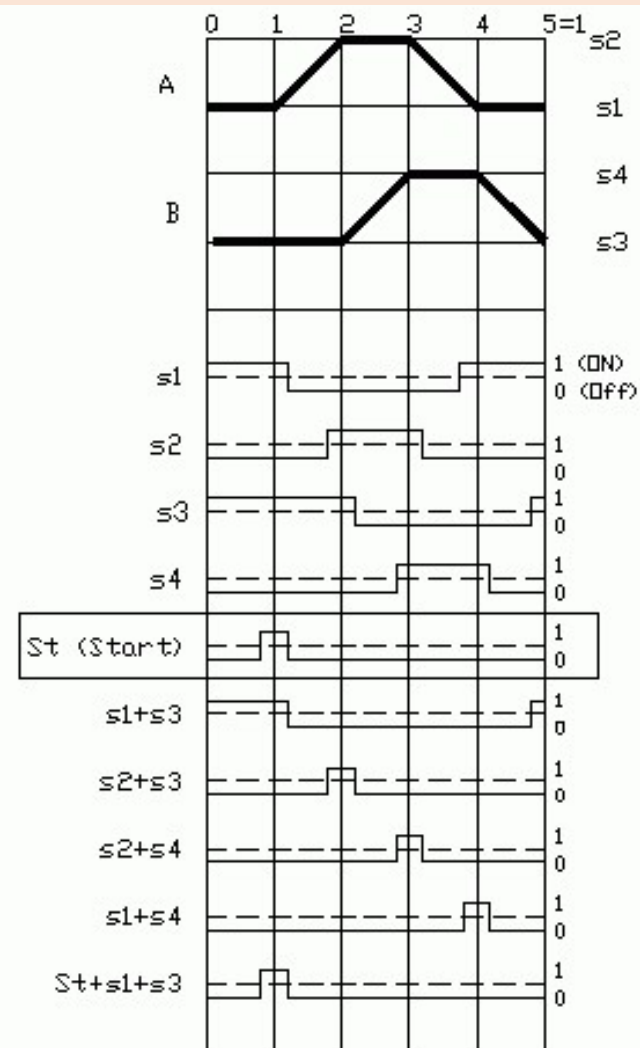
☐ Horizontal

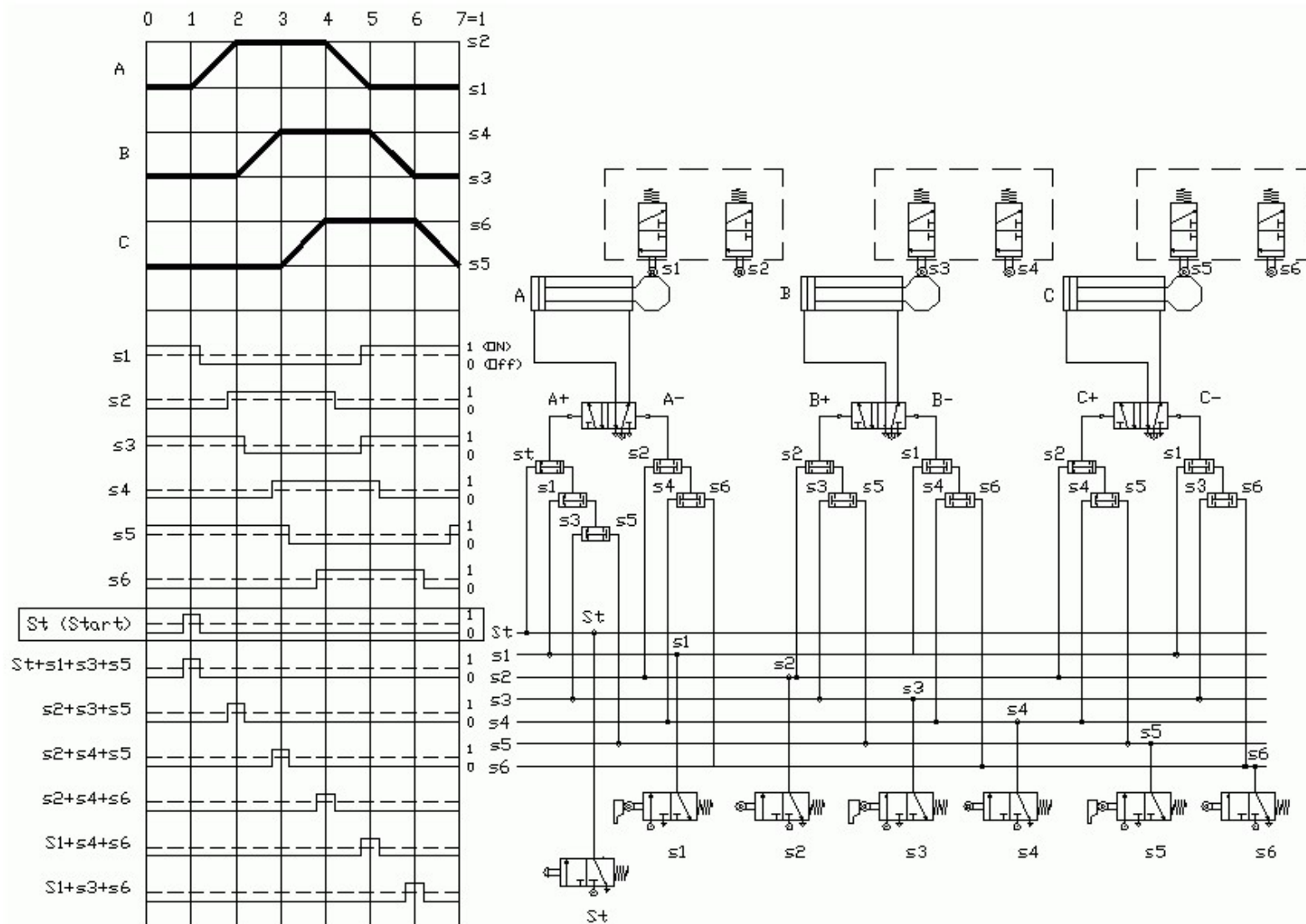
☐ Vertical

OK Cancel Help

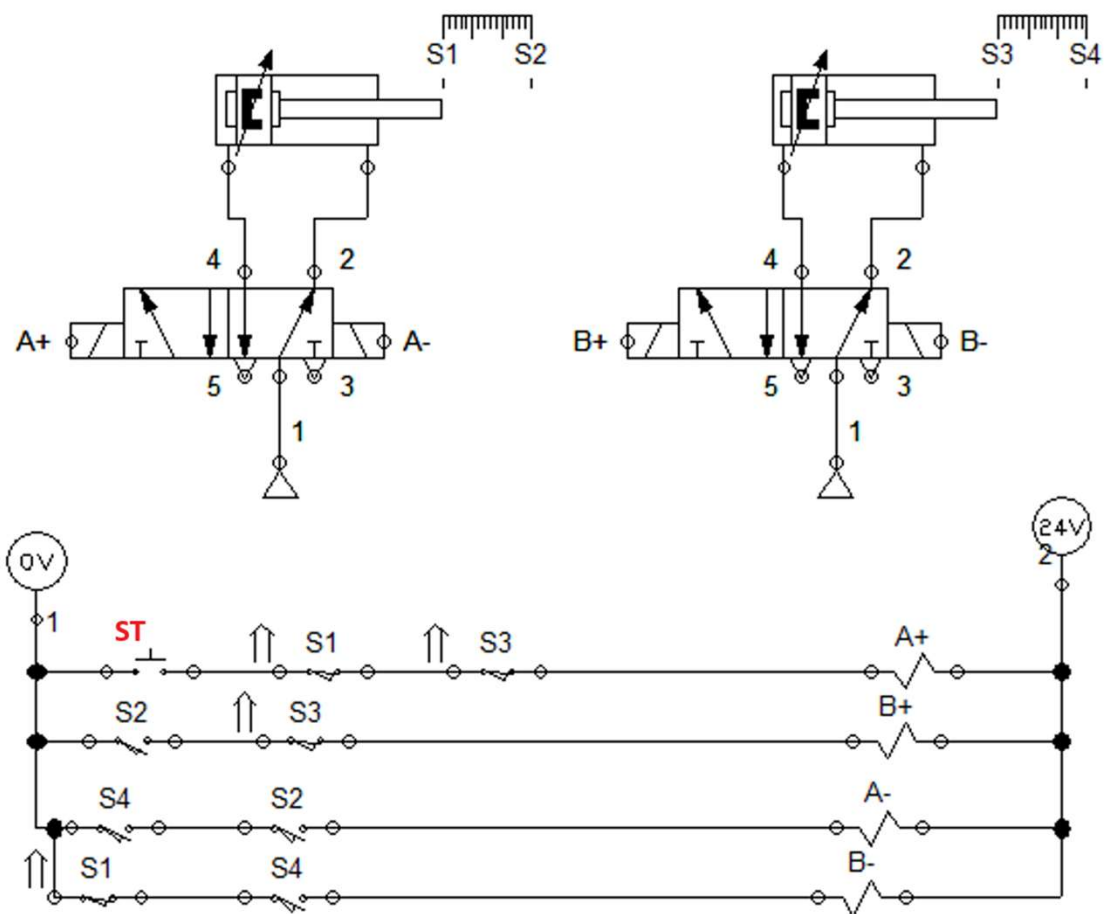
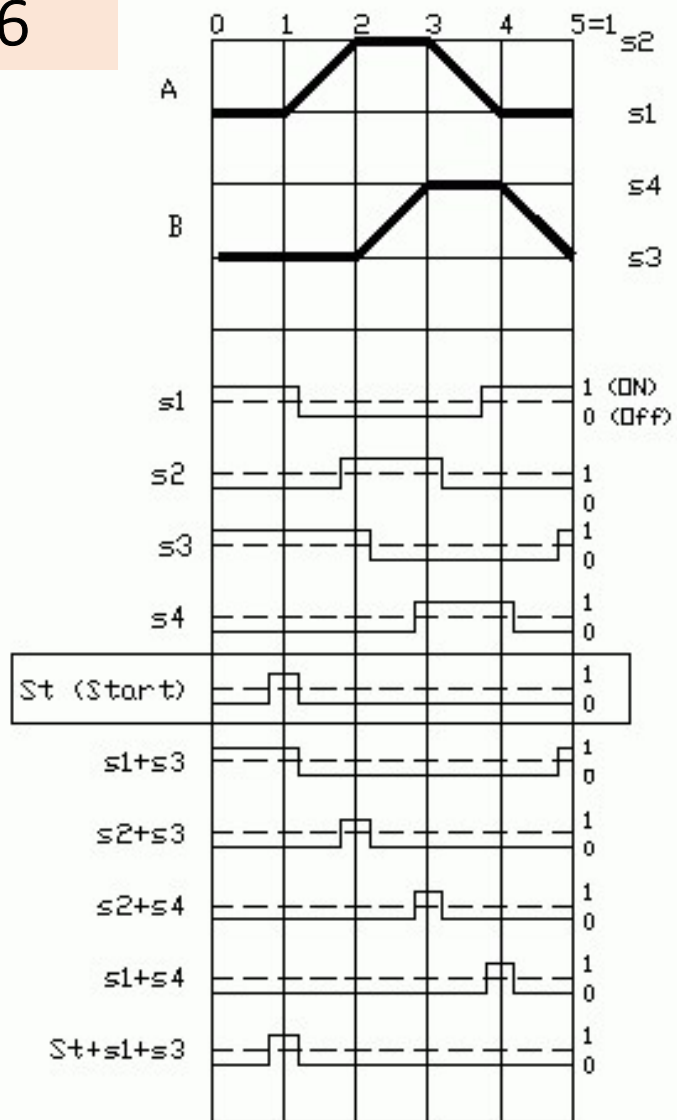


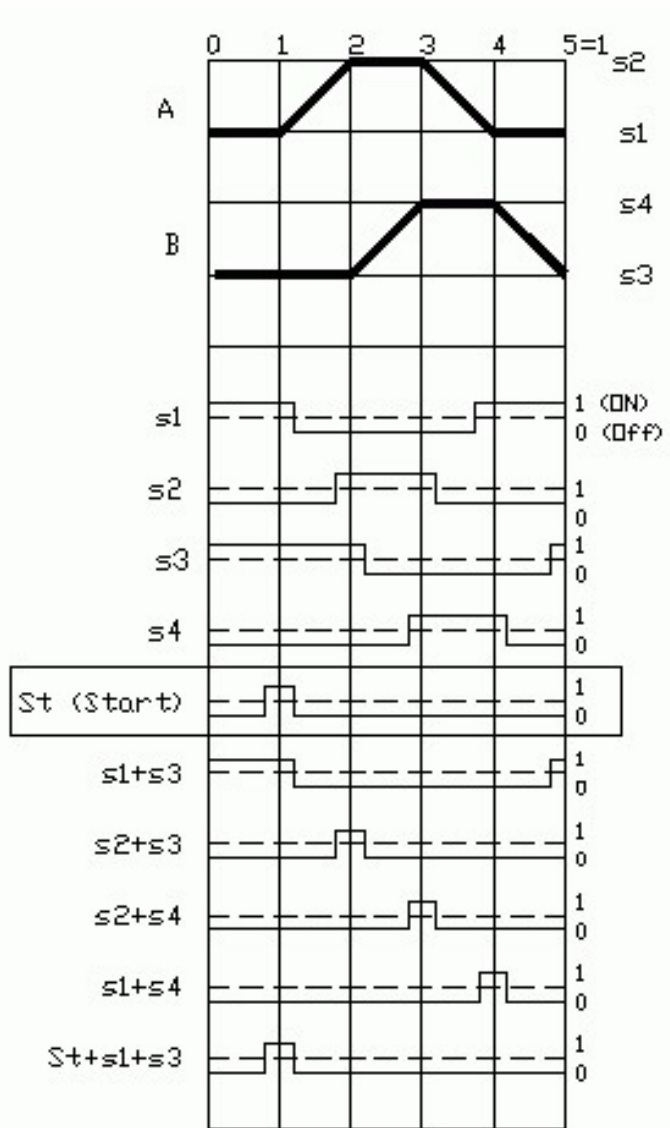
Lab 5 Step Diagram





Lab 6

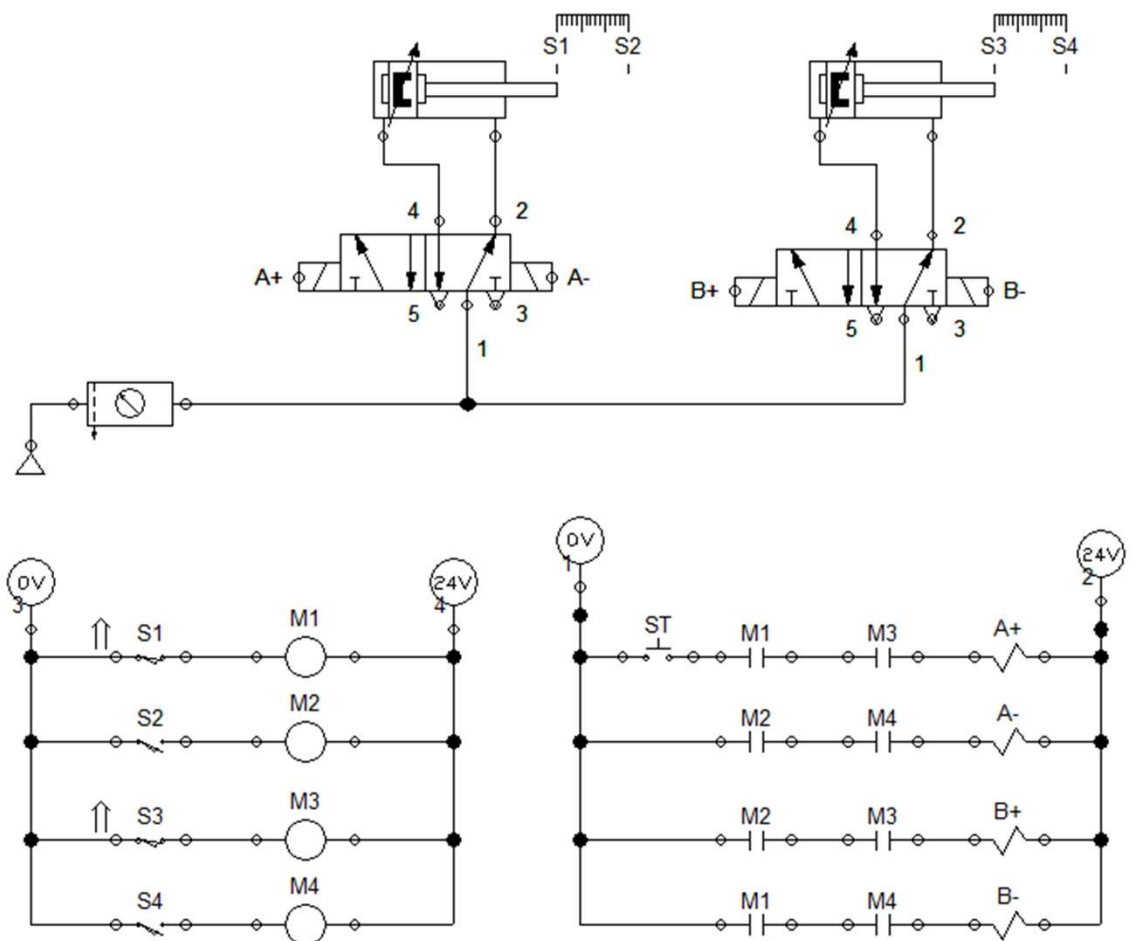


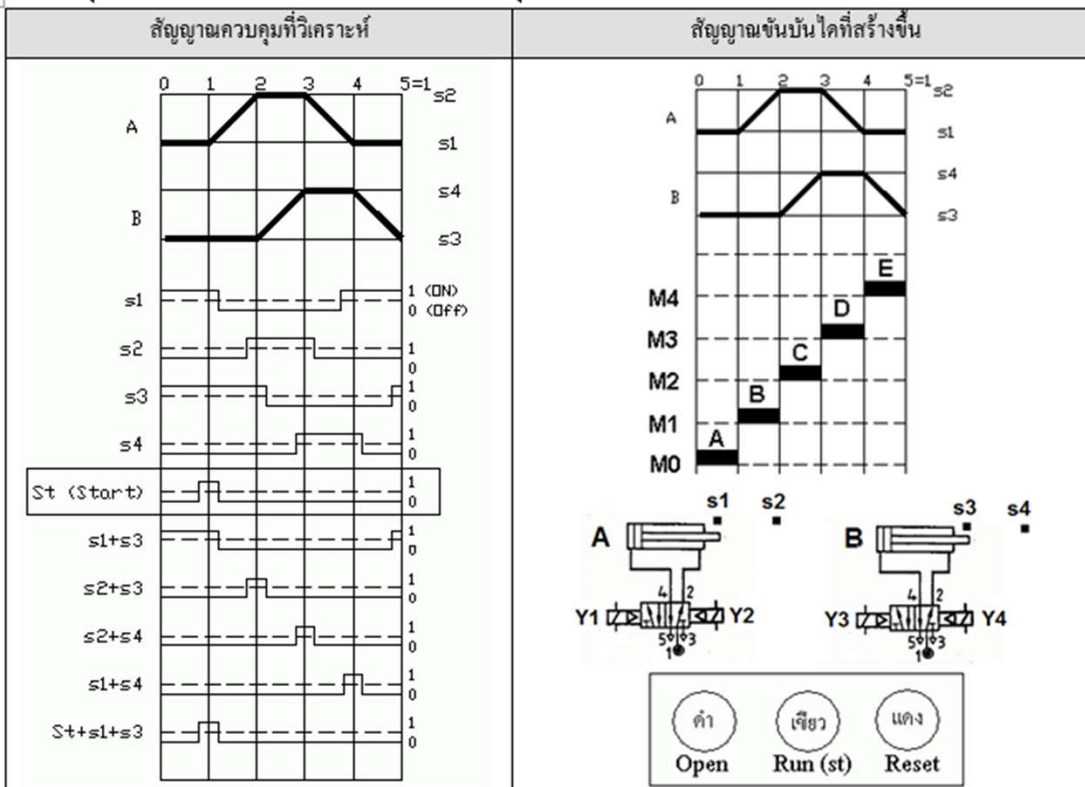


Configure Cylinder

Configuration Parameters External load Force profile Actuating Labels

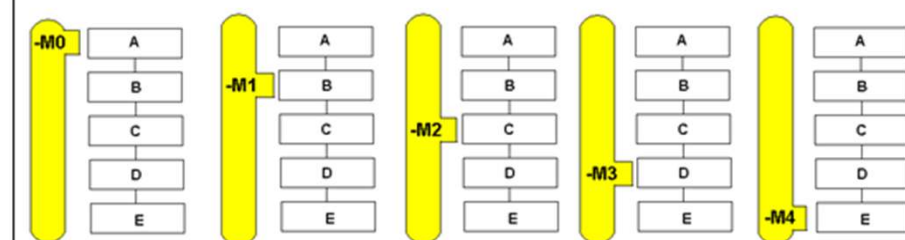
Label	Begin	End	Unit
S1	0	0	mm (0..100)
S2	100	100	mm (0..100)



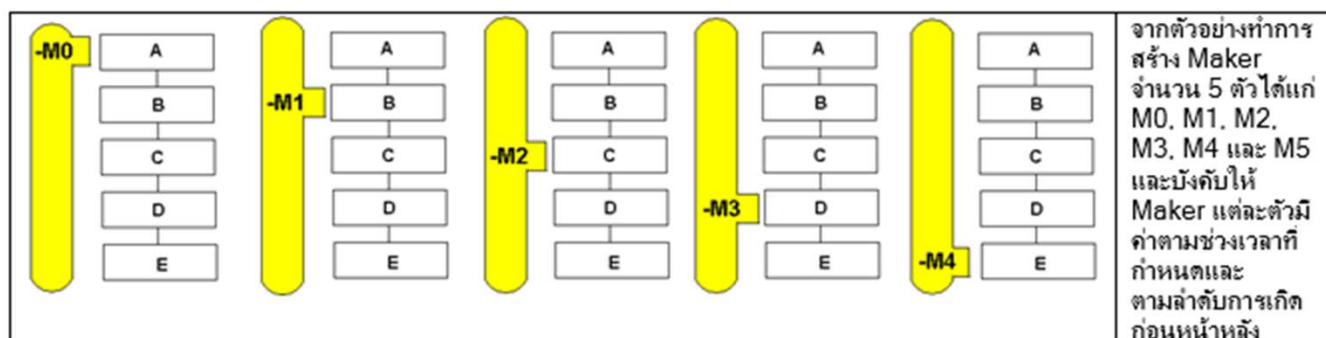
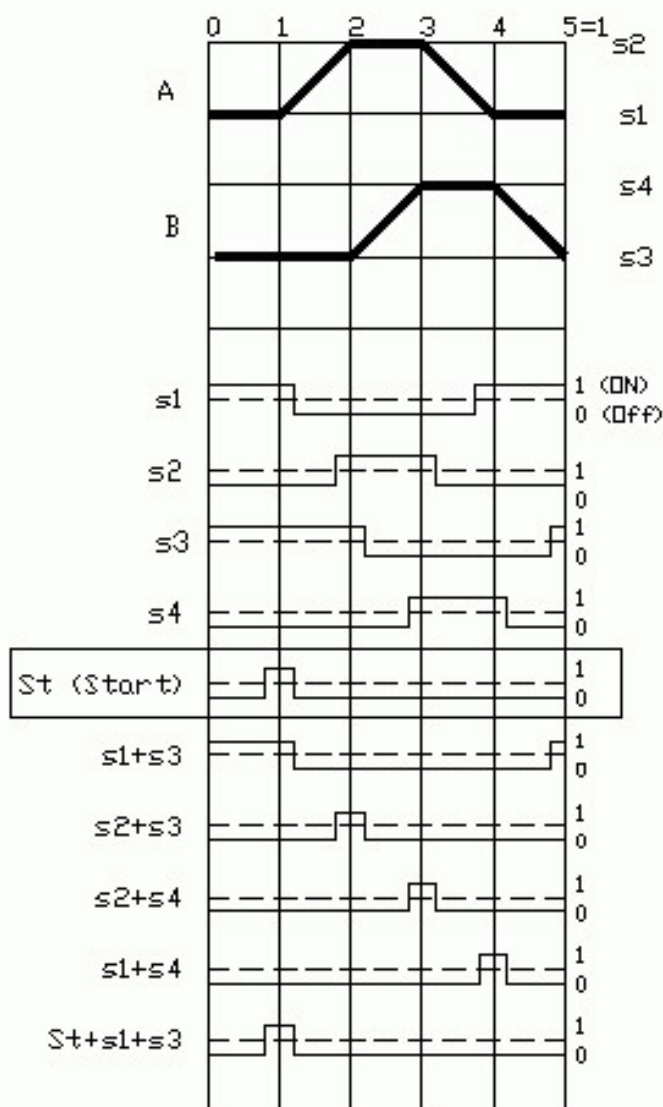


ลำดับ	Start Condition	End Condition	Reset	Set (สั่งงาน)
A	Open or Reset	$M0 + S1 + S3$	$M1, M2, M3, M4, Y1, Y3$	$M0, Y2 (A -), Y4 (B -)$
B	$M1 + St$	$M1 + S2 + S3$	$Y2$	$Y1 (A +)$
C	$M2$	$M2 + S2 + S4$	$M1$	$M2$
D	$M3$	$M3 + S1 + S4$	$Y4$	$Y3 (B +)$
E	$M4$	$M4 + S1 + S3$	$M2$	$M3$
			$Y1$	$Y2 (A -)$
			$M3$	$M4$
			$Y3$	$Y4 (B -)$
			$M4$	$M1$

+



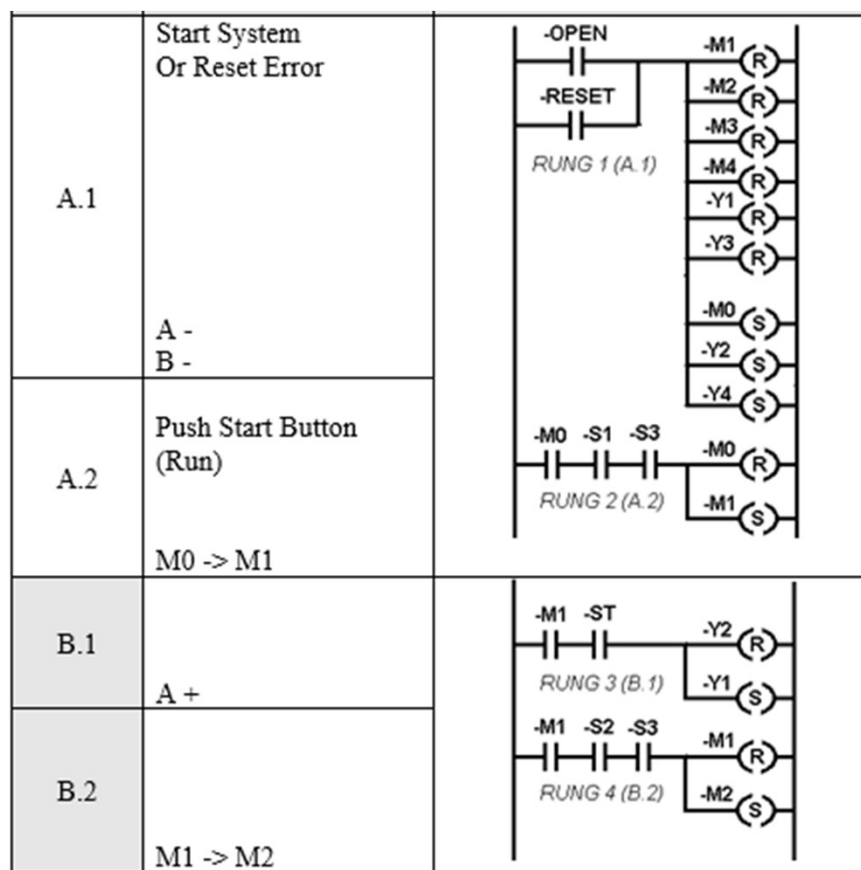
จากตัวอย่างทำการสร้าง Maker จำนวน 5 ตัวได้แก่ M0, M1, M2, M3, M4 และ M5 และบังคับให้ Maker แต่ละตัวมีค่าตามช่วงเวลาที่กำหนดและตามลำดับการเกิดก่อนหน้าหลัง



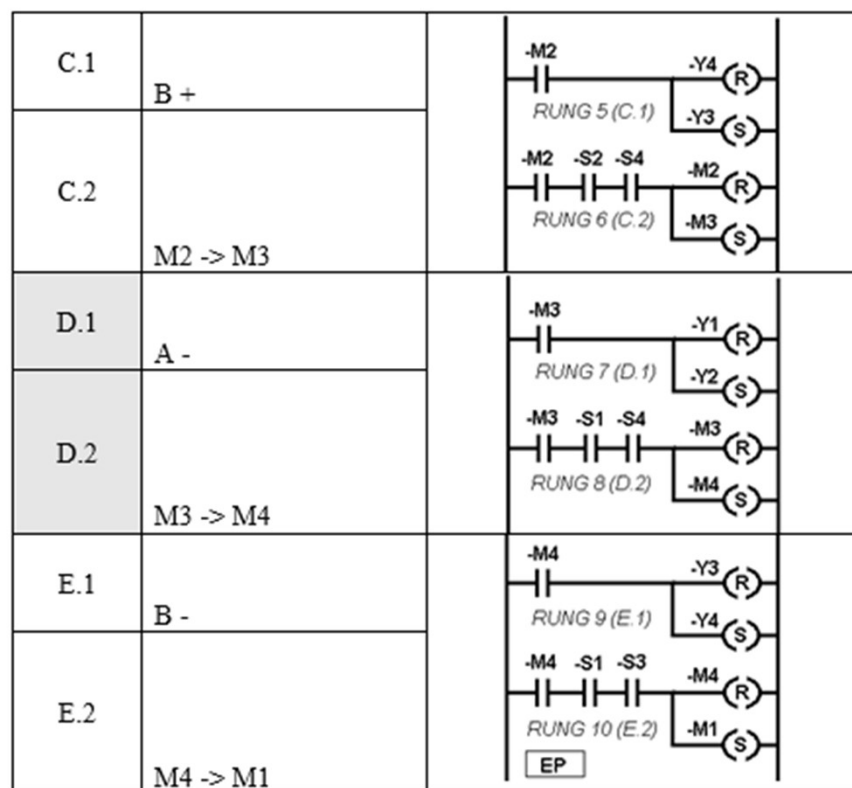
จากตัวอย่างทำการ
สร้าง Maker
จำนวน 5 ตัวได้แก่
M0, M1, M2,
M3, M4 และ M5
และบังคับให้
Maker แต่ละตัวมี
ค่าตามช่วงเวลา
ที่กำหนดและ
ตามลำดับการเกิด
ก่อนหน้าหลัง

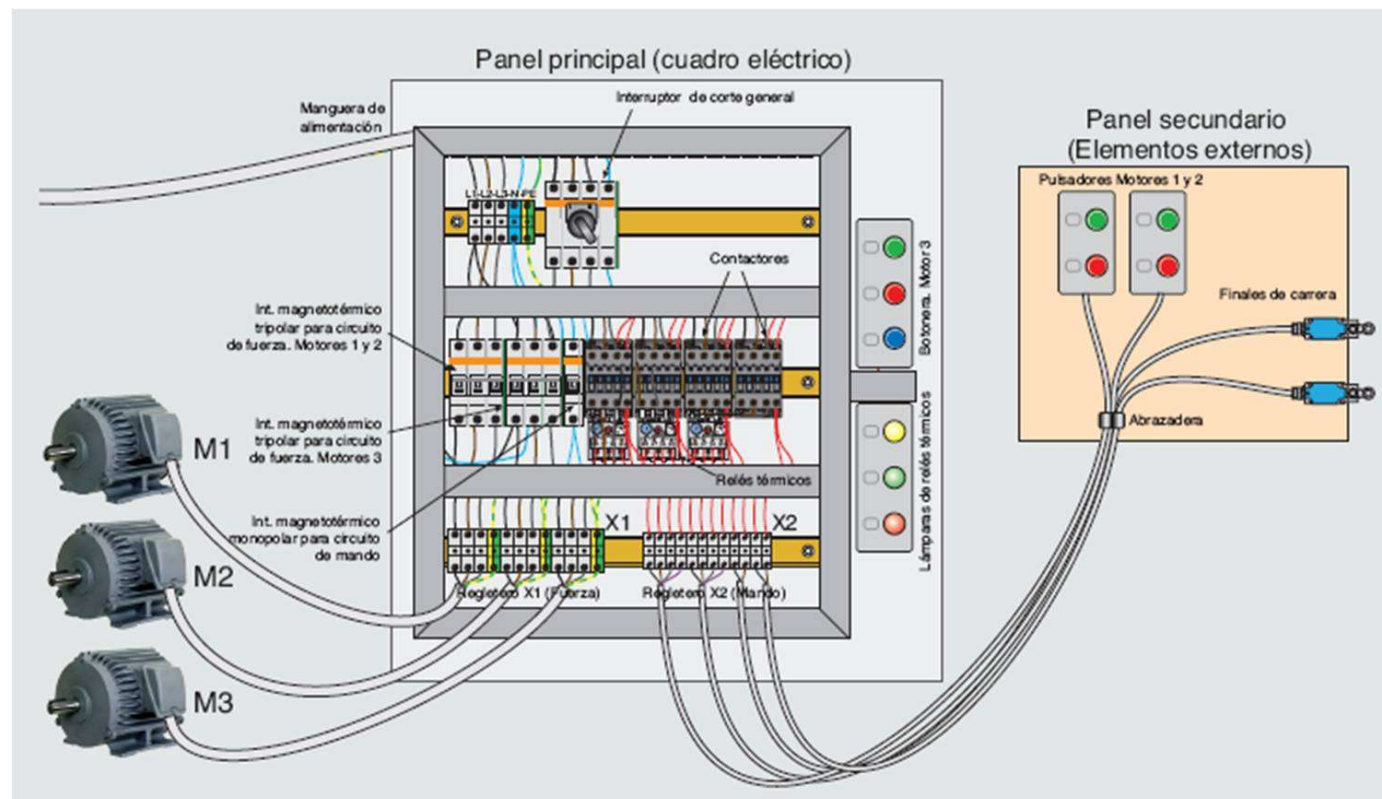
ลำดับ	Start Condition	End Condition	Reset	Set (สั่งงาน)
A	Open or <u>Reset</u>	M0+ S1 + S3	M1,M2,M3,M4, Y1,Y3	M0, Y2 (A -), Y4(B -)
B	M1+ St	M1 + S2 + S3	Y2	Y1 (A +)
C	M2	M2 + S2 + S4	Y4	Y3 (B +)
D	M3	M3 + S1 + S4	Y1	Y2 (A -)
E	M4	M4 + S1 + S3	Y3	Y4 (B -)
			M4	M1

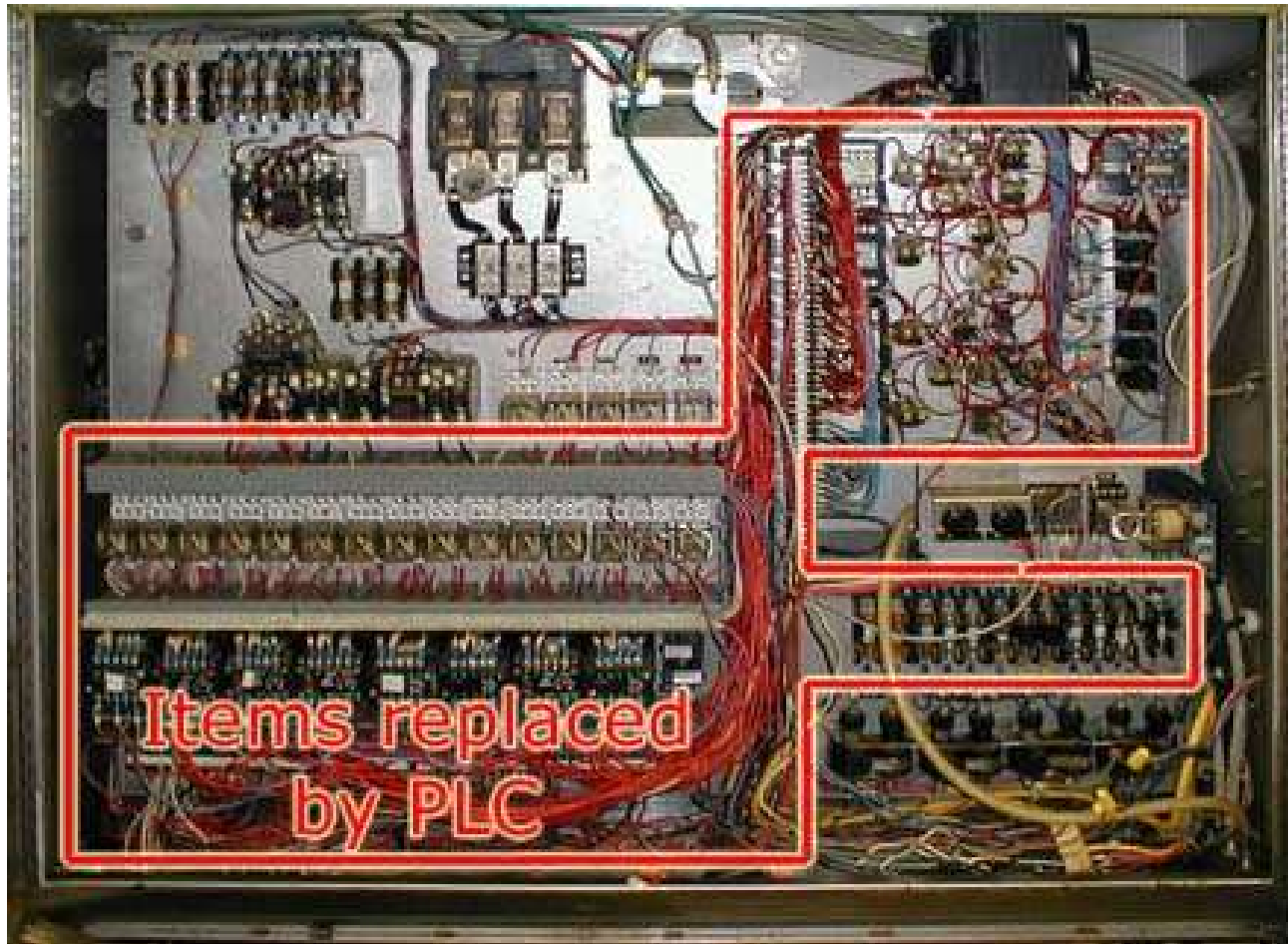
A	Open or <u>Reset</u>		M1, M2, M3, M4, Y1, Y3	M0, Y2 (A -), Y4 (B -)
		M0 + S1 + S3	M0	M1
B	M1 + St		Y2	Y1 (A +)
		M1 + S2 + S3	M1	M2



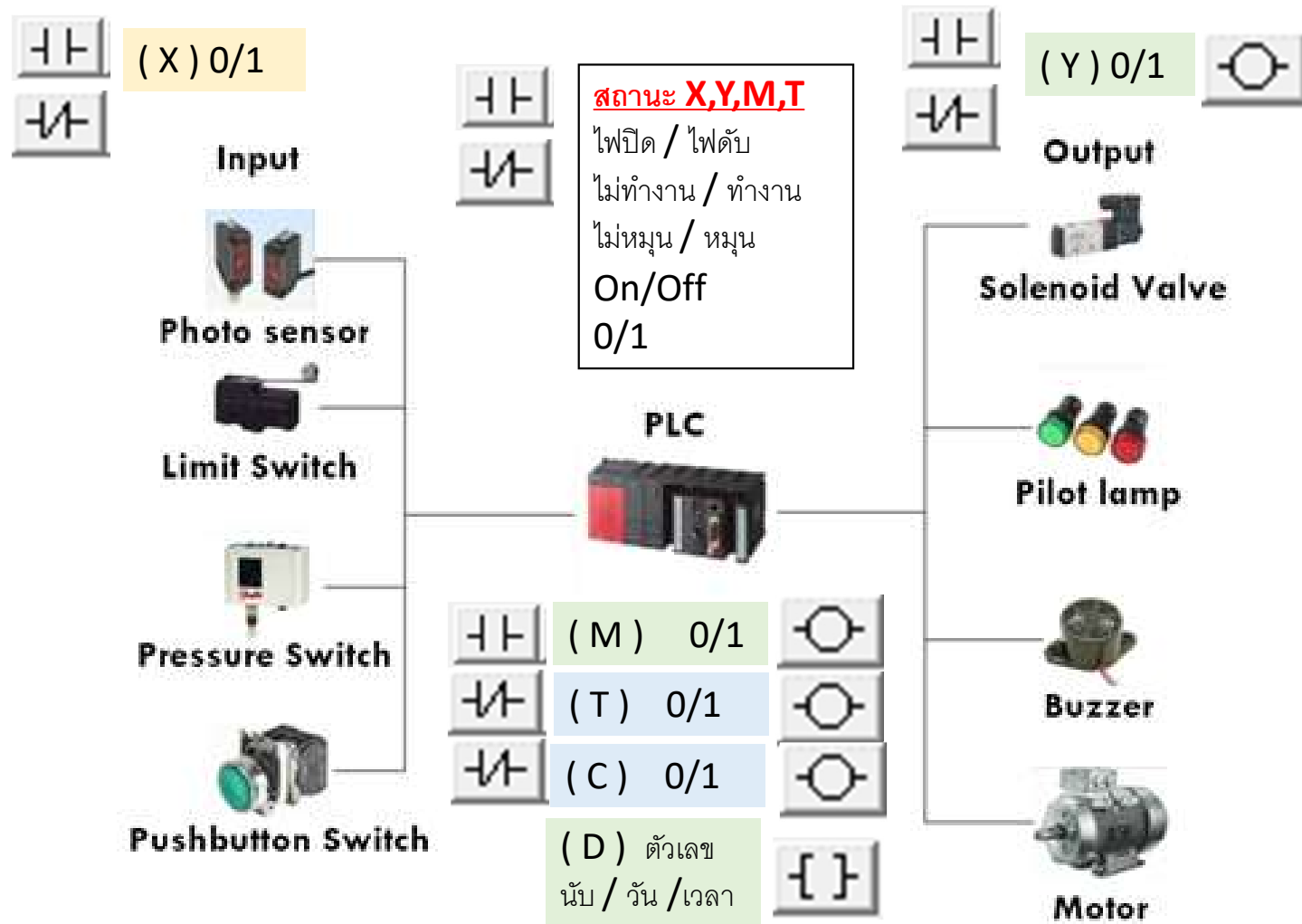
C	M2		Y4	Y3 (B +)
		M2 + S2 + S4	M2	M3
D	M3		Y1	Y2 (A -)
		M3 + S1 + S4	M3	M4
E	M4		Y3	Y4 (B -)
		M4 + S1 + S3	M4	M1







PLC (Programable Logic Control)



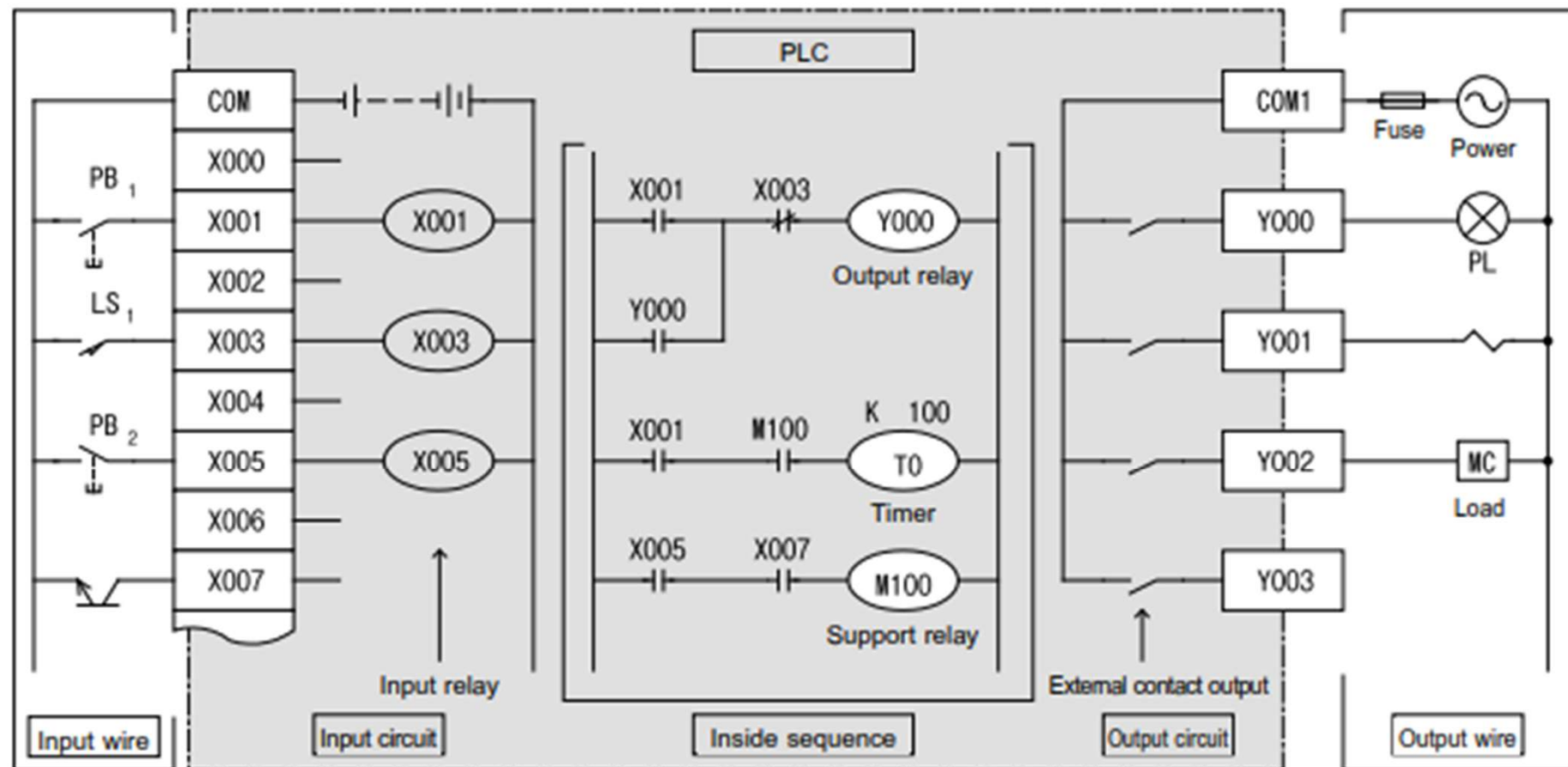
- (X) Input
- (Y) Output
- (M) Mem Relay
- (T) Timer
- (C) Counter
- (D) Memory

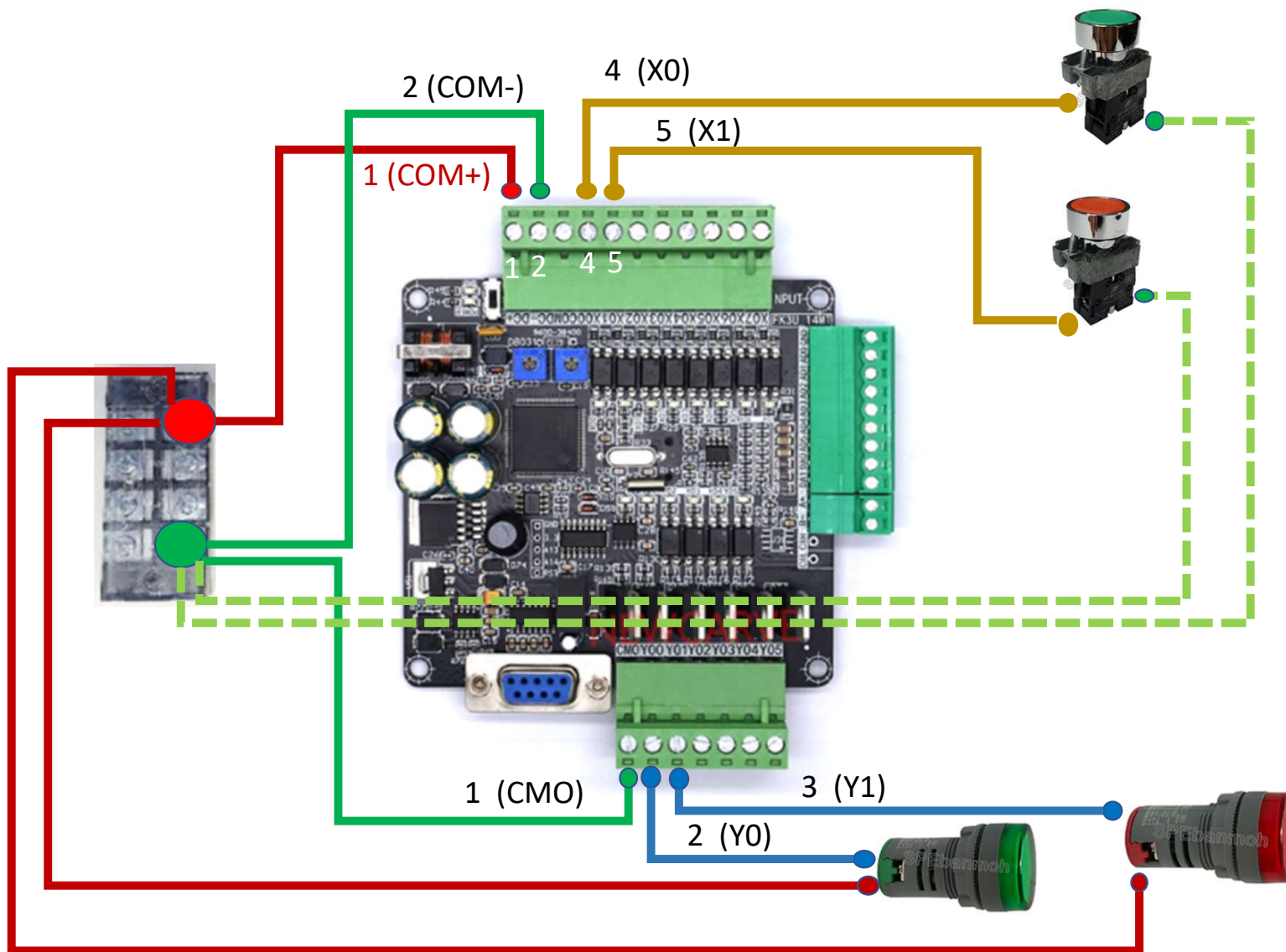
Input relay ON/OFF
ตามสัญญาณจากภายนอก

ซีควเอนซ์โปรแกรมทำงานด้วย
หน้าสัมผัสของ Input relay

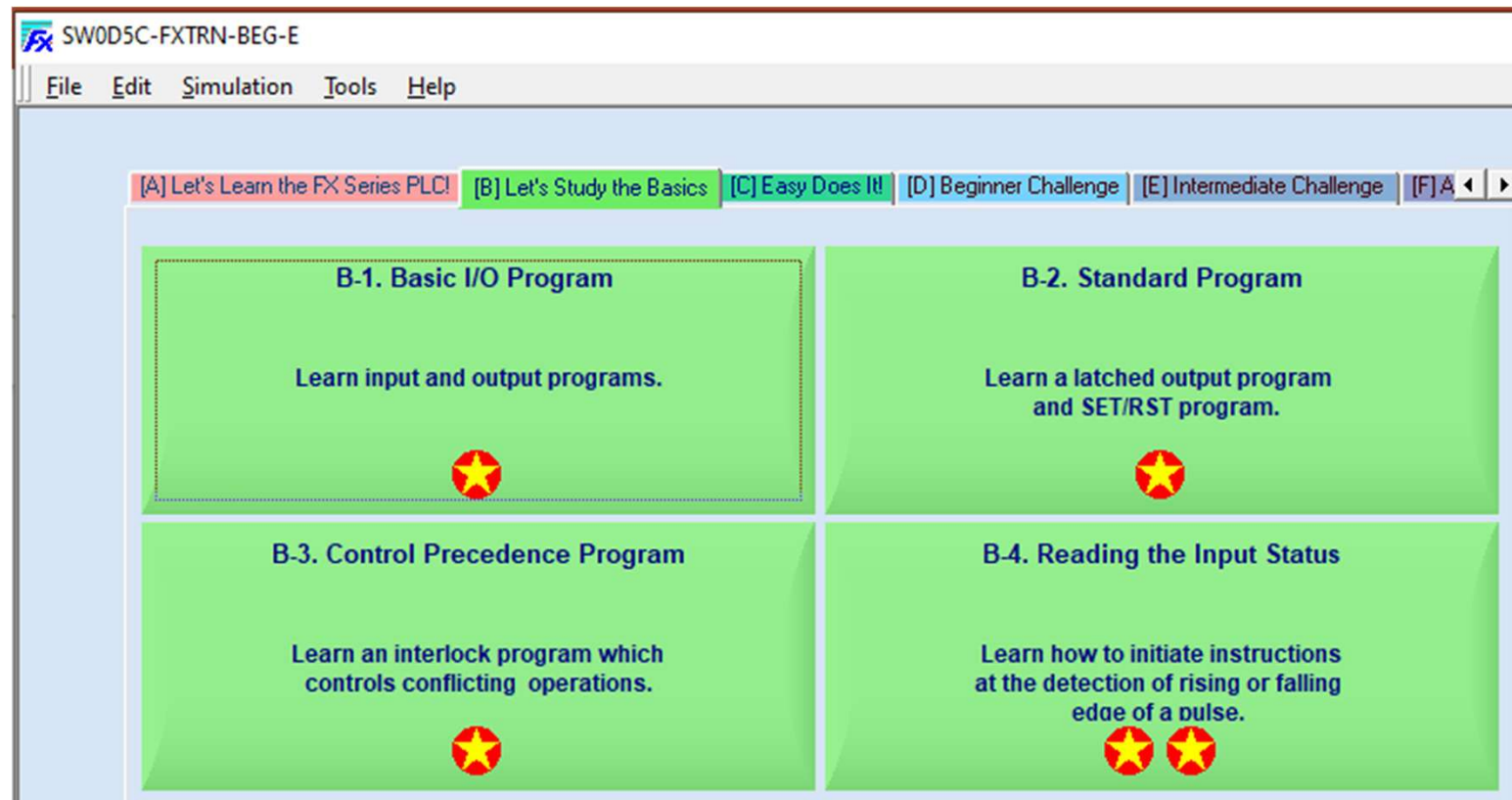
Output relay ON/OFF
ทำงานและส่งต่อ

อุปกรณ์ภายนอก Load
ทำงานหรือเคลื่อนที่





FXTRN



NO NC Contact [youtube](#)

SW0D5C-FXTRN-BEG-E

File Edit Simulation Tools Help

click

Edit Ladder
Write to PLC
Reset
F T S
Main
RUN

Operating Y0
Stop Y1
Error Y2

Basic I/O Program

X020
0 | | (Y000)
X021
2 | / (Y001)

Project Edit Convert View Online Tools

X020
0 | | (Y000)
X021
2 | / (Y001)
[END]

Write data to the PLC RUN

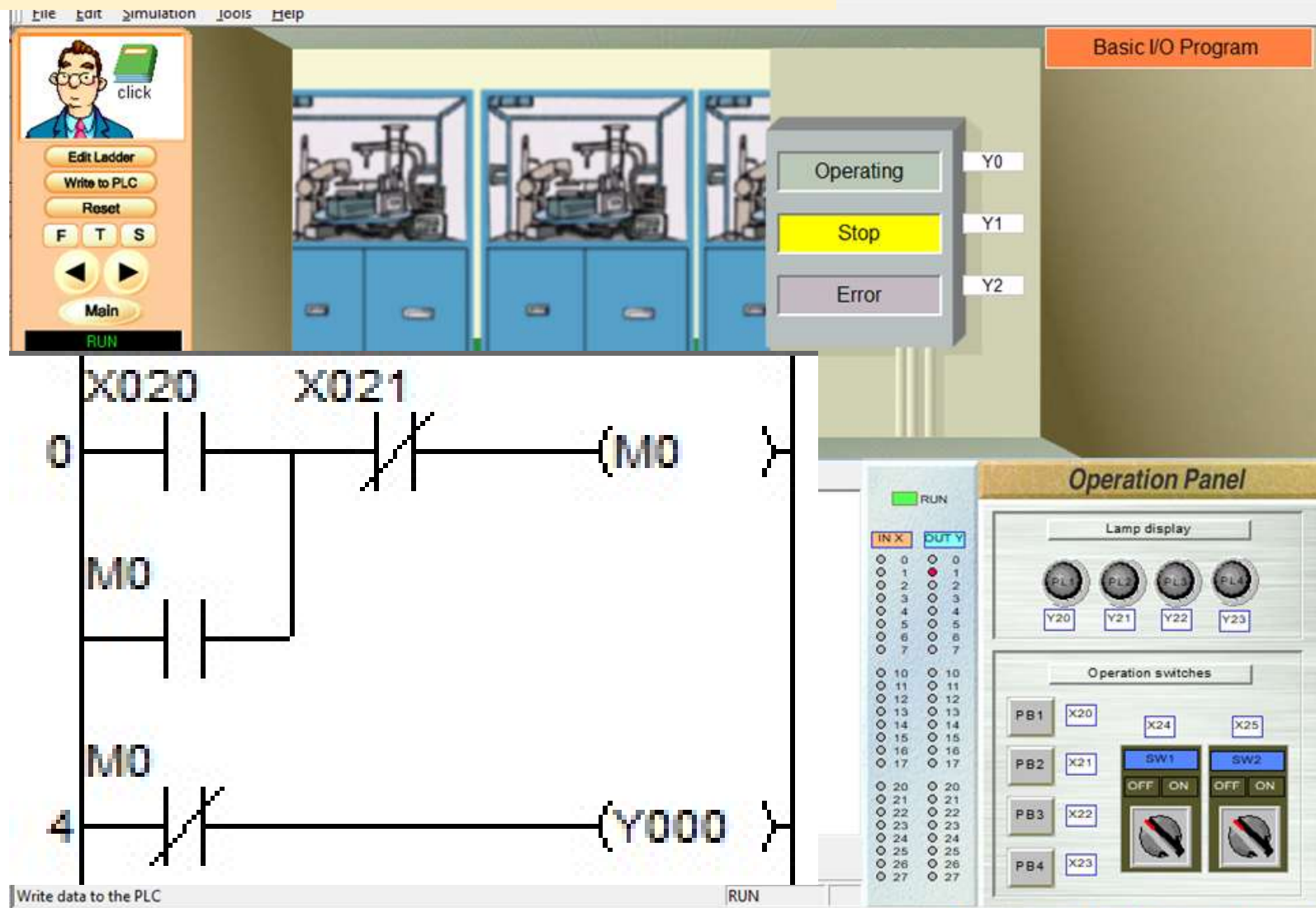
Operation Panel

Lamp display
PL1 PL2 PL3 PL4
Y20 Y21 Y22 Y23

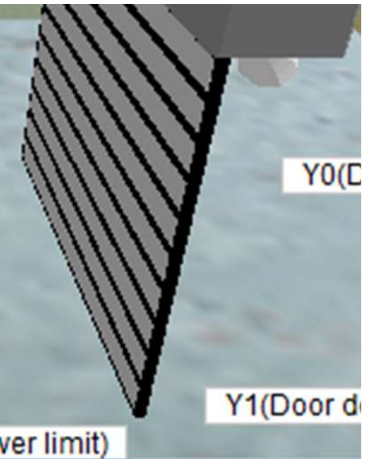
Operation switches
PB1 X20 X24 X25
PB2 X21 SW1 SW2
PB3 X22 OFF ON OFF ON
PB4 X23

IN X OUT Y
0 0 0 0
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
5 5 5 5
6 6 6 6
7 7 7 7
10 10 10 10
11 11 11 11
12 12 12 12
13 13 13 13
14 14 14 14
15 15 15 15
16 16 16 16
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24 24 24 24
25 25 25 25
26 26 26 26
27 27 27 27

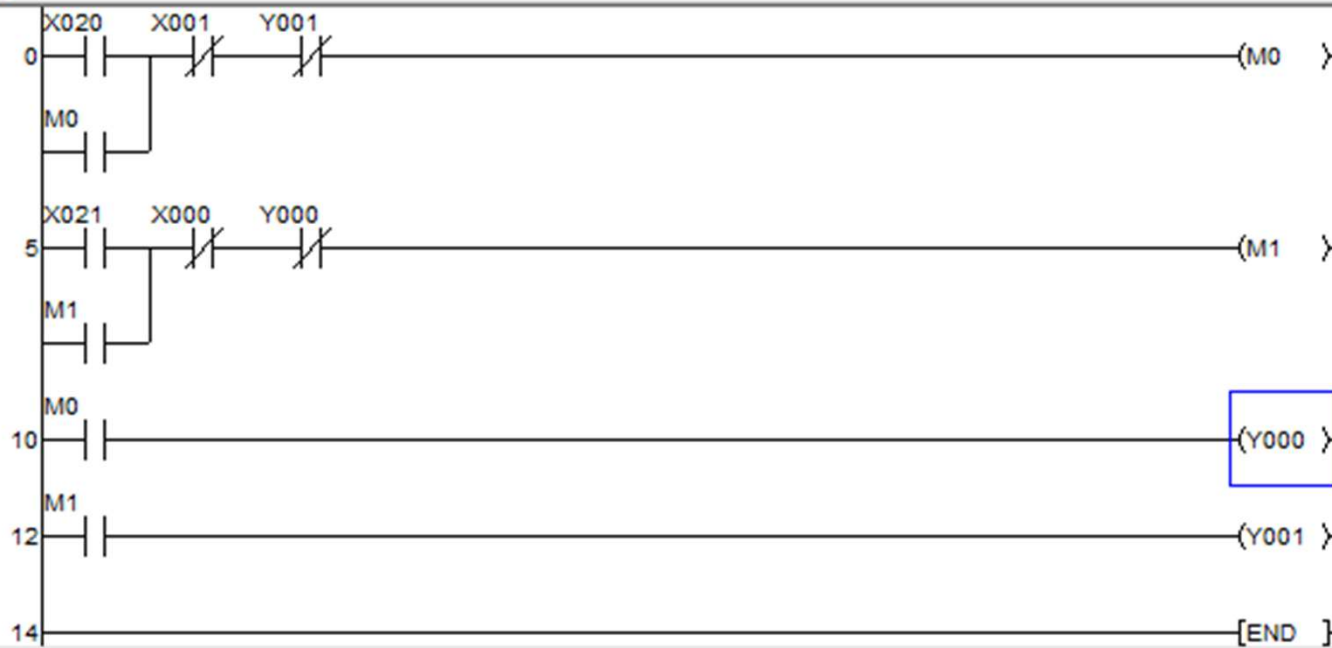
Self Holding



F T S
 ◀ ▶
 Main
 PROGRAM



Project Edit Convert View Online Tools



☐ RUN

IN X		OUT Y	
☒	0	☐	0
☐	1	☐	1
☐	2	☐	2
☐	3	☐	3
☐	4	☐	4
☐	5	☐	5
☐	6	☐	6
☐	7	☐	7
☐	10	☐	10
☐	11	☐	11
☐	12	☐	12
☐	13	☐	13
☐	14	☐	14
☐	15	☐	15
☐	16	☐	16
☐	17	☐	17
☐	20	☐	20
☐	21	☐	21
☐	22	☐	22
☐	23	☐	23
☐	24	☐	24

[A] Let's Learn the FX Series PLC!

[B] Let's Study the Basics

[C] Easy Does It!

[D] Beginner Challenge

[E] Intermediate Challenge

[F] A



C-1. Basic Timer Operation

Learn the On-delay time function.



C-2. Application Timer Program -1

Learn the Off-delay time function
and the one shot timer.



C-3. Application Timer Program - 2

Learn a "flicker" program
executed by timers.



C-4. Basic Counter Program

Learn control methods using counters.



Lab 07 Open and Close Door

SW0D5C-FXTRN-BEG-E

File Edit Simulation Tools Help

click

Edit Ladder

Write to PLC

Reset

F T S

Main

RUN

0 X020 X001 (Y000)

Y000

4 X021 X000 (Y001)

Y001

Basic Timer Operation

X1(Upper limit)

Y5(Red)
Y6(Green)
Y7(Yellow)

Y0(Door up command)

Y1(Door down command)

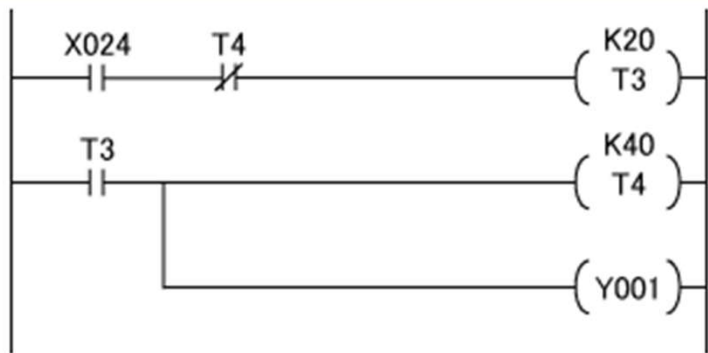
X0(Lower limit)

PB1 X20

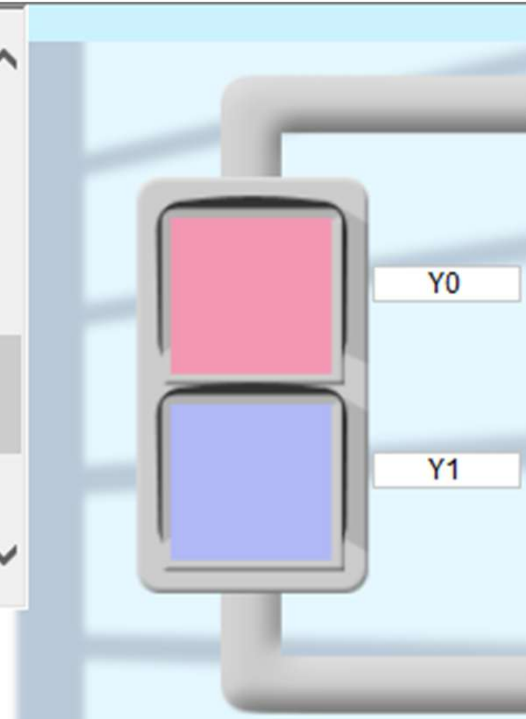
PB2 X21

Timer Cycle [YouTube](#)

In the 'Enter symbol' dialog box, enter **T3 K20** or **T4 K40** and click **[OK]** when writing the timer setting.
(Use a space between **T3** and **K20**)



3 Press the **[F4]** key to convert the program you have input.



Lab 8 PLC 03 – Basic Timer

SW003C-FAIRIN-BEG-E

File Edit Simulation Tools Help

click

Edit Ladder
Write to PLC
Reset
F T S
Main
RUN

3 Press the [F4] key to convert the program you have input.

Basic Timer Operation

X1(Upper limit)
Y5(Red)
Y6(Green)
Y7(Yellow)

Y0(Door up command)
PB1 X20
PB2 X21

Y1(Door down command)

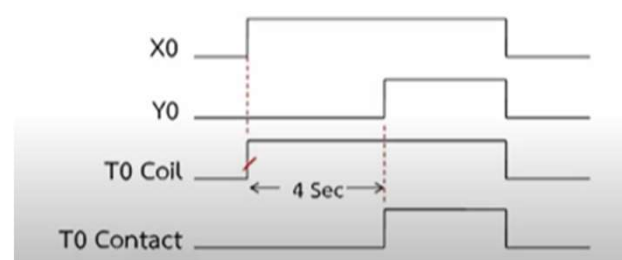
X0(Lower limit)

Timer

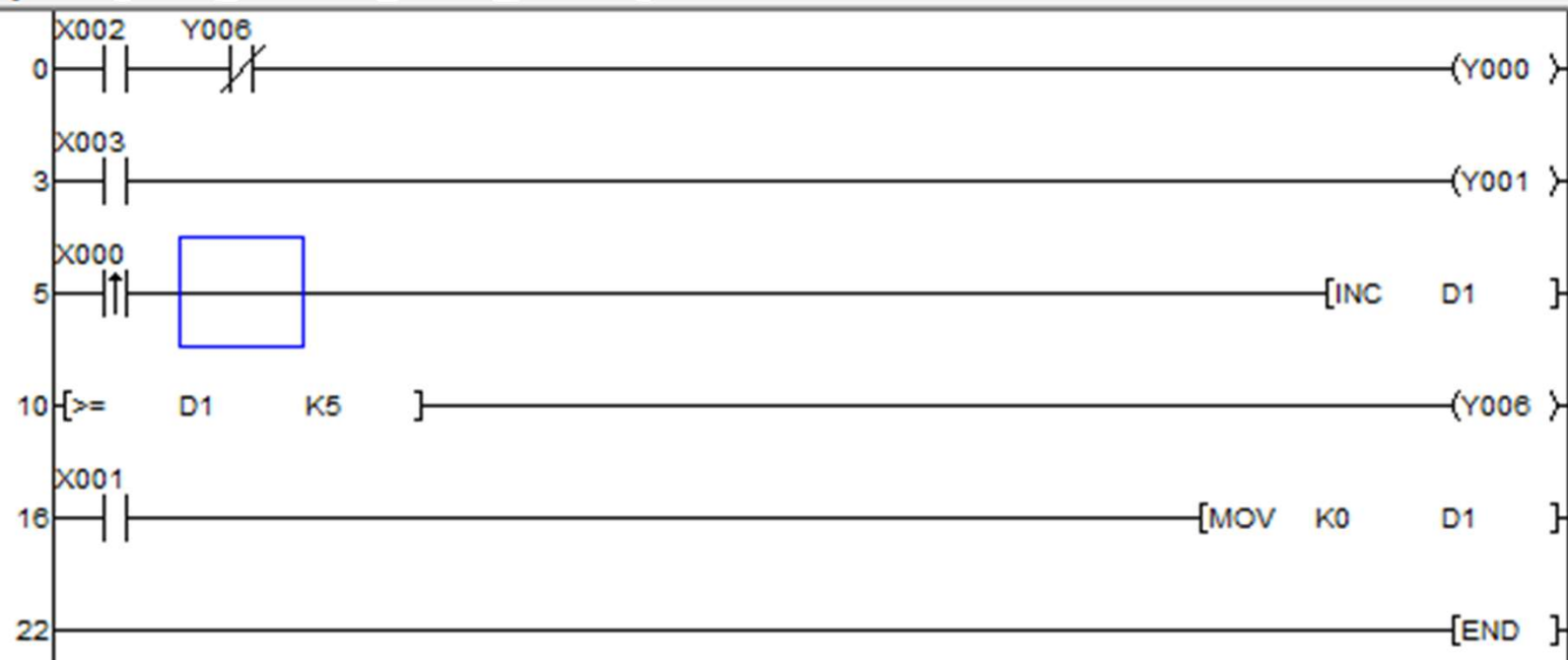
Enter symbol

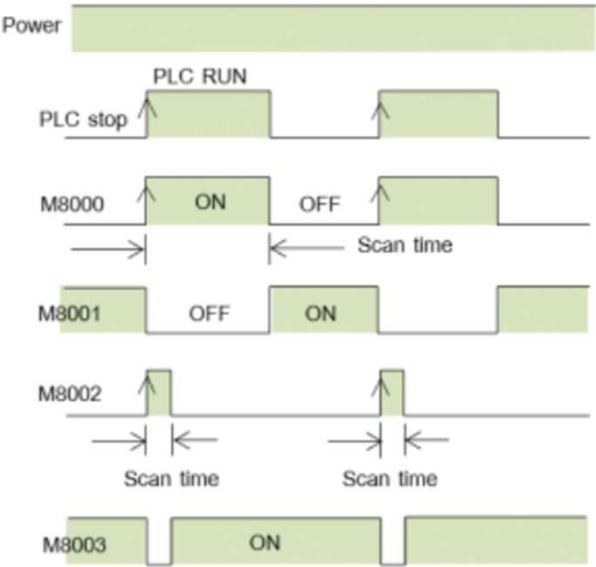
[-()-]

T0 K30

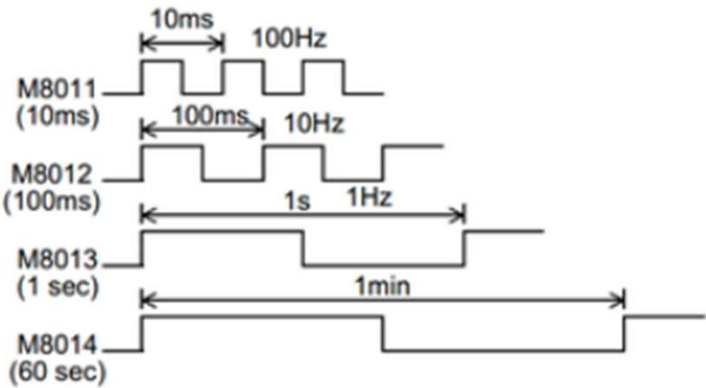


Project Edit Convert View Online Tools

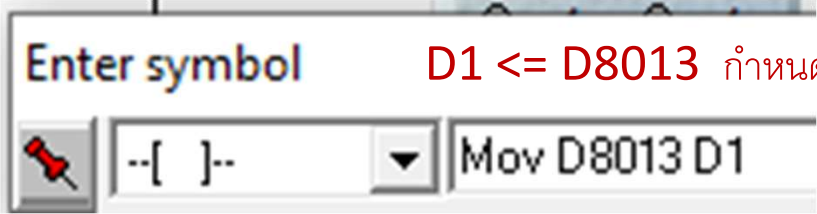




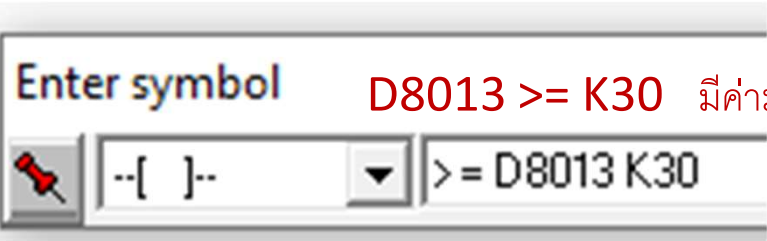
M8011 = 10 ms. (0.01 วินาที)
M8012 = 100 ms. (0.1 วินาที)
M8013 = 1 s. (1 วินาที)
M8014 = 60 s. (1 นาที)



Number	Name	Set value range
D8013	Second data	0 to 59
D8014	Minute data	0 to 59
D8015	Hour data	0 to 23
D8016	Day data	1 to 31
D8017	Month data	1 to 12
D8018	Year data	00 to 99 (last two digits of year)
D8019	Day-of-the-week data	0 to 6 (which corresponds to Sunday to Saturday)



D1 <= D8013 กำหนดค่า D1



D8013 >= K30 มีค่ามากกว่า 30 หรือไม่

Lab 9 Time Schedule

Write to PLC

Reset

F T S

◀ ▶

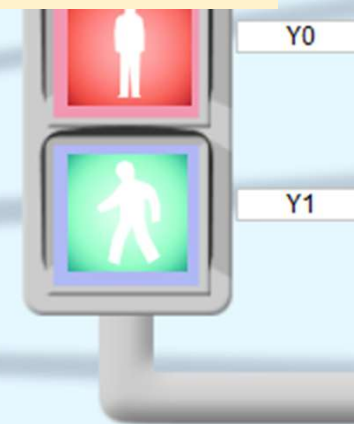
Main

PROGRAM

The following timers are prepared in the special auxiliary relays for use by the PLC.
These relays can be used for simple operations such as flickering a lamp.

- M8011 10ms (0.01sec)
- M8012 100ms (0.1sec)
- M8013 1sec
- M8014 1min

ไฟสว่างระหว่าง วินาที ที่
0 - 15 และ 30 - 45



Enter symbol

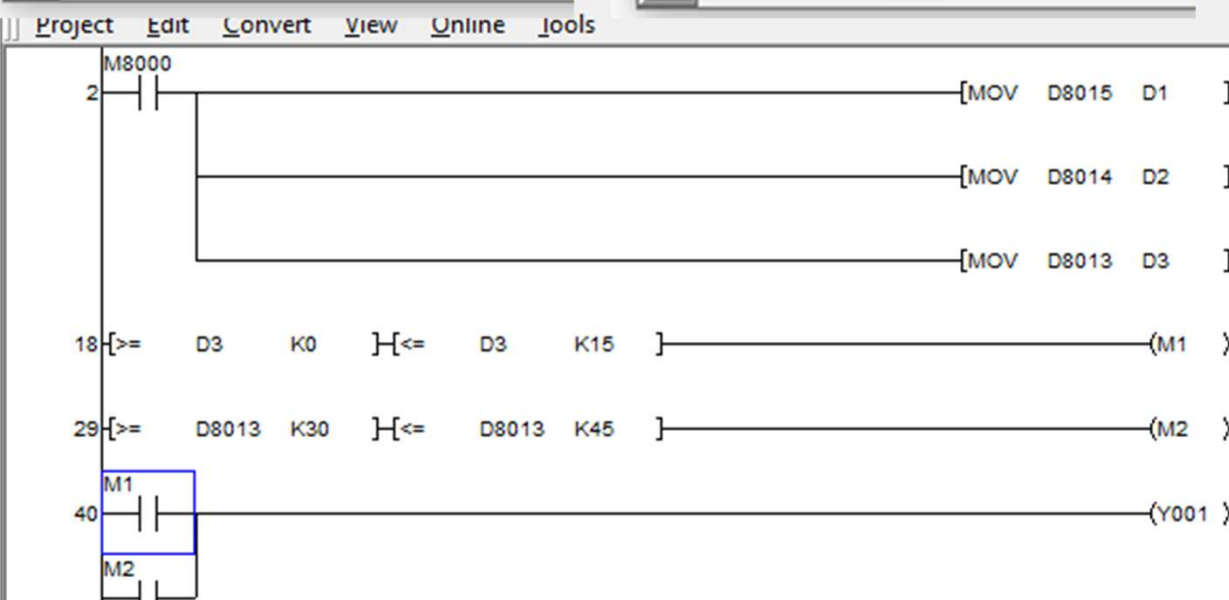
--[]--

MOV D8015 D1

Enter symbol

--[]--

>= D8013 K30



RUN

IN X OUT Y

0 0

1 1

2 2

3 3

4 4

5 5

6 6

7 7

10 10

11 11

12 12

13 13

14 14

15 15

16 16

17 17

20 20

21 21

22 22

23 23

0

1

2

3

4

5

6

7

10

11

12

13

14

15

16

17

20

21

22

23

Lamp display

PL1

PL2

PL3

PL4

Y20

Y21

Y22

Y23

Operation switches

PB1

PB2

PB3

X20

X21

X22

X24

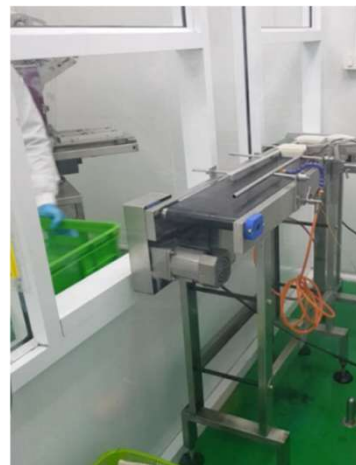
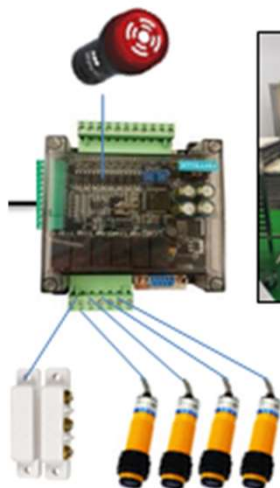
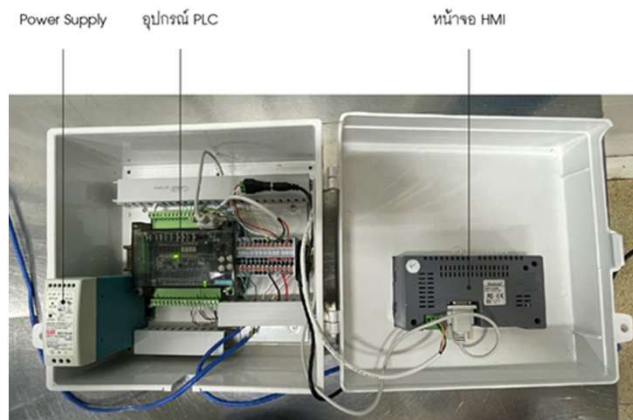
X25

SW1

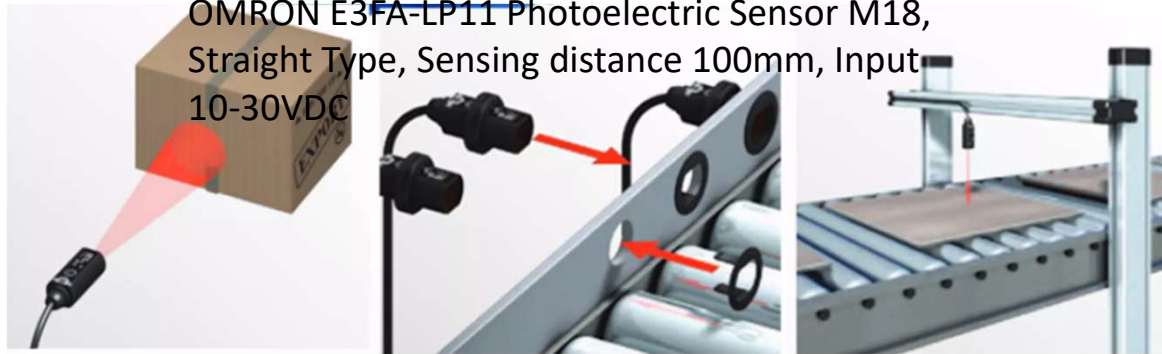
SW2

OFF ON

OFF ON



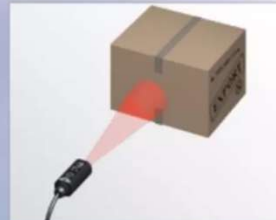
OMRON E3FA-LP11 Photoelectric Sensor M18,
Straight Type, Sensing distance 100mm, Input
10-30VDC



Unrivalled Detection with Simplicity in Setup and Installation



The short body of the E3FA/E3RA fits in tighter mounting spaces.



Visible red LED light for easy alignment.



Transparent object detection sensors utilize Omron's unique technology for detecting objects with birefringent (double refraction) properties.



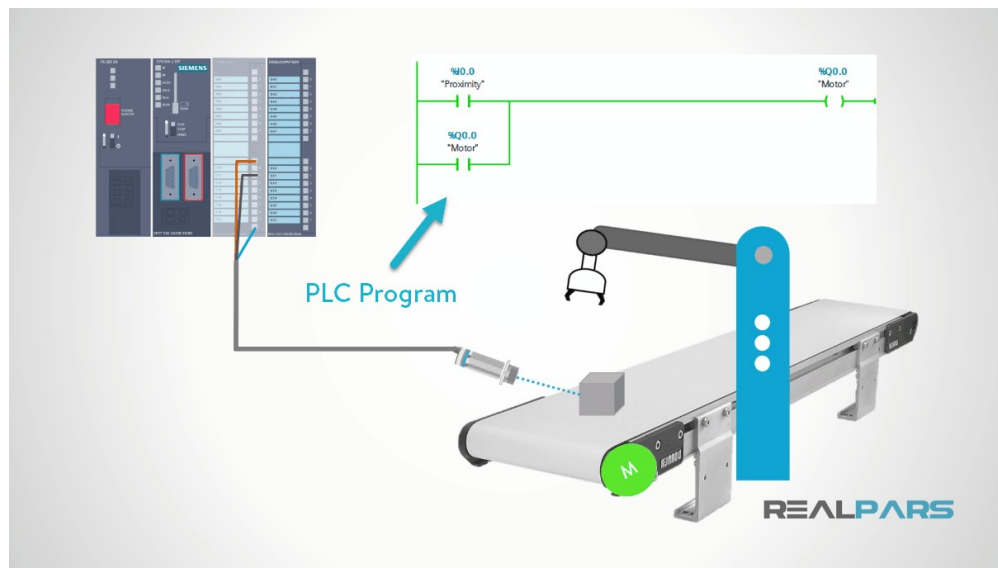
Bright LED indicators for status visibility and large sensor adjusters for use with a standard size screwdriver.



Flush, snap mounting option for quick and easy installation.



High power LED to compensate for dirt and misalignment.



Lab 10 Counter

Basic Counter Program

Y3(Red)
Y4(Green)
Y5(Yellow)

Supply allowed

X0

Y1(Conveyor forward)

click

Edit Ladder
Write to PLC
Reset
F T S
Main
PROGRAM



tion Panel

Reset counter
X1

Move conveyor
OFF ON
X3

Ladder Logic:

```
0 X002 C0 [ ] (Y000)
3 X003 (Y001)
5 X000 K5 (C0)
9 C0 (Y006)
11 X001 [RST C0]
14 [END]
```

Lab 11 Automation

- M8002 เปิดเครื่อง -> **Reset M1,M2,M3,M4, Set M0**
Set M0 , Y1 (Door Down)
- (M0 AND X0) รอประตูปิด -> **Reset M0 , Y1**
Set M1
- (M1 +X2) รถมาถึง -> **Reset M1,Y1**
Set M2 , Y0 (Door Up)
- (M2 AND X1) รอประตูเปิด -> **Reset M2, Y0**
Set M3
- (M3 + X3 - X2) รถรุดผ่าน -> **Reset M3**
Set M4 , Y1 (Door Down)
- (M4 + X0) รอประตูปิด -> **Reset M4 ,Y1**
Set M5
- (M5 - X3) รถรุดออก -> **Reset M5**
Set M1 (รถรุ่นใหม่มา)

